
Supplementary information

Small targeted dietary changes can yield substantial gains for human health and the environment

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Supplementary Information for

Prioritization of healthy and sustainable foods for small targeted dietary changes can yield substantial gains for human health and the environment

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This document contains the detailed methodology and materials used in this analysis, as well as supplementary text, figures, and tables. The document is divided into five sections. More specifically, the methods and material used to develop the nutritional health burden characterization factors are described in detail in section S1 and the Health Nutritional Index in section S2. In section S2, supplementary discussion, figures, and tables are provided in regards to the nutritional performance of individual foods in the US diet. The methodology followed to determine the life cycle assessment of individual foods is summarized in section S3, and in section S4 nutritional and environmental performances of the 167 commonly consumed foods in the US diet are discussed. Finally, section S5 summarizes the methods used for the food substitution analysis.

S1. Dietary risk Factors (DRFs)

S1.1. Description of dietary risk components

Building on the Global Burden of Disease (GBD) studies, dietary risk factors (DRFs) were developed to quantify the health burden associated with a marginal dietary change (e.g., one-gram consumption increase) of dietary risks for 15 GBD-identified dietary risks¹. The definitions and characteristics of all dietary risks are available in Suppl. Table 1.

Suppl. Table 1. Definition, description, and characteristics of dietary risk factors.

Dietary risk component	Description*	Effective intake*	Associated health outcomes (effect)
Calcium	Calcium from all sources	<1.25 g/day	Colorectal cancer
Fiber [†]	Fiber from all sources	<23.5 g/day	IHD; colorectal cancer
Polyunsaturated Fatty Acids (PUFA)	Omega-6 fatty acids from all sources	<11% total energy intake	IHD
Omega-3 Fatty Acids from seafood	Eicosapentaenoic & docosahexaenoic acids	<250 mg/day	IHD
Sodium [‡]	Dietary sodium from all sources	>3.49 g/day	15 [§]
Trans fatty acids (TFA)	Trans fat from all sources	>0.5% total energy intake	IHD
Fruits	Fresh, frozen, cooked, canned, or dried, excluding fruit juices and salted or pickled fruits	< 250 g/day	10 ^{**}
Milk	Milk (including non-fat, low-fat, and full-fat milk) but excluding plant derivatives	<435 g/day	Colorectal cancer
Nuts and seeds	Nut and seed foods	<20.5 g/day	T2DM; IHD
Processed meat	Meat preserved by smoking, curing, salting, or addition of chemical preservatives	>2 g/day	T2DM; IHD; Colorectal cancer
Legumes	Fresh, frozen, cooked, canned, or dried legumes	<60 g/day	IHD
Red meat	Beef, pork, lamb, and goat but excluding poultry, fish, eggs, and all processed meats	>22.5 g/day	T2DM; Colorectal cancer
Sugar-sweetened beverages ^{††} (SSB)	Beverages >50 kcal per 226.8 g serving, including carbonated beverages, sodas, energy drinks, and fruit drinks, but excluding 100% fruit and vegetable juices	>2.5 g/day	38 ^{‡‡}
Vegetables	Fresh, frozen, cooked, canned, or dried vegetables, excluding legumes, salted or pickled vegetables, juices, and starchy vegetables	<360 g/day	Hemorrhagic stroke; IHD; IS
Whole grains	Whole grains from breakfast cereals, bread, rice, pasta, biscuits, muffins, tortillas, pancakes, and other sources	<125 g/day	T2DM; Hemorrhagic stroke; IHD; IS

Risks highlighted in green indicate a protective effect on health, whereas red indicates detrimental health effects.

T2DM=Type 2 Diabetes mellitus; IHD=Ischemic heart disease; IS=Ischemic stroke; Other CVD=Other cardiovascular and circulatory disease

*Description obtained from Gakidou et al. 2017

[†]The effect on IHD is 100% mediated through fruits, vegetables, legumes, and whole grains¹

[‡]Based on "Diet high in sodium" described as 24 h urinary sodium in g/day. Health effects are mediated through systolic blood pressure. Effective intake calculated as $3 \frac{\text{grams urinary sodium}}{\text{day}} / 0.85 \frac{\text{grams dietary sodium}}{\text{grams urinary sodium}} = 3.49 \frac{\text{grams dietary sodium}}{\text{day}}$

[§]The 15 health outcomes associated with sodium are: Aortic aneurysm; Atrial fibrillation and flutter; Cardiomyopathy and myocarditis; Chronic kidney disease due to T2DM; Chronic kidney disease due to glomerulonephritis; Chronic kidney disease due to hypertension; Chronic kidney disease due to other causes; Endocarditis; Hemorrhagic stroke; Hypertensive heart disease; IHD; IS; other CVD; Peripheral artery disease; Rheumatic heart disease; Stomach cancer

^{**}The ten health outcomes associated with fruits are: T2DM; Esophageal cancer; Hemorrhagic stroke; IHD; IS; Larynx cancer; Lip and oral cavity cancer; Nasopharynx cancer; Other pharynx cancer; Tracheal, bronchus, and lung cancer

^{††}Health effects are mediated through body mass index

^{‡‡}The 38 health outcomes associated with SSB, mediated through body mass index, are: Atrial fibrillation and flutter; Breast cancer; Cardiomyopathy and myocarditis; Chronic kidney disease due to T2DM; Chronic kidney disease due to glomerulonephritis; Chronic kidney disease due to hypertension; Chronic kidney disease due to other causes; colorectal cancer; T2DM; Endocarditis; Esophageal cancer; Gallbladder and biliary tract cancer; Hemorrhagic stroke; Hypertensive heart disease; IHD; IS; Kidney cancer; Leukemia; Liver cancer; Low back pain; other CVD; Ovarian cancer; Pancreatic cancer; Peripheral artery disease; Thyroid cancer; Uterine cancer

S1.2. Mediated dietary risks

The health effects of two dietary risks (SSB and sodium) were determined by the GBD to be mediated by metabolic risks. Therefore, DRFs for these risks were calculated using a two-step approach. First, each dietary risk (e.g., SSB) was linked to a change in the metabolic risk (e.g., BMI). Second, the corresponding metabolic risk change (e.g., BMI) was linked to an attributable disease burden (e.g., diabetes). Consequently, this approach indirectly associated the three dietary risks with all the health outcomes associated with their respective metabolic risks.

S1.2.1. Sodium and systolic blood pressure

Evidence supports that the health effect of sodium is mediated through SBP^{1,2}, with race (black versus non-black) and hypertension status (hypertensive versus non-hypertensive) being significant effect modifiers³. In addition, sodium exposure is reported using 24-hour urinary sodium estimates in g/day¹. Therefore, the DRFs of dietary sodium for a given health outcome o , age group a , gender g , effect modifier class m ($DRF_{d.sodium,o,b}^{a,g,m}$) were estimated using:

$$DRF_{d.sodium,o,b}^{a,g,m} = \frac{\ln RR_{SBP,o,b}^{a,g}}{Ref_{SBP}} \cdot \frac{\Delta SBP^{a,g,m}}{\Delta u.sodium} \cdot f_{u.sodium \rightarrow d.sodium} \cdot \widehat{BR_{d.sodium,o,b}^{a,g}} \quad (Eq. S1)$$

$$117 \quad \text{with } \widehat{BR_{d.sodium,o,b}^{a,g}} = \frac{BR_{o,b}^{a,g}}{\sum_x P_{x_{d.sodium}}^{a,g} \cdot RR_{SBP,o,b}^{a,g} \left(\frac{x_{d.sodium}}{Ref_{SBP}} \cdot \frac{\Delta SBP^{a,g}}{\Delta u.sodium} f_{u.sodium \rightarrow d.sodium} \right)}$$

118 where $RR_{SBP,o,b}^{a,g}$ is the RR for SBP and outcome o for a given age group a, gender g, and
 119 burden b, corresponding to a reference exposure (Ref_{SBP}) of 10 mmHg¹. $\frac{\Delta SBP^{a,g,m}}{\Delta u.sodium}$ is an
 120 age-gender-modifier-specific conversion coefficient that characterizes the change in SBP
 121 levels in mmHg for a given change in urinary sodium concentrations, assumed to be 2.3
 122 g/day⁴. Coefficient estimates were derived from the 2016 GBD¹. The conversion factor
 123 $f_{u.sodium \rightarrow d.sodium}$ links urinary to dietary sodium. For the US population this factor was
 124 estimated at 86% (standard error SE: 1.6%), i.e. for every 1 g of sodium consumed
 125 (d.sodium) 0.86 g of sodium are excreted in urine (u.sodium)⁵. $\widehat{BR_{d.sodium,o,b}^{a,g}}$ is the age-
 126 gender-burden-specific BR of an outcome o adjusted for the background dietary sodium
 127 population exposure ($x_{d.sodium}$, in g) in μ DALYs/person/day.

128 S1.2.2. Sugar-sweetened beverages and body mass index

129 According to the GBD, the health effects associated with SSB consumption is
 130 mediated via BMI¹, with BMI status (BMI $\geq$ 25 vs. BMI<25) reported to modify this
 131 relationship^{1,3,6}. The following equation was used to estimate the SSB-DRFs for a given
 132 health outcome o, age group a, gender g, effect modifier class m ($DRF_{SSB,o,b}^{a,g,m}$):

$$133 \quad DRF_{SSB,o,b}^{a,g,m} = \frac{\ln RR_{BMI,o,b}^{a,g}}{Ref_{BMI}} \cdot \frac{\Delta weight^{a,g,m}}{height^g \cdot \Delta SSB} \cdot \widehat{BR_{SSB,o,b}^{a,g}} \quad (\text{Eq. S2})$$

$$134 \quad \text{with } \widehat{BR_{SSB,o,b}^{a,g}} = \frac{BR_{o,b}^{a,g}}{\sum_x P_{x_{SSB}}^{a,g} \cdot RR_{BMI,o,b}^{a,g} \left(\frac{x_{SSB}}{Ref_{BMI}} \cdot \frac{\Delta weight^{a,g,m}}{height^g \cdot \Delta SSB} \right)}$$

135 where $RR_{BMI,o,b}^{a,g}$ is the RR for BMI and outcome o for a given age group a, gender g, and
 136 burden b, corresponding to a reference exposure (Ref_{BMI}) of 5 $\frac{kg}{m^2}$. $\frac{\Delta weight^{a,g,m}}{height^g \cdot \Delta SSB}$ is an age-

gender-modifier-specific conversion coefficient that characterizes the change in body weight in kg for a given change in SSB consumption, assumed to be 226.8 g/day¹, converted to BMI units using the age-adjusted gender-specific height ($\overline{height^g}$) for US adults, which were determined using the NHANES 2011-2016⁷. Coefficient estimates were derived from the 2016 GBD¹. Finally, $\widehat{BR}_{SSB,o,b}^{a,g}$ is the age-gender-burden-specific BR of an outcome o adjusted for the background SSB consumption in the US (x_{SSB}), expressed in μ DALYs/person/day.

S1.3. Cumulative DRF estimates

Suppl. Table 3 summarizes the cumulative age-gender-adjusted DRFs for US adults. DRF estimates typically vary between 17% for sodium and whole grains and 58% for red meat and milk round the mean estimates. The DRFs for PUFA and legumes are also relatively uncertain. Finally, it was observed that males and 50 to 79 years olds experience the majority of the burden associated with these dietary risks (Suppl. Fig. 1). DRF estimates are not directly comparable between risk components, as DRFs need to be multiplied by exposure levels (e.g., daily intake of sodium or fruits in g/day) that vary substantially between nutrients and food groups.

Suppl. Table 2. Physical activity-adjusted estimates of energy requirements (kcal/person/day).

Age group	Physical activity-adjusted energy requirement (kcal/person/day)	
	Males	Females
25-29	2861	2163
30-34	2833	2051
35-39	2696	2047
40-44	2650	2028
45-49	2594	2014
50-54	2554	1862
55-59	2445	1825
60-64	2375	1781
65-69	2302	1795
70-74	2267	1760
75-79	2198	1736
80+	2140	1702

Energy requirement estimates were obtained from the 2015-2020 Dietary Guidelines for Americans⁸ and physical activity from NHANES 2011-2016⁷.

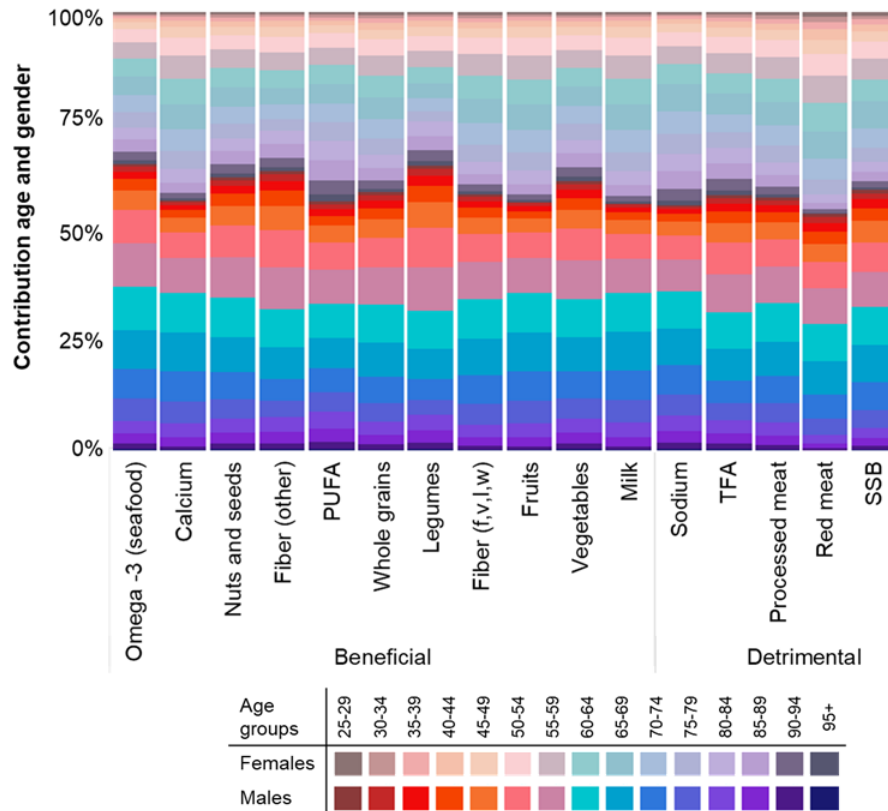
Suppl. Table 3. 95% confidence interval (CI) characterization of dietary risk factors (DRFs) in μ DALYs/g.

Dietary risk	DRF (μ DALYs/g)		
	Mean	Lower	Upper
<i>Omega-3 (seafood)</i>	-81	-37	-110
<i>Calcium</i>	-5.1	-4.0	-6.2
<i>Nuts and seeds</i>	-1.5	-1.1	-1.9
<i>Fiber_{other}</i>	-0.99	-0.71	-1.3
<i>Polyunsaturated fatty acids (PUFA)</i>	-0.60	-0.26	-0.94
<i>Whole grains</i>	-0.34	-0.28	-0.40
<i>Legumes</i>	-0.23	-0.10	-0.34
<i>Fiber_{f,v,l,w}</i>	-0.19	-0.11	-0.26
<i>Fruits</i>	-0.18	-0.12	-0.22
<i>Vegetables</i>	-0.083	-0.042	-0.11
<i>Milk</i>	-0.0077	-0.0027	-0.012
<i>Sugar-sweetened beverages (SSB)</i>	0.066	0.043	0.089
<i>Red meat</i>	0.099	0.038	0.15
<i>Processed meat</i>	0.86	0.41	1.1
<i>Trans fatty acids (TFA)</i>	4.4	3.3	5.6
<i>Sodium</i>	13.9	11.5	16.1

Fiber_{other}= fiber obtained from sources other than fruits, vegetables, legumes, and whole grains

Fiber_{f,v,l,w}=fiber obtained from fruit, vegetables, legumes, and whole grains

Omega-3 fatty acids are restricted to those that originate from seafood sources

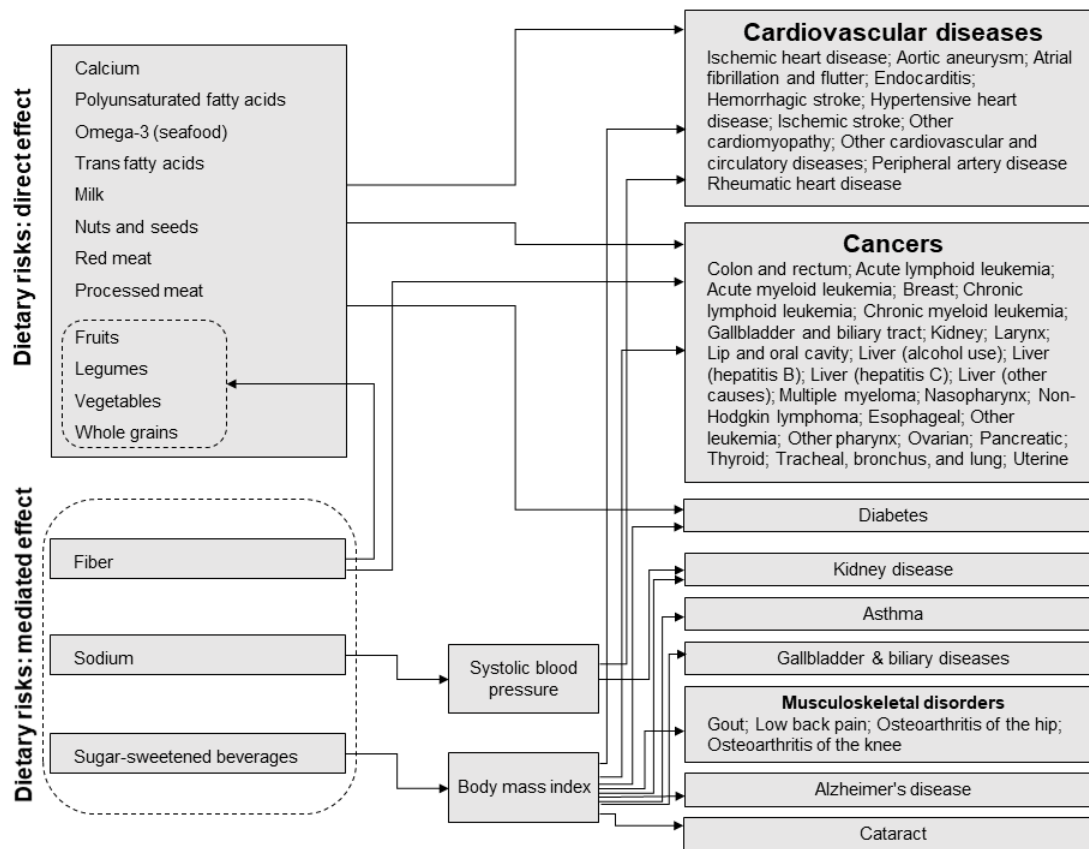


Suppl. Fig. 1. Age group and gender contribution to cumulative gender- and age-adjusted dietary risk factors (DRFs) for US adults (25+ yr). Fiber is divided into fruit, vegetables, legumes, and whole grains (f,v,l,w) and other sources. PUFA: polyunsaturated fatty acids. TFA: trans fatty acids.

171 **Suppl. Table 4.** Disease grouping used in this analysis as seen in Fig. 1 of the main text.

Disease group	Disease
Other cardiovascular diseases	Aortic aneurysm Atrial fibrillation and flutter Endocarditis Hemorrhagic stroke Hypertensive heart disease Ischemic stroke Other cardiomyopathy Other cardiovascular and circulatory diseases Peripheral artery disease Rheumatic heart disease
Other diseases	Alzheimer disease and other dementias Asthma Cataract Chronic kidney disease due to diabetes mellitus Chronic kidney disease due to glomerulonephritis Chronic kidney disease due to hypertension Chronic kidney disease due to other causes Gallbladder and biliary diseases Gout Low back pain Osteoarthritis Osteoarthritis
Other neoplasms	Acute lymphoid leukemia Acute myeloid leukemia Breast cancer Chronic lymphoid leukemia Chronic myeloid leukemia Esophageal cancer Gallbladder and biliary tract cancer Kidney cancer Larynx cancer Lip and oral cavity cancer Liver cancer due to alcohol use Liver cancer due to hepatitis B Liver cancer due to hepatitis C Liver cancer due to other causes Multiple myeloma Nasopharynx cancer Non-Hodgkin lymphoma Other leukemia Other pharynx cancer Ovarian cancer Pancreatic cancer Thyroid cancer Tracheal, bronchus, and lung cancer Uterine cancer

S2. Health Nutritional Index (HENI)



Suppl. Fig. 2. Schematic diagram of the Healthy Nutritional Index (HENI) framework

S2.1. Food composition in dietary risk components

The amount of each dietary risks present in a given food ($d_{i,r}$) in the What We Eat in America (WWEIA) 2011-2016 database⁷ was determined using the approach by Fulgoni et al.⁹. More specifically, a combination of publically available databases developed by the US Department of Agriculture (USDA) were customized to comply with the dietary risk definitions from the GBD. Variables from the Food Patterns Equivalents Database (FPED)¹⁰ were adapted for dietary risks that represent food groups. For dietary risks representing nutrients, the USDA National Nutrient Database for Standard Reference, Release 28 (SR)¹¹ was combined with the Food and Nutrient Database for Dietary Studies (FNDDS)¹². TFA estimates required special treatment since 63% of the foods in the

185 WWEIA database had missing or incomplete TFA profiles. To avoid underestimating the
186 effect of TFA, TFA estimates were imputed using a regression model that took into account
187 food category and available nutrient information ($R^2=0.69$). For milk and yogurt, the
188 regression model was further adapted based on the food category-specific ratio of known
189 TFA/total fat.

S2.2. HENI calculations for a serving of chicken wing

In this section the detailed HENI calculations are demonstrated using an 85 g serving of chicken wings as an example. First, the composition of the food for the 16 dietary risks in g/serving is determined. A serving of chicken wings contains four HENI components in different amounts, i.e., 1.85 g PUFA and 0.0281 g calcium on the health beneficial side, as well as 0.492 g sodium, and 0.139 g TFA on the detrimental health side. The remaining amount that is primarily from poultry is considered to have neutral health effects. The food composition is multiplied with the corresponding DRFs (Suppl. Table 3) and rescale results from μ DALY to minutes of life gained to estimate the all-cause health burden attributable to each dietary risk, i.e., 0.59 and 0.076 minutes of healthy life gained for PUFA and calcium, respectively, and 3.6, and 0.33 minutes lost attributable to sodium, and TFA, respectively. Aggregating these estimates results in a (net) HENI score of 3.3 minutes of healthy life lost per serving chicken wings (Eq. S3).

$$\begin{aligned}
 HENI_{chicken\ wings} &= -0.53 \cdot \sum_r DRF_{r,DALY} \cdot d_{chicken\ wings,r} \quad (Eq. S3) \\
 &= -0.53 \frac{\text{minutes}}{\mu\text{DALY}} \left[1.85 \frac{g_{PUFA}}{\text{serving}} \cdot \left(-0.60 \frac{\mu\text{DALY}}{g_{PUFA}} \right) + 0.0281 \frac{g_{calcium}}{\text{serving}} \right. \\
 &\quad \cdot \left(-5.1 \frac{\mu\text{DALY}}{g_{calcium}} \right) + 0.492 \frac{g_{sodium}}{\text{serving}} \cdot 13.9 \frac{\mu\text{DALY}}{g_{TFA}} + 0.139 \frac{g_{TFA}}{\text{serving}} \cdot 4.4 \frac{\mu\text{DALY}}{g_{TFA}} \Big] \\
 &= -3.3 \frac{\text{minutes}}{\text{serving}}
 \end{aligned}$$

S2.3. HENI scores for all food items

The HENI scores per serving of the 5,853 foods in the WWEIA 2011-2016 are summarized by food category (Suppl. Table 5). The energy density ($R^2=0.019$, Suppl. Fig. 3A) and serving size ($R^2=0.040$, Suppl. Fig. 3B) of foods could have little influence on HENI scores. On average, processed meat and red meat dishes, as well as sweetened beverages, yield the highest health losses, whereas nuts and seeds, legumes, fruits, and seafood yield the highest health gains. In particular, the food categories with the most detrimental foods are hot dog sandwiches (35 minutes lost/serving, interquartile range (IQR): 31 to 41), cured meats (26 minutes lost/serving, IQR: 22 to 29), egg/breakfast sandwiches (11 minutes lost/serving, IQR: 8 to 18), sweetened beverages (8.1 minutes lost/serving, IQR: 7.0 to 9.0), and red meat (6.5 minutes lost/serving, IQR: 5.5 to 7.2). The food categories with the most health beneficial foods are peanut butter and jelly sandwiches (33 minutes gained/serving, IQR: 29 to 34), nuts and seeds (25 minutes gained/serving, IQR: 24 to 27), fruits (12 minutes gained/serving, IQR: 7.0 to 14), legumes and soy products (10 minutes gained/serving, IQR: 9.1 to 11), and seafood (5.2 minutes gained/serving, IQR: 1.5 to 14.0). Milk and dairy (excluding cheese), oils, starchy vegetables, breads, and sugars tend to have neutral HENI scores.

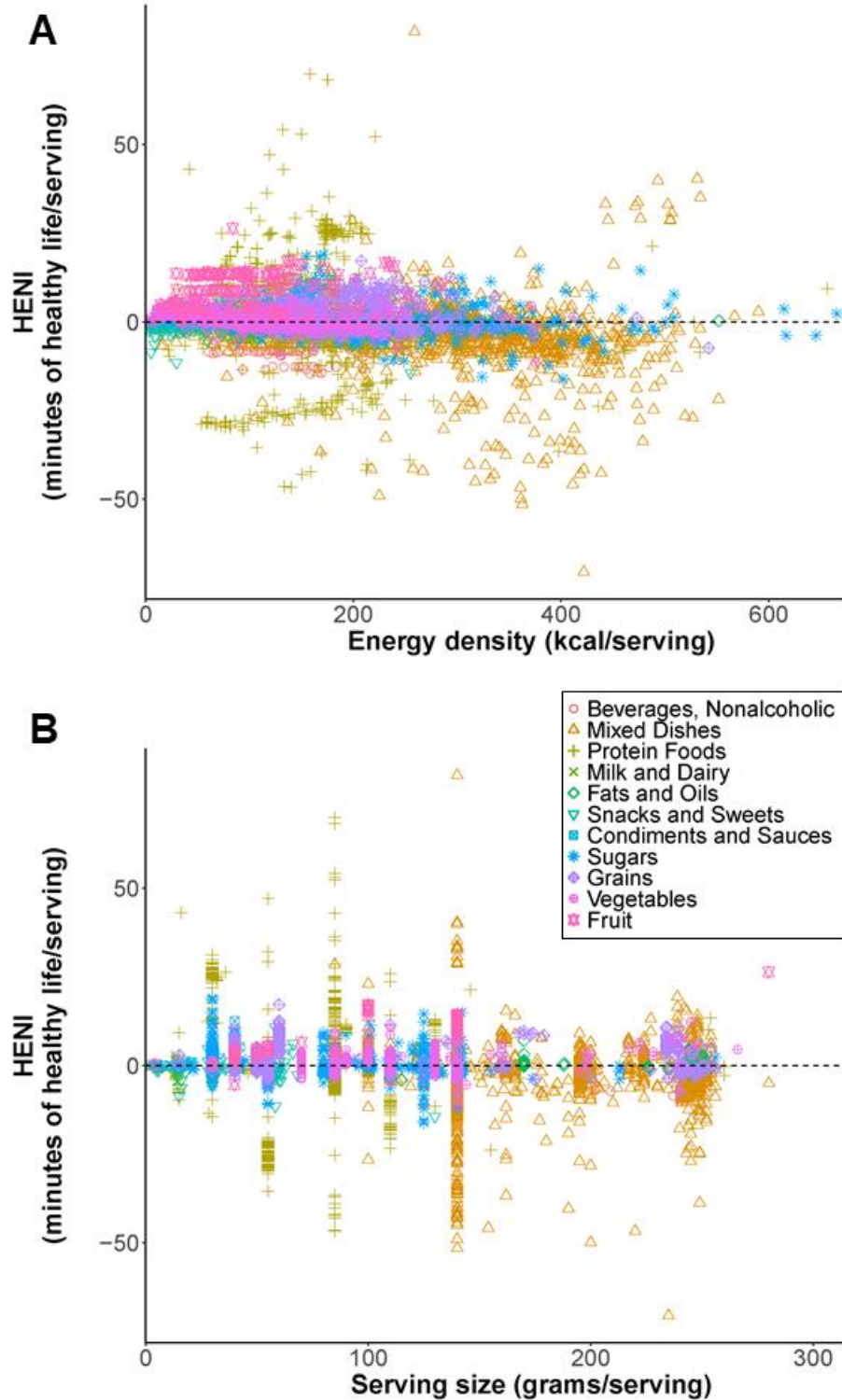
HENI scores can vary considerably for a given food category. This could be due to a large amount of a single ingredient (e.g., the omega-3 content of seafood dishes, the food category with the highest IQR of 12.7 minutes/serving) or by big differences in the composition of similar foods (e.g., regular versus low sodium or low fat foods). The latter is particularly true for mixed dishes that typically have multiple ingredients. Nevertheless, the influence of ingredients on the HENI scores can fluctuate. For example, HENI scores

are inversely proportional to the amount of processed meat in frankfurter sandwiches (due to their detrimental health effects), whereas for peanut butter and jelly sandwiches HENI scores are directly proportional to the amount of peanut butter (due to the beneficial health effect of nuts. In contrast, the type of bread, which may differ in the amount of whole grains that are beneficial for health, seems to have minimal influence on the HENI score. More specifically, the HENI score for “white buns” is 0.88 minutes lost/serving (95% CI: -1.25 to -0.51) and for “wholegrain buns” is 1.32 minutes gained/serving (95% CI: 0.75 to 1.84). In other foods, such as cereals (cooked cereals and ready-to-eat cereals) the contribution whole grains to the HENI scores is more significant and can reach up to 11 minutes of healthy life gained/serving.

Suppl. Table 5. HENI score summary statistics per serving for 48 food categories.

HENI categories	N	Min	Q1 (25th pctl)	Median	Q3 (75th pctl)	Max	Mean
Frankfurter sandwiches	37	-49.0	-41.4	-35.4	-31.3	-0.627	-34.2
Cured Meats/Poultry	123	-46.7	-28.5	-25.6	-22.0	-0.212	-24.3
Egg/breakfast sandwiches	40	-33.7	-17.8	-10.8	-8.31	-1.28	-13.6
Sweetened Beverages	111	-14.6	-8.96	-8.13	-7.02	0.0306	-7.95
Red meat	158	-36.5	-7.18	-6.49	-5.55	-0.138	-6.65
Burgers	104	-12.2	-8.46	-6.20	-4.34	2.21	-6.16
Other sandwiches	36	-35.0	-10.9	-5.88	0.137	9.88	-7.12
Meat mixed dishes	200	-70.5	-8.91	-5.85	-2.39	6.16	-8.44
Macaroni and cheese	15	-24.7	-4.85	-4.07	-2.47	0.728	-5.25
Pizza	77	-13.2	-7.63	-3.64	-1.36	2.23	-4.35
Soups	187	-27.0	-4.83	-3.07	-0.00257	10.2	-2.66
Eggs	146	-23.8	-10.7	-2.60	-0.495	2.94	-5.71
Poultry sandwiches	30	-15.8	-4.13	-2.43	-0.713	4.64	-3.09
Poultry	300	-28.5	-2.60	-2.07	-1.65	0.920	-2.32
Margarine	21	-4.62	-2.79	-1.95	-0.929	-0.523	-2.11
Poultry mixed dishes	118	-13.9	-3.35	-1.67	0.386	8.86	-1.76
Cheese sandwiches	22	-6.11	-4.68	-1.66	-0.384	2.31	-2.16
Cheese	67	-4.13	-2.24	-1.41	-0.653	4.34	-1.41
Quick Breads and Bread Products	94	-7.39	-2.21	-0.98	0.827	11.4	-0.676
Asian Mixed Dishes	127	-11.5	-2.72	-0.784	2.20	16.2	0.214
Other grain-based mixed dishes	134	-21.3	-3.92	-0.738	3.19	19.4	-0.510
Oils	38	-2.72	-1.61	-0.615	0.343	2.91	-0.481
Sweet Bakery Products	293	-16.1	-2.34	-0.473	0.535	14.5	-0.830
Butter, Animal Products and Fats	66	-2.79	-0.880	-0.472	-0.231	2.14	-0.546
Other desserts	133	-3.81	-1.11	-0.353	0.612	14.9	0.340
Milk	26	-2.57	-0.859	-0.316	-0.147	0.223	-0.608
Starchy vegetables	309	-9.51	-1.00	-0.284	0.416	10.7	-0.381
Other pasta mixed dishes	126	-16.7	-3.67	-0.202	2.11	14.1	-0.486
Condiments and Sauces	165	-14.3	-1.51	-0.190	0.326	10.7	-0.404

HENI categories	N	Min	Q1 (25th pctl)	Median	Q3 (75th pctl)	Max	Mean
Other Dairy Drinks and Substitutes	78	-1.27	-0.576	-0.147	0.375	4.68	0.218
Yogurt	42	-0.277	-0.0739	-0.0464	0.794	7.06	0.512
Sugars	64	-1.11	-0.125	-4.55E-05	0.226	12.6	0.388
Breads, Rolls, Tortillas	167	-2.20	-0.888	0.0643	1.39	5.78	0.456
Mexican Mixed Dishes	110	-9.60	-2.90	0.0756	2.41	9.07	0.0605
Candy	106	-2.32	-0.0379	0.460	4.90	19.1	3.18
Cooked Grains	69	-3.95	-1.61	0.576	7.05	9.70	2.27
Crackers and savory snacks	211	-5.00	-0.334	0.606	3.00	8.58	1.24
Cooked Cereals	123	-3.82	-1.06	1.77	5.19	10.9	2.19
Mixed vegetables	386	-11.4	1.41	2.17	2.59	12.3	2.04
Vegetables, excluding starchy	301	-0.349	2.15	2.56	3.40	7.99	2.88
Snack/meal bars	37	-0.373	2.33	2.78	4.51	10.3	3.79
Seafood mixed dishes	69	-27.6	1.57	3.34	5.46	81.8	4.83
Ready-to-Eat Cereals	158	-3.03	1.30	3.57	6.36	17.2	3.89
Seafood	221	-9.67	1.48	5.21	14.2	70.0	9.64
Legumes and soy products	135	-11.5	9.08	10.3	10.5	21.4	8.37
Fruits	167	-5.36	7.05	11.9	13.5	26.4	10.0
Nuts and seeds	89	0.95	24.4	25.3	26.9	31.3	24.1
Peanut butter and jelly sandwiches	17	3.92	29.2	33.1	33.7	40.4	31.7



243

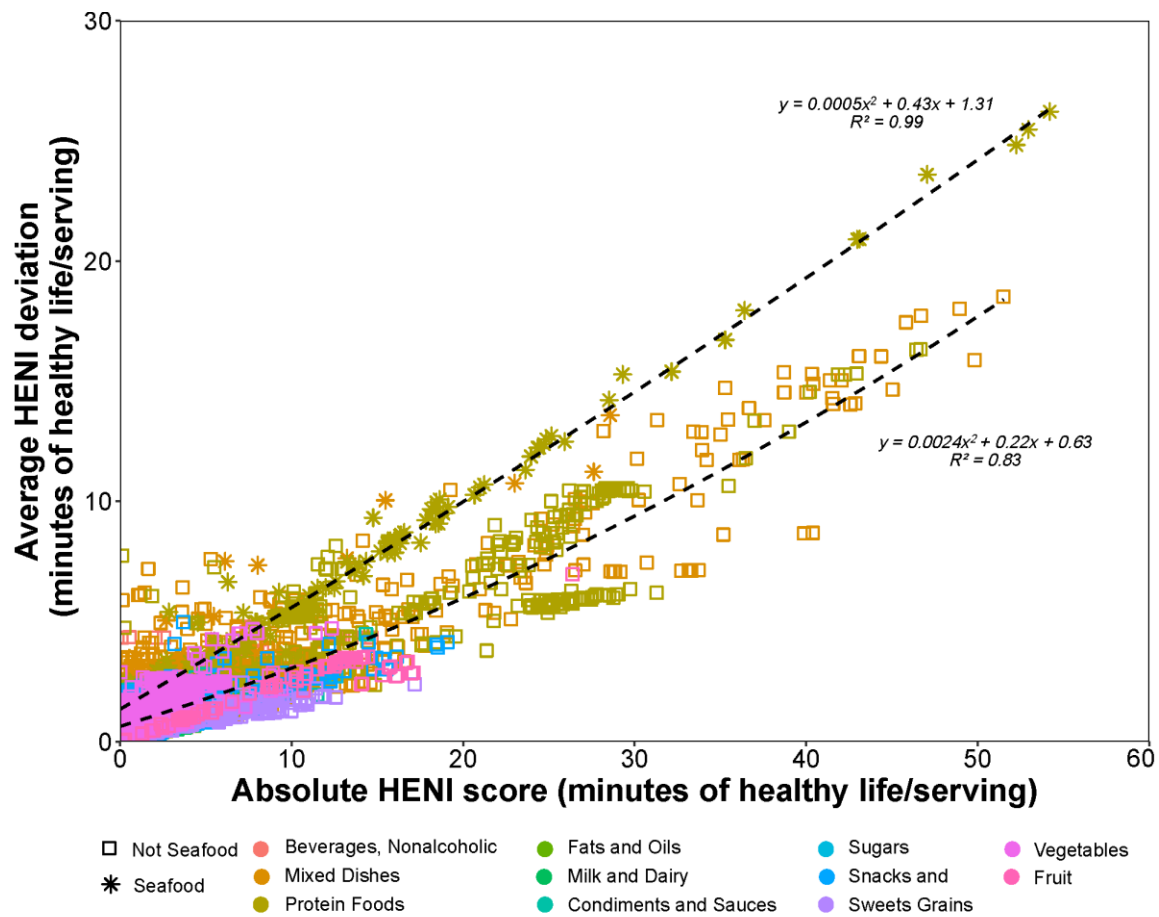
244 **Suppl. Fig. 3.** HENI per serving as a function of energy density (**A**) and serving size (**B**)

245 by food group for 5,853 food items in the US diet.

S2.4. Uncertainty analysis of HENI scores for all food items

Supplementary figure 4 analyses the uncertainty associated with HENI scores per serving for all food items considered in this analysis, plotting the average deviation on each HENI score (calculated as the (upper limit-lower limit)/2) as a function of the absolute value of the corresponding HENI score.

For foods with HENI scores between -5 and 5 minutes, uncertainty is restricted to a median variation of ± 1 minutes. The uncertainty increases with the absolute value of HENI to a median variation of about ± 2.5 minutes for absolute HENI scores between 5 and 15 minutes. The trends observed are different for seafood and non-seafood dishes (Suppl. Fig. 4). For non-seafood dishes, the regression curve ($y_{\text{uncertainty}} = 0.0024 \cdot \text{HENI}^2 + 0.22 \cdot \text{HENI} + 0.63$, $R^2 = 0.83$) shows that the uncertainty of HENI score continues to increase, reaching ± 14 minutes for an absolute HENI score of 40 minutes of healthy life (e.g., for one serving of hot-dog). For seafood, uncertainty is higher ($y_{\text{uncertainty}} = 0.005 \cdot \text{HENI}^2 + 0.43 \cdot \text{HENI} + 1.31$, $R^2 = 0.99$) and is reaching ± 20 minutes for an absolute HENI score of 40 minutes of healthy life.



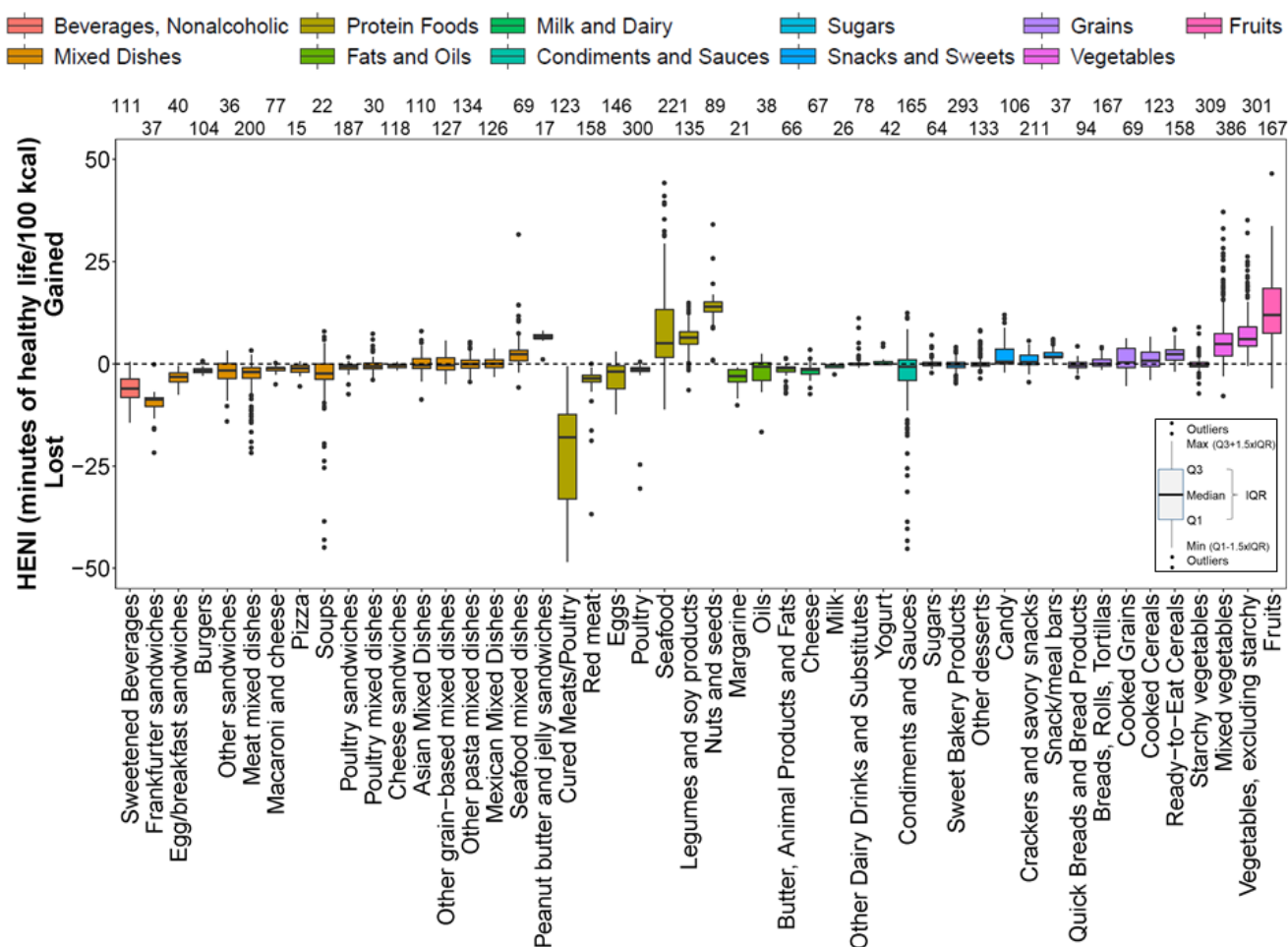
Suppl. Fig. 4. Average HENI deviation as a function of absolute HENI per serving for 5,853 food items in the US diet. The average HENI deviation was calculated as the mean difference between upper and lower estimates to each HENI score. Trend lines were fitted for foods with and without seafood separately. Five foods are not shown (additional outliers).

S2.5. Sensitivity analysis: HENI scores per 100 kcal and 100 g

RACC served as the primary comparison basis in this analysis. Nevertheless, HENI can also be determined per 100 kcal (Suppl. Fig. 5) or 100 g of food (Suppl. Fig. 6).

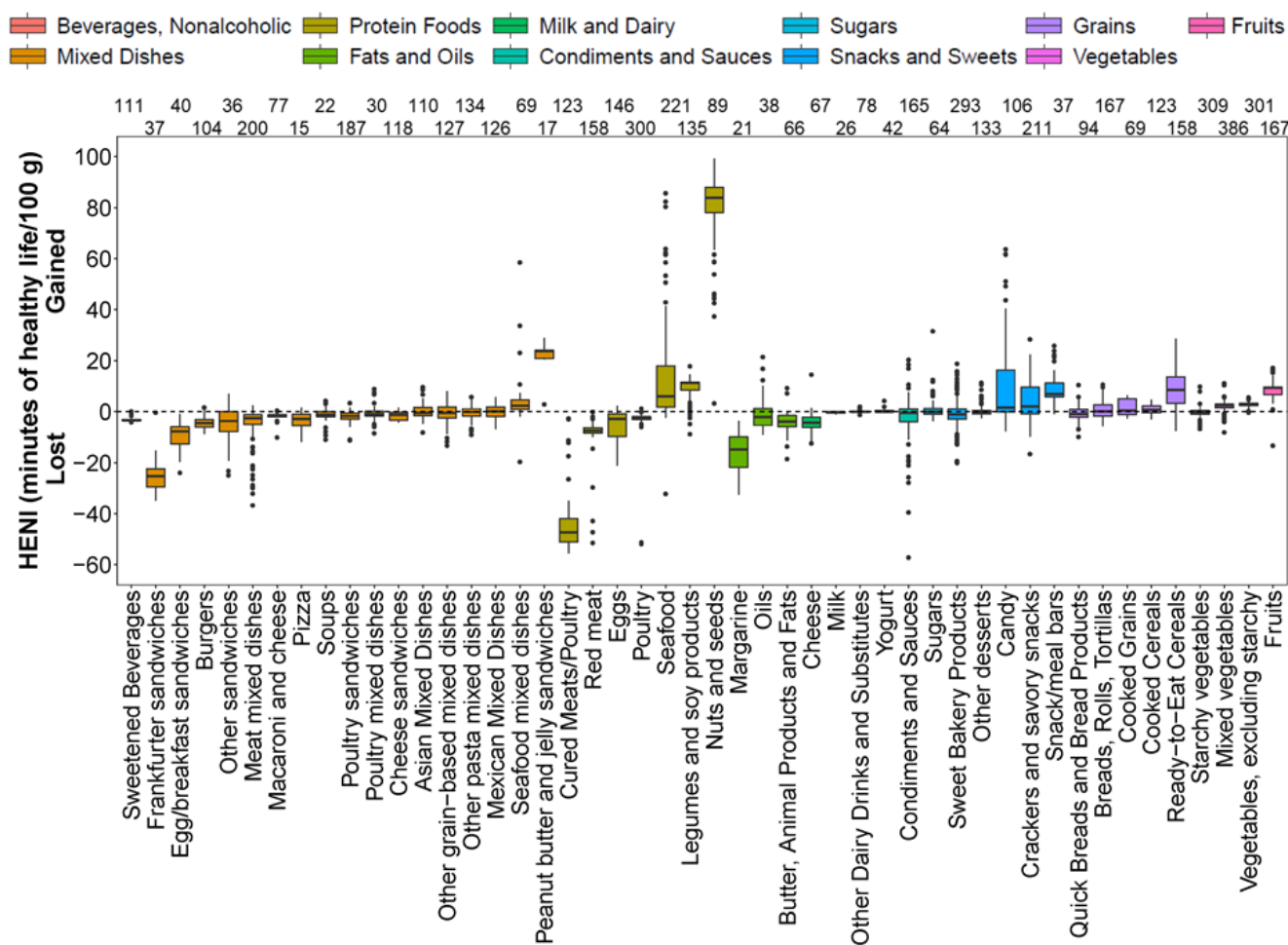
The 100 kcal-based HENIs give somewhat similar food category rankings as the serving-based, but with less extreme scores for energy-dense foods (e.g., nuts, meat) and higher scores for foods with low energy density (e.g., fruit and non-starchy vegetables, see Suppl. Fig. 7A). HENI comparisons on a caloric basis are informative for foods with positive scores as it enables the identification of foods that offer maximum health benefits per kcal. On certain occasions, such as low caloric density fruits and vegetables, 100 kcal of foods require unrealistic mass equivalents (e.g., 100 kcal of cucumbers corresponds to ~10 servings). On the other hand, a caloric-based comparison can be misleading for foods with negative scores that are energy-dense, generating results that contradict the serving-based results. More specifically, HENI scores per serving suggest that “low-fat” foods - foods lower in unhealthy fats such as TFA - tend to perform better than their conventional counterparts. However, HENI scores per 100 kcal identify conventional fat foods as better-performing due to their higher caloric content. Integrating conventional fat foods in the overall diet can lead to excess daily caloric intake, which increases body mass index and is acknowledged as an important health risk factor¹.

HENI scores per 100 g result in considerable differences in inter and intra food category ranking, especially for foods with small serving sizes such as fats, oils, sugars, snacks, sweets, condiments, nuts, seeds, and cheese (Suppl. Figs. 6 and 7B). Hence, 100 g-based HENI scores should be interpreted with caution.



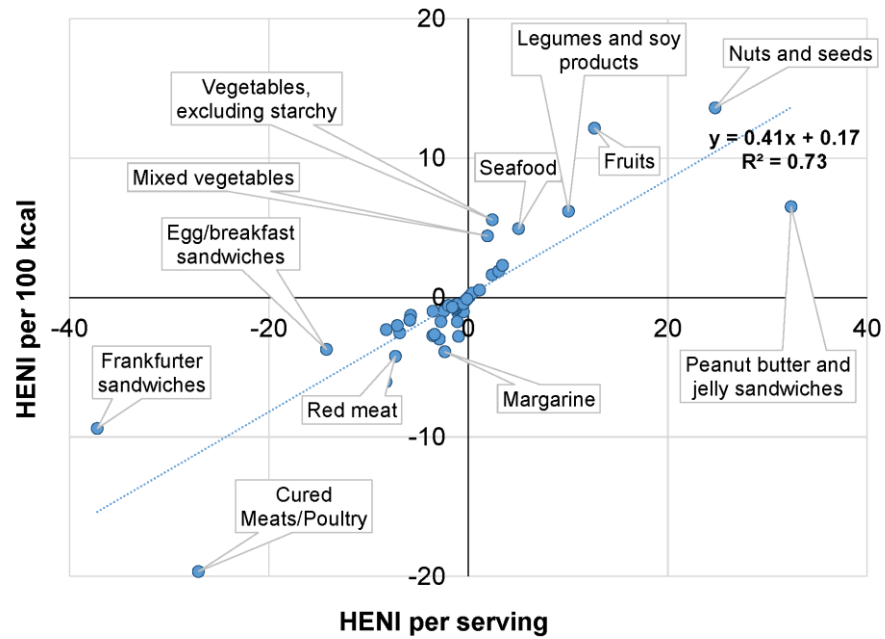
291

292 **Suppl. Fig. 5. HENI per 100 kcal for 5,853 foods in the US diet by food category.** Positive scores indicate health benefits. Nine foods
 293 are not shown (additional outliers). Numbers on top denote the number of foods in each category.

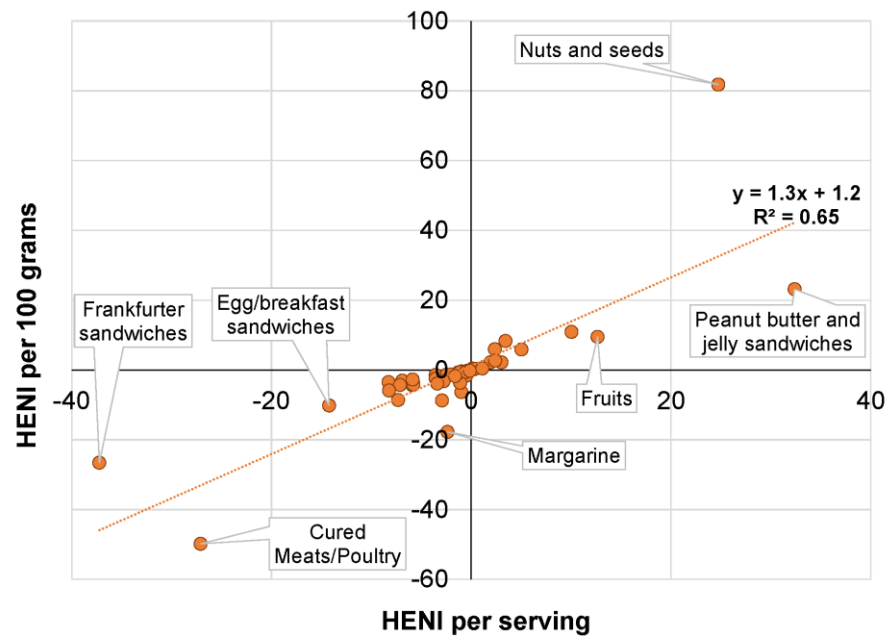


Suppl. Fig. 6. HENI per 100 g for 5,853 foods in the US diet by food category. Positive scores indicate health benefits. Five foods are not shown (additional outliers). Numbers on top denote the number of foods in each category.

A



B



Suppl. Fig. 7. Association between and median HENI scores per serving by food category and (A) median HENI scores per 100 kcal, and (B) median HENI scores per 100 grams.

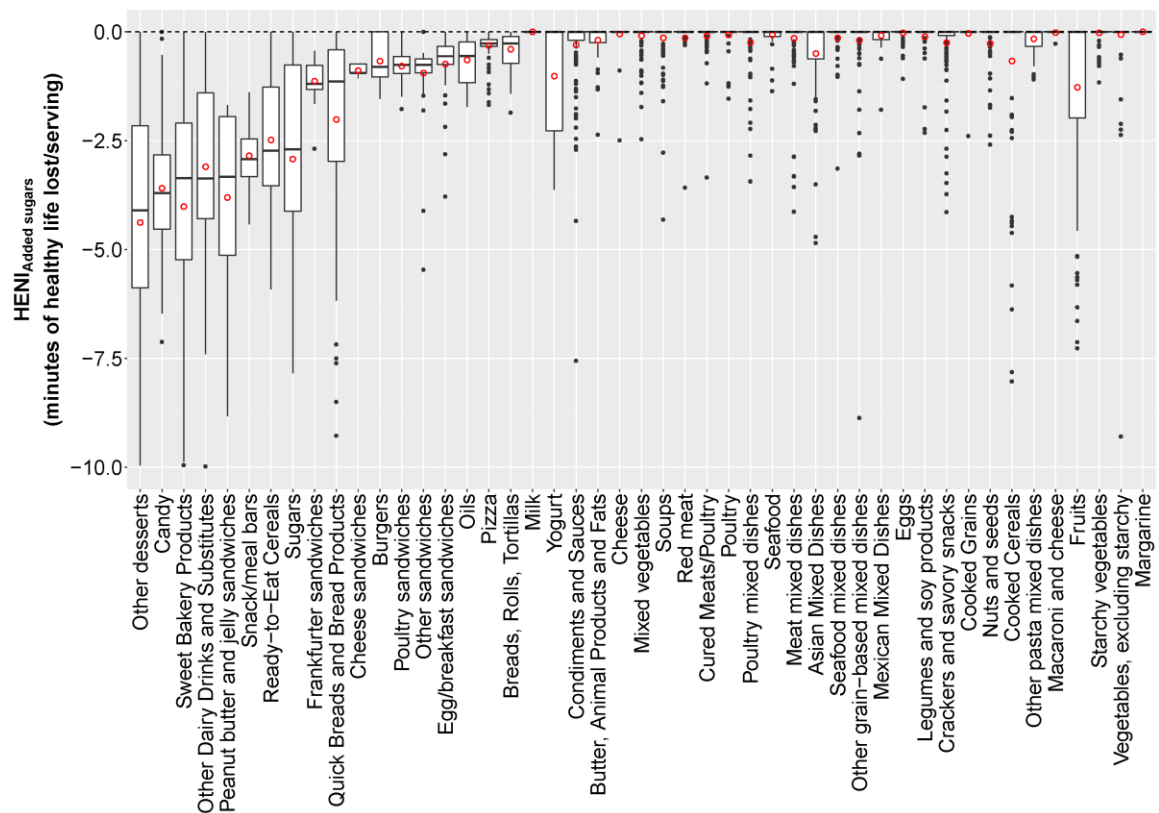
S2.6. Sensitivity analysis: Added sugar as a HENI risk component

Dietary guidelines propose the restriction of added sugars to <10% of total daily energy^{8,13}. Evidence supports an association between added sugars and weight gain¹⁴, diabetes¹⁵, and cardiovascular diseases¹⁶. However, these associations are not consistent¹⁷, and as a result, added sugar is not considered a dietary risk in the GBD. In a sensitivity study, the change in HENI scores per serving as a result of considering added sugars as a risk component in HENI is investigated. Firstly, a cumulative age-gender-adjusted DRF for added sugar ($\overline{DRF}_{added\ sugar}$) was calculated based on an extrapolation from SSB. It was assumed that the health burden attributable to a marginal increase in added sugar intake equaled to half of the health burden attributable to the average added sugar equivalent of SSBs. According to WWEIA, the mean added sugar content of the sweetened beverages is estimated at $0.065 \frac{g_{added\ sugar}}{g_{SSB}}$ (Data S1). Therefore, the $\overline{DRF}_{added\ sugar}$ was calculated as follows:

$$\overline{DRF}_{added\ sugar} = 0.5 \cdot \frac{\overline{DRF}_{SSB}}{added\ sugar\ in\ SSB} = \frac{0.5 \cdot 0.066 \frac{\mu DALY}{g_{SSB}}}{0.065 \frac{g_{added\ sugar}}{g_{SSB}}} = 0.51 \frac{\mu DALY}{g_{added\ sugar}} \quad (Eq. S4)$$

The change in HENI per serving due to added sugar (i.e., $HENI_{added\ sugar}$, Suppl. Fig. 8) was estimated by multiplying the $\overline{DRF}_{added\ sugar}$ of $0.51 \frac{\mu DALY}{g_{added\ sugar}}$ (95% CI: 0.33 to 0.69) with the added sugar content of each food item as reported in the FPED¹⁰, except SSBs to avoid double counting. This analysis suggests that added sugar is not a major contributor to health burden for most food categories, with $HENI_{added\ sugar}$ for ~85% of foods lower than 1 minute lost/serving. However, certain categories are substantially affected by the consideration of this additional risk component. In particular, the median HENI change of 3-4 minutes lost per serving for other desserts, candy, sweet bakery

products, other dairy drinks and substitutes, and ready-to-eat cereal results in considerable shift in the nutritional performance of these foods, as they tend to have close to neutral scores when impacts from added sugar are not considered (Suppl. Table 5). For example, the median HENI for ready-to-eat cereal is reduced from 3.6 minutes gained/serving to 0.92 minutes gained/serving when added sugar is considered as a model component (median $HENI_{added\ sugar}=2.7$ minutes lost/serving, IQR: 2.5 to 5.9). For foods with high levels of added sugar and large serving size, such as the “Milk dessert or milk candy, Puerto Rican style” (155 g/serving size), $HENI_{added\ sugar}$ can reach up to 18 minutes of minutes lost per serving.



Suppl. Fig. 8. HENI scores per serving in minutes of healthy life attributable to added sugar for 5,742 foods (excluding sweetened beverages) by food category. Estimates are

334 based on $\overline{DRF}_{added\ sugar}$ estimate of $0.51 \frac{\mu DALY}{g_{added\ sugar}}$. Red markers represent mean values.

335 Eighteen foods are not shown (additional outliers). Food categories are ranked from highest
336 to lowest median HENI change.

337 **S2.7. Sensitivity analysis: Saturated fatty acids as a HENI risk component**

338 The GBD recognizes high total serum cholesterol as an important metabolic risk
339 factor for cardiovascular diseases¹. Diet has been found to influence total serum cholesterol
340 levels, with evidence supporting an association between saturated fatty acids (SFA) and
341 total serum cholesterol^{1,18,19}. This relationship seems to be influenced by the nutrient that
342 SFA substitutes in the diet (e.g., carbohydrates versus PUFA replaced with SFA)¹⁹.

343 While the health burden of SFA might be captured in GBD via total cholesterol,
344 given recent evidence that does not support detrimental health effects associated with all
345 types of SFA intake²⁰. Thus, SFA was not included in the default HENI model and rather
346 the potential change of HENI scores when SFA is considered as a risk component was
347 examined in this sensitivity study.

348 Using an effect mediation via total serum cholesterol, the SFA-DRFs for a given
349 health outcome o , age group a , and gender g ($DRF_{SFA,o,b}^{a,g,m}$) was calculated using:

$$350 \quad DRF_{SFA,o,b}^{a,g,m} = \frac{\ln RR_{cholesterol,o,b}^{a,g}}{Ref_{cholesterol}} \cdot \frac{\Delta_{cholesterol}}{\Delta_{SFA \cdot EER^{a,g}}} \cdot f_{fat,kcal \rightarrow g} \cdot \widehat{BR_{SFA,o,b}^{a,g}} \quad (Eq. S5)$$

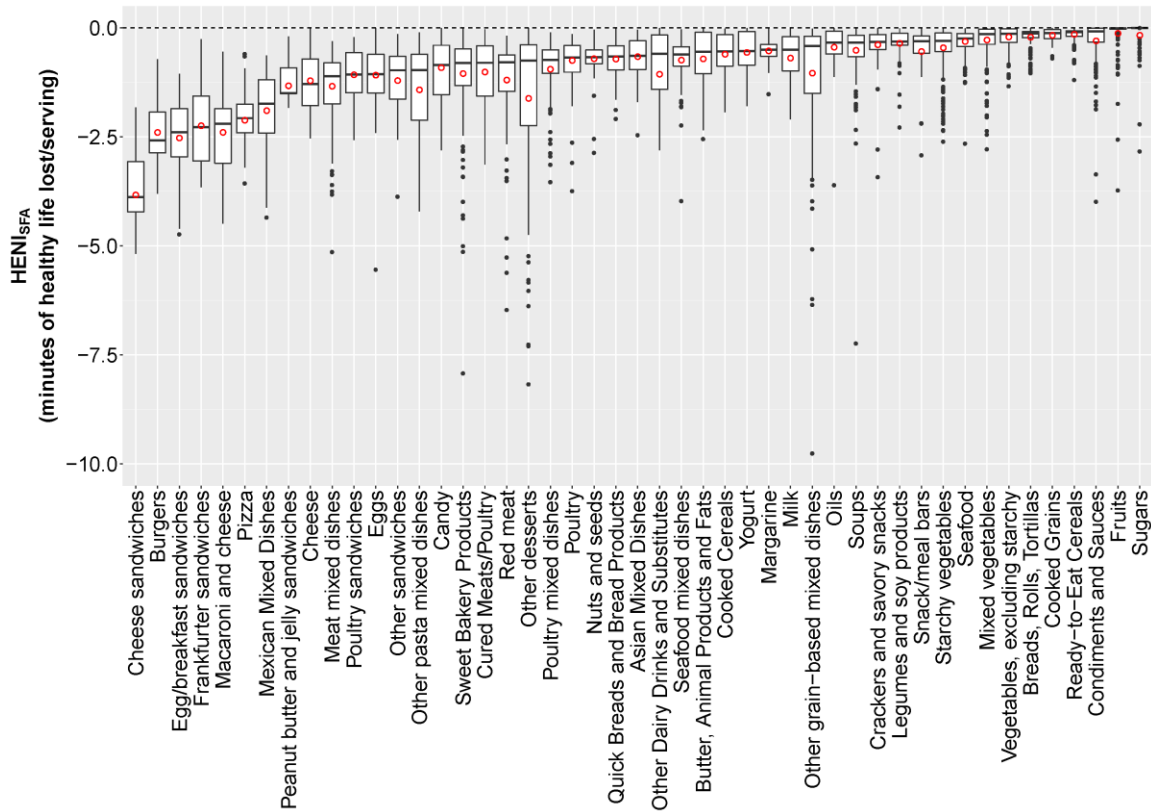
$$351 \quad \text{with } \widehat{BR_{SFA,o,b}^{a,g}} = \frac{BR_{o,b}^{a,g}}{\sum_x P_{x_{SFA}}^{a,g} \cdot RR_{cholesterol,o,b}^{a,g} \left(\frac{x_{SFA}}{Ref_{cholesterol}} \cdot \frac{\Delta_{cholesterol}}{\Delta_{SFA \cdot EER^{a,g}}} \cdot f_{fat,kcal \rightarrow g} \right)}$$

352 where $RR_{cholesterol,o,b}^{a,g}$ is the RR for total serum cholesterol and outcome o for a given age
353 group a , gender g , and burden b , corresponding to a reference exposure ($Ref_{cholesterol}$) of

354 $1 \frac{mmol}{L}$. $\frac{\Delta_{cholesterol}}{\Delta_{SFA \cdot EER^{a,g}}}$ is a conversion coefficient representing the change in total serum

cholesterol per 1% energy increase from SFA, adjusted for the age- and gender-specific estimated energy requirements ($EER^{a,g}$). The RRs for total serum cholesterol and health outcomes were obtained from the GBD¹. The conversion coefficient used was 0.045 mmol/L total cholesterol increase (95% confidence interval (CI): 0.038 to 0.051)¹⁹. EER estimates were obtained from Suppl. Table 2. To convert SFA from kcal to grams, the conversion factor ($f_{fat,kcal \rightarrow g}$) of 9.25 kcal/g_{fat}²¹ was used. Lastly, $\widehat{BR}_{SFA,o,b}^{a,g}$ is the age-gender-burden-specific BR of an outcome o adjusted for the background SFA consumption in the US (x_{SFA} , in g), expressed in μ DALYs/person/day.

The median $HENI_{Saturated\ fat}$ ranges from 0.008 to 3.9 minutes lost per serving of fruits and cheese sandwiches, respectively (Suppl. Fig. 9). Considerable $HENI_{Saturated\ fat}$ estimates (>5 minutes lost/serving) is observed in only ~30 foods from a variety of food categories such as red meat, cheese sandwiches, desserts, and foods containing coconut milk. The highest $HENI_{Saturated\ fat}$ is estimated for a serving of coconut milk at 17 minutes of healthy life lost. However, evidence support that SFA obtained from coconut may not share the adverse health effects associated with SFA obtained from other sources^{22,23}, which it has not been considered here. Similarly, emerging evidence supports a food-dependent effect of SFA on health, with SFA from dairy having a cardiovascular neutral or protective effect²⁴⁻²⁶. As a result, the estimated $HENI_{Saturated\ fat}$ for coconut milk and dairy products (Suppl. Table 6) should be interpreted with caution as it is likely to be reduced when these food-specific associations are considered.



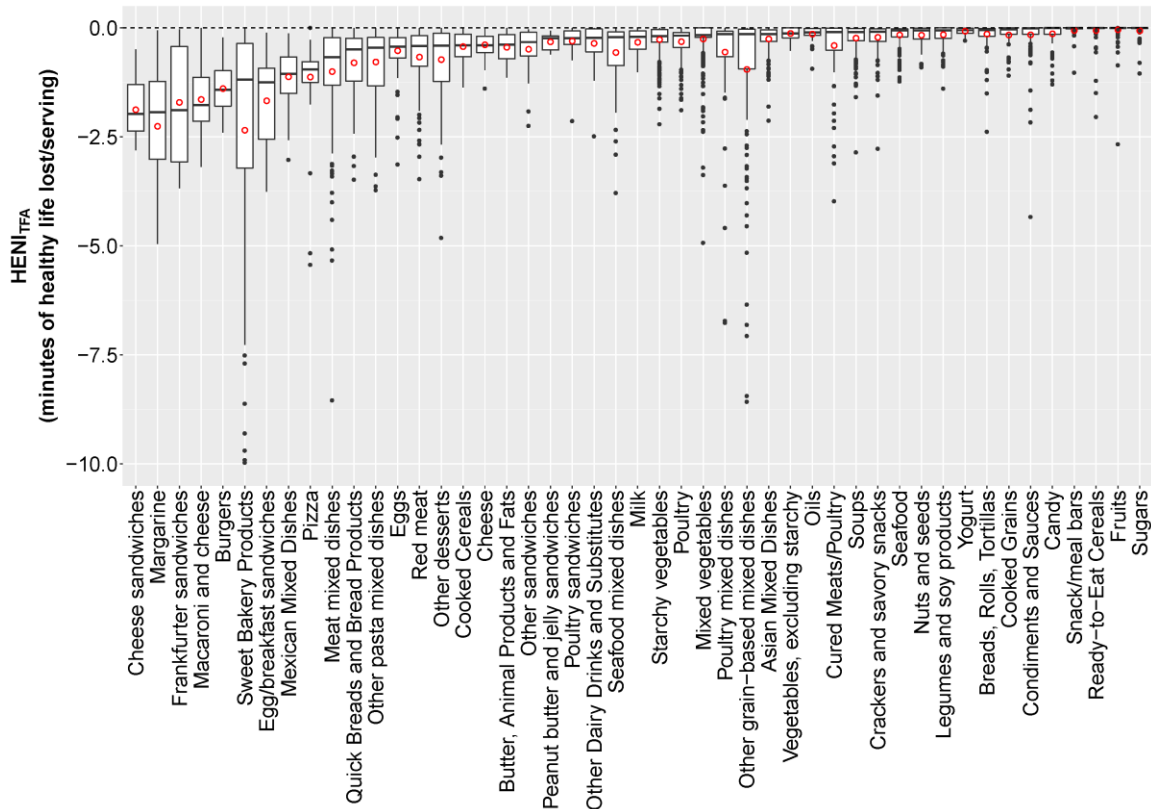
Suppl. Fig. 9. HENI scores per serving in minutes of healthy life attributable to saturated fats for 5,853 foods by food category. Red markers represent mean values. Two foods are not shown (additional outliers). Food categories are ranked from highest to lowest median HENI change.

S2.8. Sensitivity analysis: Contribution of trans to HENI

The influence of TFA on HENI was evaluated in a separate sensitivity study since their contributions could be refined in the future. The reason is that TFA contributions to HENI scores, $\text{HENI}_{\text{Trans fat}}$, for ~60% of the foods in this analysis depend on imputed TFA content estimates⁹ and should be interpreted with caution. Moreover, these contributions are estimated to decline in the near future as TFA is starting to be eliminated or reduced in many food products since 2013²⁷. There is also recent evidence supporting that the health

effect of natural ruminant TFA could be source-specific^{25,28,29}, which was not considered in this analysis.

Overall, TFA appears to have a relatively small impact on HENI scores per serving for most food categories (Suppl. Fig. 10). Median HENI_{Trans fat} varies from 0.0009 minutes lost/serving of up to 2.0 minutes lost/serving of cheese sandwiches. A limited number of foods (N=53) that are primarily sweet bakery products show considerable HENI_{Trans fat} estimates (> 5 minutes lost/serving). For foods high in natural ruminant TFA such as red meat, cheese, and milk, the median HENI_{TFA} is 0.41, 0.40, and 0.20 minutes lost/serving, respectively (Suppl. Table 6).



Suppl. Fig. 10. HENI scores per serving in minutes of healthy life attributable to trans fats for 5,853 foods by food category. Red markers represent mean values. Nine foods are not shown (additional outliers). Food categories are ranked from highest to lowest median HENI change.

Suppl. Table 6. Median HENI per serving attributable to trans and saturated fatty acids for select food categories.

Food category	N	Median HENI (minutes of healthy life lost/serving)	
		Trans fat	Saturated fat
<i>Cheese</i>	67	-0.40	-1.3
<i>Food containing coconut milk</i>	4	-1.8	-11.1
<i>Milk</i>	26	-0.20	-0.50
<i>Other Dairy Drinks and Substitutes</i>	78	-0.22	-0.62
<i>Red meat</i>	158	-0.41	-0.79
<i>Yogurt</i>	42	-0.051	-0.53

S2.9. Discussion

Essential attributes for nutrition indices include considering the health burden associated with each dietary risk component, expanding beyond nutrients, applicability to all food items, and having the ability to be regularly updated to reflect new scientific information as it becomes available³⁰. By meeting each of these criteria, HENI represents a substantial improvement over presently used tools to evaluate the nutritional performance of food and for policy making, which are primarily based on nutrient intake recommendations and adherence to dietary guidelines.

HENI is based on both food groups and nutrients, which is consistent with recent nutrition advancements³¹. HENI covers components that both promote and damage health, as identified by the GBD, avoiding inherent bias in nutrient profiling indices due to component selection (e.g., which nutrients to consider) and reference intake values used in current approaches³². HENI directly links the nutrition composition of foods to human health burden by leveraging epidemiological evidence, whereas current approaches might correlate well or be indirectly associated with disease burden^{33–38}. It is the first nutritional index that is able to evaluate the nutrition impacts of individual foods on health and complement environmental health impact assessments in a common burden-based metric expressed in DALYs³⁹. The weighting between the different risk components considered is evidence-based, and a direct function of the risk magnitude and the background disease burden of the various considered outcomes in the population of study, providing a more transparent basis and rationale than many of the present nutritional evaluations^{33–38,40,41}.

Using HENI to evaluate the nutritional performance of 5,853 foods provides insights on distinctions between and within food groups as well as trade-offs between

430 dietary components associated with health burden. Consequently, HENI scores enable
431 improvements to the current understanding of food performance and that can be utilized to
432 refine dietary recommendations. In particular, HENI can be used to identify and prioritize
433 foods that should be increased and decreased in the diet as a priority, empowering small
434 dietary changes to have large benefits. Additionally, HENI evaluates nutritional
435 performances of any food with known nutritional composition, in minutes of healthy life
436 gained or lost. Unlike current evaluations, which can be hard to interpret and use for daily
437 food choices and substitutions^{42,43}, this simple metric does not require training or additional
438 knowledge for consumers to understand. Building on the present study, it would be
439 interesting to assess the level of acceptability adherence of the recommended substitutions,
440 and how HENI scores could further stimulate consumer behavior change towards more
441 healthy foods.

442 The validity of HENI relies on several assumptions and limitations. First, the
443 intended scope of the marginal HENI was to characterize the marginal health burden
444 associated with individual foods. For the evaluation of the marginal nutritional impacts of
445 foods, the risk components are assumed to affect human health independently and
446 additively. This assumption is valid for the type of restricted changes tested in this analysis
447 that are limited to ~10% change daily caloric intake. As a food-specific tool, HENI scores
448 are meant to be used for relative comparisons and are primarily meaningful under the
449 assumption of isocaloric changes. Marginal changes are expected to be indicative of diet-
450 level changes in terms of direction, but with important variations in the absolute impact
451 magnitude of these changes. For example, from the foods analyzed in the present study that
452 include multiple foods from a diversity of traditions and cultures, Asian mixed dishes are

tangentially healthier than other mixed dish food groups that are popular in the American diet such as burgers and pizzas. This finding is in line with studies that support that Asian-American follow diets of higher diet quality compared to other Americans^{44,45}. To further extend the validity of the approach to larger diet-level shifts, future diet-level studies would need to consider multiplicative cumulative risk effect³⁰ and also to capture additional health benefits or impacts associated with caloric intake changes. The approach could be further extended to compare and contrast various cultures, traditions, and dietary patterns.

Second, HENI estimates presented in this work were developed using US-specific data for background disease prevalence, disease severity, and food composition, which limits its validity to US food consumption, but is flexible and could be easily adapted to other countries. Additionally, marginal HENI scores apply to the considered dietary risks when the background ‘exposure’ levels are above (detrimental effect) or below (protective effect) certain levels of consumption, known as theoretical minimum risk level (Suppl. Table 1). This condition is met for the majority of US adults⁸. For individuals with dietary patterns that deviate substantially from the average US diet, it is recommended to restrict the application of the DRFs to those above or below these theoretical minimum risk levels.

Finally, results depend on the accuracy and resolution of the epidemiological evidence provided by the GBD studies. The GBD studies follow a systematic approach for risk-outcome pair selection that is based on four criteria: 1) the importance of risk factor to disease burden and/or policy; 2) availability of sufficient risk factor exposure data; 3) support of a causal relationship based on epidemiological studies and ability to estimate effect magnitude per exposure unit increase; and 4) evidence supporting that the association can be generalized. To evaluate these criteria, the GBD has adopted the World

Cancer Research Fund (WCRF) grading system, and risk-outcome pairs are included only when it is determined that they have of convincing or probable evidence¹ according to WCRF⁴⁶.

While the GBD follows a systematic approach to identify risk-outcome pairs and quantify their pooled associations, recent evidence of additional associations and effect modifiers, such the joint effect of sodium and potassium on cardiovascular and all-cause mortality⁴⁷, have not been considered. In the future, HENI could be regularly updated to incorporate such new evidence, as they will be either considered in future versions of the GBD or broadly confirmed by a large body of studies. Similarly, nutrients of public health concern (e.g., vitamin D, and potassium) and shortfall nutrients (e.g., magnesium, vitamin C or B12) that are identified by the US Department of Agriculture (USDA) and are not included in the GBD studies⁸ could also be incorporated in HENI in the future. Finally, this analysis does not capture possible additional damaging health effects from cooking methods⁴⁸ and discretionary salt added at the dining table.

To avoid or limit double counting, the following approach was applied according to the GBD used method for mediation:

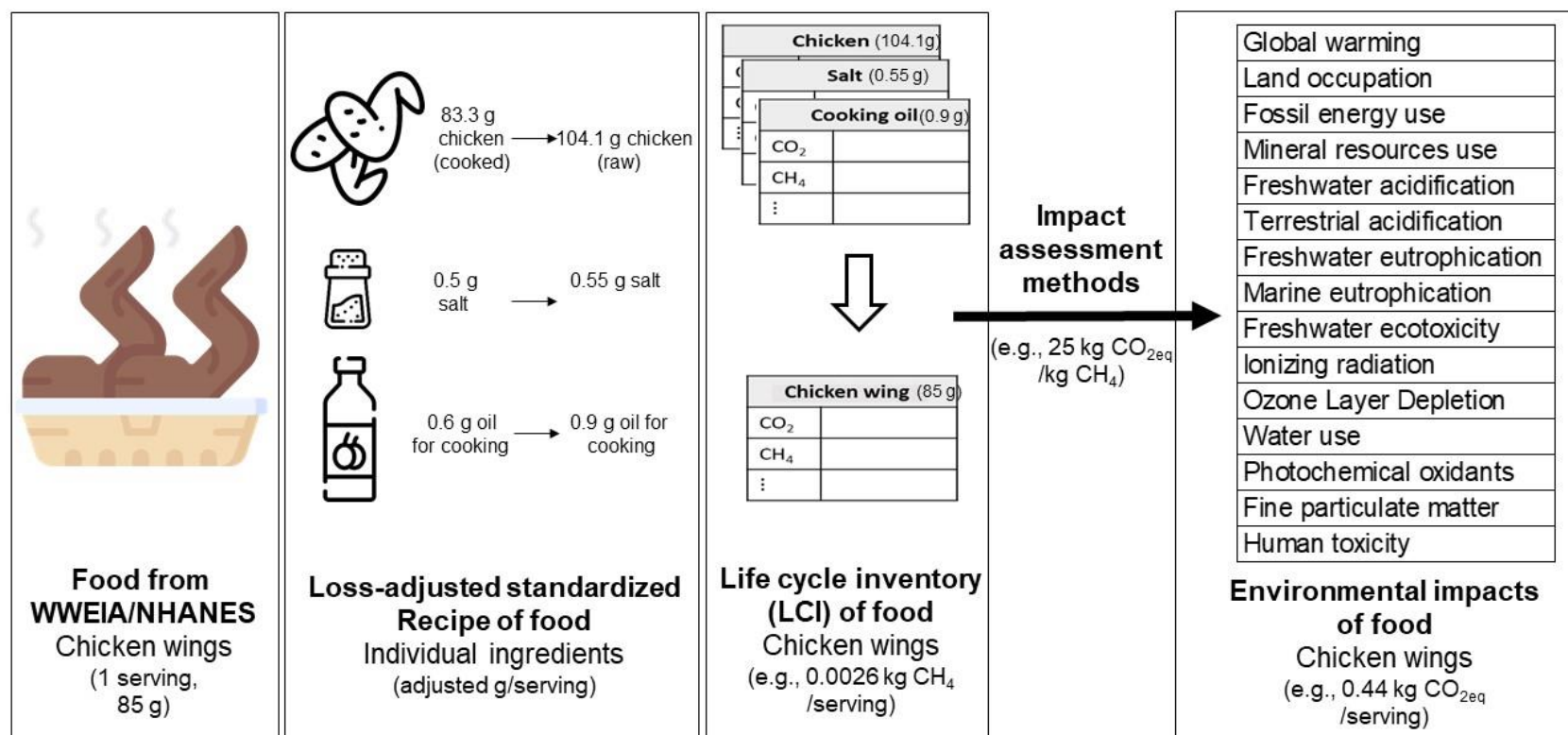
- For the health benefits of milk and calcium, the DRF_{milk} was only applied to foods classified as “milk” and “flavored milk”. For all other foods, only the benefits from calcium were considered.

- For the health benefits of fiber and fruits/vegetables/legumes/whole grains as there is a mediation effect, two source-specific fiber DRFs (and corresponding food composition profiles) were developed. Therefore, HENI considered the health burden of “fiber from fruits, vegetables, legumes, and whole grains”, e.g. $DRF_{fiber(f,v,l,w)}$, and “fiber

499 from other sources”, e.g. $DRF_{\text{fiber (other)}}$, separately as each source-specific fiber is
500 associated with different outcomes. More specifically, the $DRF_{\text{fiber (f,v,l,w)}}$ only considers the
501 health benefits on colorectal cancer (as the mediation is for ischemic heart disease) while
502 the $DRF_{\text{fiber (other)}}$ captures health benefits on colorectal cancer and ischemic heart disease.

503 • The individual contribution of TFA on HENI was investigated in the above
504 sensitivity study (Suppl. Fig. 10) to evaluate the magnitude of such effect and potential
505 double counting.

506 **S3. Environmental life cycle impact assessment of food items**



507

508 **Suppl. Fig. 11.** Schematic diagram of process followed to characterize environmental life cycle impacts for individual food items.

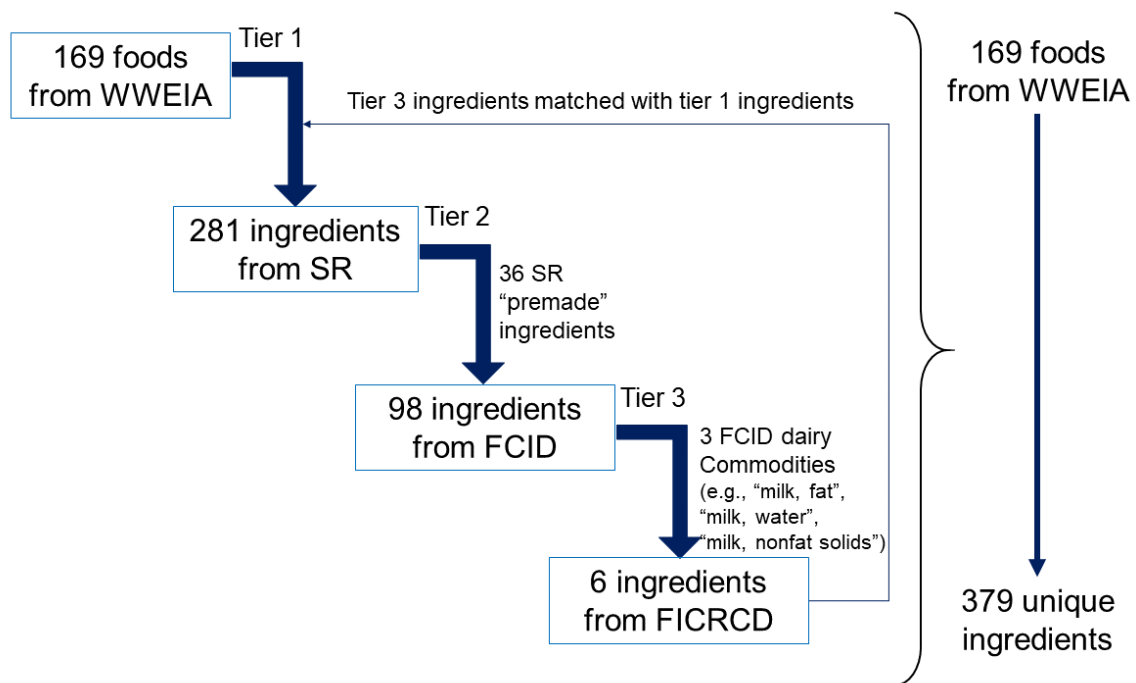
509 Diagram illustrates a serving of chicken wings as an example.⁴⁹

S3.1. Food reference flows

To determine the reference flows of foods, “standard recipes” were used to identify the ingredients that compose a food following a tiered approach (Suppl. Fig. 12), starting from the Standard Reference (SR) v.28 database¹¹. Developed by USDA, the SR links individual foods in WWEIA to a list of ingredients through standardized recipes reported in proportional masses for 100 g food. For example, the list of ingredients of “macaroni or noodles, with cheese, canned” includes cooked macaroni (29%), water (60%), cheese (3%), white flour (3%), and traces of (<1%) margarine, animal fat, sodium, whey, and dry nonfat milk.

Overall, the SR database covers 8,790 raw, processed, and prepared ingredients. The 169 foods considered in this analysis were decomposed into 281 ingredients (tier 1). For 36 of these ingredients, primarily those premade and “mixed,” further deconstruction was required to identify ingredients that matched with available LCIs. For this second deconstruction tier, the Food Commodities Intake Database (FCID) developed by US EPA was used to link WWEIA/NHANES foods items with 421 agricultural commodities (excluding baby food commodities)⁵⁰. For the deconstruction 98 FCID commodities were identified (tier 2 ingredients). An example of a tier 1 to tier 2 ingredient decomposition is “mayonnaise” that was further deconstructed into soybean oil (78%), egg (5%), vinegar (5%), egg yolk (3%), lemon juice (1%) and sugar (1%). However, the FCID database is limited in identifying dairy ingredients as these are reported in the database as aggregated forms of three commodities: “milk, fat,” “milk, water,” and “milk, nonfat solids.” To provide a suitable decomposition of dairy ingredients, the total mass of these three commodities was therefore assigned to one of eight dairy commodities (tier 3 ingredients)

reported for the same food item in the Food Intakes Converted to Retail Commodities Database (FICRCD)⁵¹. The eight dairy commodities in FICRCD include whole fluid milk, fluid 2% milk, fluid 1% milk, fluid skim milk, butter, cheese, yogurt, and other dairy products. In this analysis six FICRCD ingredients were identified. Tier 3 ingredients were then matched with the equivalent tier 1 ingredient. For example, “fluid whole milk” was matched with “milk, whole, 3.25% milkfat”. For the purpose of this paper, a total of 344 unique ingredients were identified, with 73% of the ingredients identified through tier 1 (including a small number of ingredients identified through tier 3). The food reference flows are available in Data S1.



Suppl. Fig. 12. Diagram of the tiered approach to determine the reference flows for foods in the WWEIA database. Foods are deconstructed to unique ingredients that can match with available life cycle inventory. SR: Standard Reference. FCID: Food

Commodities Intake Database. FICRCD: Food Intakes Converted to Retail Commodities Database.

S3.2. Adjustments for food losses and consumable amounts

The amount of all ingredients reported in standardized recipes was adjusted to reflect food loss and waste using the Loss-Adjusted Food Availability (LAFA) data series⁵². Depending on the ingredient's nature (processed or not) and the available LCI to map it with, three levels of loss factors were used. For unprocessed ingredients (e.g., fresh fruits and vegetables) loss factors included primary (manufacturing, milling, carcass weight), retail (supermarkets and restaurants), and consumer (nonedible share, cooking losses, and uneaten food) losses. Loss factors for processed ingredients (e.g., dairy products, condiments, and flours) covered retail and consumer losses. The loss factors used for cooked pasta ingredients (e.g., cooked macaroni) only considered consumer-level losses. For ingredients that an exact LAFA commodity match was not available, a similar commodity (e.g., fresh potatoes for cassava) or an average commodity (e.g., legumes for soybean seeds) were used as proxies.

An additional adjustment factor was applied when needed to convert the recipe reported amounts which are typically reported in consumable amounts (e.g., cooked or excluding inedible share such as skins, peels, seeds) to raw equivalent amounts. Conversion factors were primarily obtained from FICRCD (FICRCD documentation appendix B⁵³). Such adjustments were necessary since most LCIs are available for raw ingredients (e.g., fresh meat, dry pasta, dry legumes, and uncooked grains). For example, 85 g of roasted chicken wings correspond to 113 g of raw chicken (cooked to raw chicken correction factor of 1.33). For “boneless” meat and seafood ingredients, an ingredient-specific multiplier

was applied to account for the carcass to boneless weight ratio when only carcass LCIs were available. These multipliers were obtained from Heller et al.⁵⁴.

S3.3. Matching ingredients to life cycle inventory processes

Library of LCIs

A library of LCIs was compiled for food items in raw and lightly processed forms (e.g., dairy products, fruit juices, flours, refined sugar, processed tomatoes, and balsamic vinegar). Three databases were used to maximize coverage (listed in the order of selection priority): ecoinvent (Cut-off) v3.3⁵⁵, World Food LCA Database (WFLDB) v3.1⁵⁶, and ESU World food LCA database⁵⁷. The system boundary for LCIs was “cradle to farm gate or processor gate,” excluding emissions and resources required during the packaging, transportation, storage, and cooking stages.

All ingredients were matched with LCIs based on similarity, prioritizing for region of origin with a preference for US LCIs, followed by LCIs from Canada and Mexico as important source countries for US foods⁵⁸, and “rest of the world” (RoW) or global (GLO) LCIs that represent the market-weighted average of LCIs for the product in the respective region based on the available LCIs. RoW LCIs were preferred in the ecoinvent database, whereas GLO was preferred in the WFLDB and ESU databases. For ingredients with missing LCIs, the matching process required assigning proxy LCIs to ingredients that were chosen based on production similarities. These approximations are not expected to substantially influence results as ingredients with currently missing LCIs have minor contributions to climate impacts from the US diet⁵⁴.

Overall, 306 ingredients were matched with 120 LCIs (Data S1). Most pairs were a direct match (~60% directly matched sometimes after the ingredient amount was adjusted

for cooking and processing). The remaining 40% of pairs were established using proxy LCIs based on LCIs of similar ingredients (e.g., type-specific hard cheeses, such as cheddar, were matched with a hard cheese LCI) and best available LCIs (typically for infrequently consumed ingredients), with the understanding that there is higher uncertainty associated with the latter matching approach. Overall, proxies were determined primarily based on production methods similarities determined by the expert judgment of the authors. For ingredient-LCI pairs that required adjustment for processing, such as a dried apple ingredient matched with a fresh apple LCI, conversion factors discussed in section S3.2. were applied to account for the required farm-gate mass for a unit of processed ingredient. Weighted average LCIs (see below) were matched with 29 ingredients. Eight ingredients were matched with LCIs of “pre-made” ingredients that were developed using the FCID decomposition. As a sensitivity analysis, seven complementary ingredient-LCI pairs were considered to represent heated greenhouse cultivation. This addition aimed to illustrate the variability of environmental impacts for the same food as a function of differences in the ingredient production methods.

For the 120 LCIs, about 34% of LCIs used were selected from ecoinvent, 32% from the WFLDB, and 34% from the ESU database. The regional distribution of these LCIs was global (30%), Switzerland (23%), US (14%), rest of the world (13%), and the remaining represented other country-specific LCIs.

Average LCIs

Five ‘new’ LCIs were developed as the weighted average of available LCIs (see Table 7 in Suppl. Data S1). This methodology was implemented for ingredients that: a) involve different food systems (e.g., group beef in the US is a mix of suckled-based and

615 dairy-based meat⁵⁹), b) the US market share covers multiple production systems for the
616 same food (e.g., fresh tomatoes in the US can be cultivated in open fields, unheated
617 greenhouses, and heated greenhouse⁶⁰), and c) LCIs were available for a group of similar
618 ingredients only (e.g., a weighted average of the different types of oils produced in the US⁵⁸
619 was calculated for sesame oil as a proxy).

620

S3.4. Impact assessment methodology

Life cycle environmental impacts were calculated using the default factors from Impact World+ v1.4⁶¹, except for fine particulate matter formation (PM_{2.5}) and water use for which improved evaluations were developed. Human health damages from PM_{2.5}-related emissions (DALYs/serving) were estimated by multiplying inventory data for primary PM_{2.5}, NH₃, NO_x, and SO₂ emissions with spatially-explicit characterization factors (CFs) for the US from Stylianou et al.^{62,63}. Agriculture-specific national CFs were utilized for primary PM_{2.5} and NH₃, while the CFs for NO_x, and SO₂ were emission-weighted (Suppl. Table 8). CFs were estimated using an average slope (between background annual PM_{2.5} concentration at receptor location and theoretical minimum risk level for PM_{2.5}) from the non-linear exposure-response function by Cohen et al.⁶⁴ and region-specific background burden rates at each receptor location obtained from the GBD. Health damages are reported in minutes of healthy life lost per serving by multiplying μ DALYs estimates with 0.53⁶⁵ so that damages are comparable with HENI scores.

Water use is the impact category with most spatialize variation, thus it is important to have for this category a maximum of spatialized data corresponding to regional or at least national characteristics of water use. In analyzing the data available for the considered ingredients, only 15% of the LCIs used in this analysis cover US-specific production systems, estimates can significantly underestimate or overestimate water usage⁶⁶, in particular for animal production. For example, the estimated consumptive water use for milk in the WFLDB v3.1 database of 15 L/kg does not represent the average blue water use in milk production in the US that ranges from 50-200 L/kg milk according to Henderson et al. (2017)⁶⁶, the latter study being derived using detailed spatial production

system information. To improve estimates, a separate assessment was performed using US-specific blue water footprint estimates for animal products⁶⁷ and global average blue water footprint estimates for crops derived with the same methodology^{68,69}. For beef, pork, and chicken, blue water footprint estimates were differentiated for the co-products from slaughtering (e.g., fresh meat, offal, bones, and fat) based on their economic allocation, using allocation factors from WFLDB 3.1⁷⁰ that were calculated following ISO 14040/ ISO 14044 and information from the European Fat Processors and Renderers Association (<http://www.efpra.eu/>). The blue water footprint for cream and yogurt was calculated by adjusting other dairy product estimates based on their caloric density and an estimate for seafood using the consumptive water use reported for tuna in ecoinvent v.3.5⁷¹ as a proxy. The ingredients considered in this analysis were then matched with a commodity with an available blue water footprint estimate. The blue water footprint estimates were not differentiated between production systems (e.g., open field vs. greenhouse crop production) or food characteristics (e.g., whole vs. 2% vs. nonfat milk) since these data were not provided in by the used database. Information was also missing for whey, mustard, salt, salad dressing, mushrooms, honey, baking powder/soda, and water. All ingredients-commodity matches and blue water footprint estimates are available in Data S1. Since consumptive water use is reported at national level at best, adding water scarcity indicators in IMPACT World+ based on the available water remaining (AWARE) would correspond to applying a constant factor to all flows at this stage, which does not bring added information. Therefore, for this analysis, only consumptive water use estimates are reported. Future studies should spatialize the water use at e.g. state or watershed level for each crops and ingredients towards improved accuracy.

Suppl. Table 7. List of environmental indicators in the life cycle impact assessment (adapted from Impact World+ v1.4⁶¹)

Impact category	Indicator (Units)	Endpoint damage*	
		HH	EQ
Global warming, shorter term	Global warming potential over first 100 years (kg CO ₂ eq, shorter)	X	X
Global warming, long term	Global temperature potential at 100 years (kg CO ₂ eq, long term)	X	X
Land occupation, biodiversity	Biodiversity loss relative to arable land (ha-yr arable land)		X
Fossil energy use	Non-renewable primary energy consumed (MJ)		
Mineral resources use	Material competition scarcity index (kg _{deprived})		
Water use	Consumptive (blue) water use (m ³)		
Freshwater acidification	Change in freshwater pH relative to sulfur dioxide (kg SO ₂ eq)		X
Terrestrial acidification	Change in soil pH relative to sulfur dioxide (kg SO ₂ eq)		X
Freshwater eutrophication	Increase in phosphorus mass in freshwater relative to phosphate (kg PO ₄ →P eq)		X
Marine eutrophication	Nitrogen released to marine environments (kg N eq)		X
Freshwater ecotoxicity	Time and volume integrated potentially affected fraction of species in freshwater (CTU _e)		X
Fine particulate matter formation	Epidemiologically based human health burden attributable to fine particulate matter (DALYs)	X	
Photochemical oxidant formation	Human health burden attributable to ozone formation (DALYs)	X	
Human toxicity (cancer and non-cancer)	Human health burden attributable to carcinogenic and non-carcinogenic effects (DALYs)	X	
Ionizing radiation	Ionizing radiation impacts relative to C14 (Bq C-14 eq)	X	X
Ozone Layer Depletion	Ozone depletion potentials (kg CFC-11 eq)	X	X

CTU_e = comparative toxic units; DALYs= disability adjusted life years; PDF= Potentially Disappeared Fraction of species

* Midpoint impact categories contribute to human health (HH) and environmental quality (EQ) endpoint damages reported in DALYs and PDF-m²-yr, respectively.

Suppl. Table 8. National US characterization factors for PM_{2.5} precursors using an average slope of a non-linear exposure-response function⁶².

Characterization factors (DALYs/kg emitted)	PM _{2.5} Agriculture	SO ₂ All	NO _x All	NH ₃ Agriculture
	1.13·10 ⁻⁴	7.52·10 ⁻⁵	3.80·10 ⁻⁵	9.37·10 ⁻⁵

677

678 **S3.5. Uncertainty analysis of the environmental impacts**

679 Most LCA approaches and software only provide uncertainties at the inventory
680 level, for a limited number of databases such as ecoinvent, and disregard the important or
681 even dominant uncertainties associated with the life cycle impact assessment factors.
682 Hong et al. (2010) demonstrated the use of a Taylor series expansion approach to account
683 for both inventory and impact assessment steps⁷². In this approach, since LCA impacts are
684 bounded at zero and can span several orders of magnitude, uncertainties at each of the
685 assessment step are characterized using a squared geometric standard deviation (GSD^2)
686 that indicates that 95% of the estimates fall within the median divided by GSD^2 and the
687 median multiplied by GSD^2 , assuming a lognormal distribution. Uncertainty is then
688 combined and propagated across the different assessment step using Taylor series
689 expansion.

690 For each food-impact category pair an overall GSD^2 was estimated by combining
691 base uncertainty (specific to the LCI flows and impact characterization of each impact
692 category) and uncertainty associated with data pedigree for three categories (LCI match,
693 Loss/Waste factor, consumable amounts) as follows:

$$\begin{aligned} 694 \quad GSD_{i,j,overall}^2 = & \exp(\ln(GSD_{i,base}^2)^2 + \ln(GSD_{j,LCI}^2)^2 + \ln(GSD_{j,L/W}^2)^2 + \\ 695 \quad & \ln(GSD_{j,consumable}^2)^2)^{1/2} \quad (Eq. S6) \end{aligned}$$

696 where i represents an impact category and j an individual food.

697 **S3.5.1. Pedigree-related uncertainties**

698 Using expert judgment, the uncertainty distribution of factors influencing the data
699 quality of ingredient LCIs used in this analysis was qualitatively characterized using a

700 pedigree approach. Uncertainties were characterized by a lognormal distribution
701 determined by a data pedigree of qualitative scores from 1-5 (lower score=higher data
702 quality) for three factors: matched LCI representation, loss/waste adjustment, and
703 additional adjustment, with higher corresponding GSD² ranging from 1 to 1.5 for LCI
704 match, 1 to 5 for loss/waste, and 1 to 1.5 for consumable amounts. The description of data
705 quality levels for each factor and the corresponding quantitative uncertainty expressed as
706 a GSD² is shown in Suppl. Table 9.

Suppl. Table 9. Description of data pedigree and relationship between data pedigree GSD².

Uncertainty factors						
LCI representativeness			Loss/waste adjustment		Additional adjustment	
	Description	GSD²_{LCI}	Description	GSD²_{L/W}	Description	GSD²_{consumable}
1	US-specific LCI is an exact match with ingredient	1	Commodity and adjustment level are an exact match for the ingredient	1	Additional adjustment not needed	1
2	LCI is an exact (or close) match with ingredient but for region other than the US (1.05)	1.05	Commodity is an exact match with ingredient and level is a proxy based on matched LCI information	1.5	Additional adjustment is an exact match to ingredient-LCI pair (from FICRCD)	1.1
3	New LCI determined using a combination of available LCIs (processing adjustment, LCI addition, ingredient disaggregation with FCID)	1.1	Commodity is a proxy for the ingredient based on shelf life/consumer behavior characteristics and level is in line with LCI	1.75	Additional adjustment is a proxy (from FICRCD or other sources)	1.25
4	Proxy LCI (similar ingredients or production systems)	1.25	Commodity and adjustment level are proxies based on shelf-life/consumer behavior characteristics	2.5	Unknown or unassigned	1.5
5	Unknown or unassigned	1.5	Unknown or unassigned	5		

S3.5.2. Base uncertainty for midpoint impact score and endpoint damage

A screening analysis of LCI uncertainties was performed based on a set of diverse foods available in ecoinvent, showing that the uncertainty across foods is primarily driven by the considered impact category. Thus, base uncertainty factors were estimated for each impact category to reflect both LCI and LCIA uncertainties on the midpoint impacts and endpoint damages, accounting for two factors: 1) The average uncertainty of the LCI elementary flows from the set of diverse foods related to this impact category (GSD^2_{LCI}), and 2) the uncertainty on the LCIA characterization factors, as determined by expert judgment of Impact World+ developer. The LCIA uncertainty of midpoint impacts was categorized using a qualitative pedigree score from 1-4 with the corresponding GSD^2_{LCIA} ranging from 1.4 to 10 (lower score = higher data quality). Similarly, the uncertainty of endpoint damages was categorized using a qualitative score from 1-5, with higher corresponding GSD^2_{LCIA} ranging from 2.5 to 1000 to account for the uncertainty on the midpoint-to-endpoint damage factors. The classification of the impact categories considered and the corresponding GSD^2_{LCI} and GSD^2_{LCIA} used in this analysis are listed in Suppl. Table 10.

For a given environmental indicator i , the base uncertainty is calculated as:

$$GSD^2_{i,base} = \exp(\ln(GSD^2_{i,LCI})^2 + \ln(GSD^2_{i,LCIA})^2)^{1/2} \quad (Eq. S7)$$

Suppl. Table 10. Summary of inventory and characterization factor uncertainty for midpoint impacts (A) and endpoint damages (B) by impact category.

A. Midpoint impacts						
	Inventory	Characterization factor				
	GSD ² _{LCI}	Class	1 ^a	2	3	4
		GSD ² _{LCIA}	1.4	1.75	2.5	10
Fossil energy use	1.24		X			
Freshwater acidification	1.20			X		
Freshwater ecotoxicity	1.56					X
Freshwater eutrophication	1.71			X		
Global warming, shorter term	1.31		X			
Global warming, long term	1.33		X			
Ionizing radiation	2.56					X
Land occupation, biodiversity	1.32				X	
Marine eutrophication	1.19				X	
Mineral resources use	1.90				X	
Ozone Layer Depletion	1.65			X		
Terrestrial acidification	1.18			X		
Water use	1.49			X		

B. Endpoint damages ^b							
	Inventory	Characterization factor					
	GSD ² _{LCI}	Class	1	2	3	4	5
		GSD ² _{LCIA}	2.5	7.5	25	100	1000
Global warming, long term, HH	1.21				X		
Global warming, shorter term, HH	1.20				X		
Human toxicity, HH							
Cancer						X	
Long-term	2.89						
Short-term	2.02						
Non-cancer							X
Long-term	6.39						
Short-term	4.71						
Ionizing radiation, HH	2.57					X	
Ozone layer depletion, HH	1.61			X			
Particulate matter formation, HH	1.19			X			
Photochemical oxidant formation, HH	1.28				X		
Freshwater acidification, EQ	1.18				X		
Freshwater ecotoxicity, EQ						X	
Long-term	1.98						
Short-term	1.36						
Freshwater eutrophication, EQ	1.37			X			
Global warming, long term, EQ	1.21				X		
Global warming, shorter term, EQ	1.31				X		
Ionizing radiation, EQ	1.67					X	
Land occupation, biodiversity, EQ	1.32				X		
Marine acidification, EQ	1.56				X		
Marine eutrophication, EQ	1.19				X		
Terrestrial acidification, EQ	1.18				X		

^a Ingredient-specific estimated derived based on emissions weighted average of CO₂ emissions from burning fossil fuels (GSD²=1) and other greenhouse gas emissions (GSD²=1.4) contributing to global warming potential.

^b Midpoint impact categories contribute to human health (HH) and environmental quality (EQ) endpoint damages.

Midpoint categories related with global warming and fossil energy use were assigned a class 1 LCIA uncertainty. The GSD^2_{LCIA} for this global warming for ingredient j was estimated using the Taylor series expansion method⁷² from the uncertainty of CO₂ and other greenhouse gases (GHGs):

$$GSD^2_{midpoint\ class\ 1,LCIA,j} = \exp(f_{CO_2\ fossil-GWP,j}^2 \cdot \ln(GSD^2_{LCIA,CO_2\ fossil-GWP})^2 + (1 - f_{CO_2\ fossil-GWP,j})^2 \cdot \ln(GSD^2_{LCIA,other\ GHG-GWP})^2)^{1/2} \quad (Eq. S8)$$

where $f_{CO_2\ fossil-GWP}$ is the fraction of global warming potential for ingredient j attributable to carbon dioxide (CO₂) emissions from burning fossil fuels. $GSD^2_{LCIA,CO_2\ fossil-GWP}$ and $GSD^2_{LCIA,other\ GHG-GWP}$ characterize the uncertainty associated with the contribution of CO₂ and other greenhouse gases, such as CH₄ and N₂O emissions contributing to global warming potential for 100-year time horizons, respectively. For CO₂ emissions, this uncertainty is assumed as 1 since this is the reference substance, while for CH₄, N₂O and other greenhouse gases a factor of 1.4 was used as reported in IPCC 2013⁷³. The CO₂ emission estimates for all ingredients were obtained from LCIs.

For human toxicity, total human health (HH) damages, and total ecosystem quality (EQ) damages the Taylor series expansion method⁷² was applied to estimate aggregated GSD^2_{base} for each ingredient j following^{72,74}:

$$GSD^2_{i_{ag.base},j} = \exp(S_{i_1}^2 \cdot \ln(GSD^2_{i_1\ base,j})^2 + S_{i_2}^2 \cdot \ln(GSD^2_{i_2\ base,j})^2 + \dots + S_{i_n}^2 \cdot \ln(GSD^2_{i_n\ base,j})^2)^{1/2} \quad (Eq. S9)$$

where $GSD^2_{i_w\ base,j}$ is the base uncertainty of impact category w that contributed to the aggregated endpoint damages of category i for ingredient j and S_{i_w} is the proportion of

aggregated impact estimates for i to the endpoint damage estimates of contributing category w , with $\sum_w S_{iw} = 1$. Human toxicity impacts were estimated by aggregating cancer and non-cancer human health damages. All impact categories that contain “HH” in their description in Suppl. Table 10 were aggregated for the total HH damages. Likewise, impact categories that contribute to total EQ damages are identified in Suppl. Table 10 by the inclusion of “EQ” in their description.

S3.6. Limitations

The approach developed to evaluate the environmental impacts of individual food items in the US diet addresses a major challenge in food LCA associated with evaluating and consistently comparing ‘as-consumed’ foods that contain multiple ingredients. This approach could be extended to all foods in the WWEIA database, allowing for better quantification of the environmental impacts associated with the US diet. Most importantly, this analysis evaluates various environmental impacts and their corresponding aggregated damages on ecosystem quality and human health. Human health damages are comparable with nutritional impacts as they are both based on the same underlying data and can be reported in the same units (DALYs or minutes of healthy life).

Several limitations need to be acknowledged and reflected in the interpretation of the food-specific environmental impacts presented in this work. Data gaps related to LCI availability, particularly for ingredient coverage and geographical representativeness, and spatially-explicit impacts methodologies have been key challenges in food-LCAs^{75–78}. To improve ingredient and geographical coverage LCIs from three databases (ecoinvent v3.3, WFLDB v3.1, and ESU World food LCA database) were utilized. Although all databases have been built according to ecoinvent-based recommendations, LCIs in each database

might have been developed using different assumptions that might influence the environmental performance of foods. Food ingredients were matched with available LCIs and used proxies to best match ingredients or production system characteristics when LCIs were not available. Furthermore, a regional prioritization of LCIs was implemented to improve the geographical representativeness of this analysis: the US, major import countries, global, and other European countries.

In this analysis, the work by Mekonnen and Hoekstra^{67,69} was used to account for US-specific water usage estimates for food production, as the majority of LCIs in the LCA databases were for regions outside the US. While the estimates by Mekonnen and Hoekstra provided adequate coverage of different ingredients for animal products such as beef, pork, and chicken, estimates were further differentiated for by-products using economic allocation. While this water footprint source improved the spatial representativeness of estimates, it should be mentioned that estimates specific to different production modes were not available for all ingredients (e.g., open field and greenhouse crops). US-specific water footprint information was also not available for certain foods such as yogurt, cream, neither for processed meat, such as sausage. For yogurt and cream, the water footprint estimates were extrapolated from other dairy products such as milk and cheese based on standard caloric densities (see Data S1). For processed meat, “fresh meat” estimates of the reported meat (pork/beef/chicken) were used, which might overestimate the water impacts associated as processed meats as these are typically cuts of lower value. Overall, water usage of food production systems is sensitive to inventory sources.

The influence and importance of spatial LCIs have been highlighted by the work of Poore and Nemecek (2018), where the authors illustrate differences in the impacts

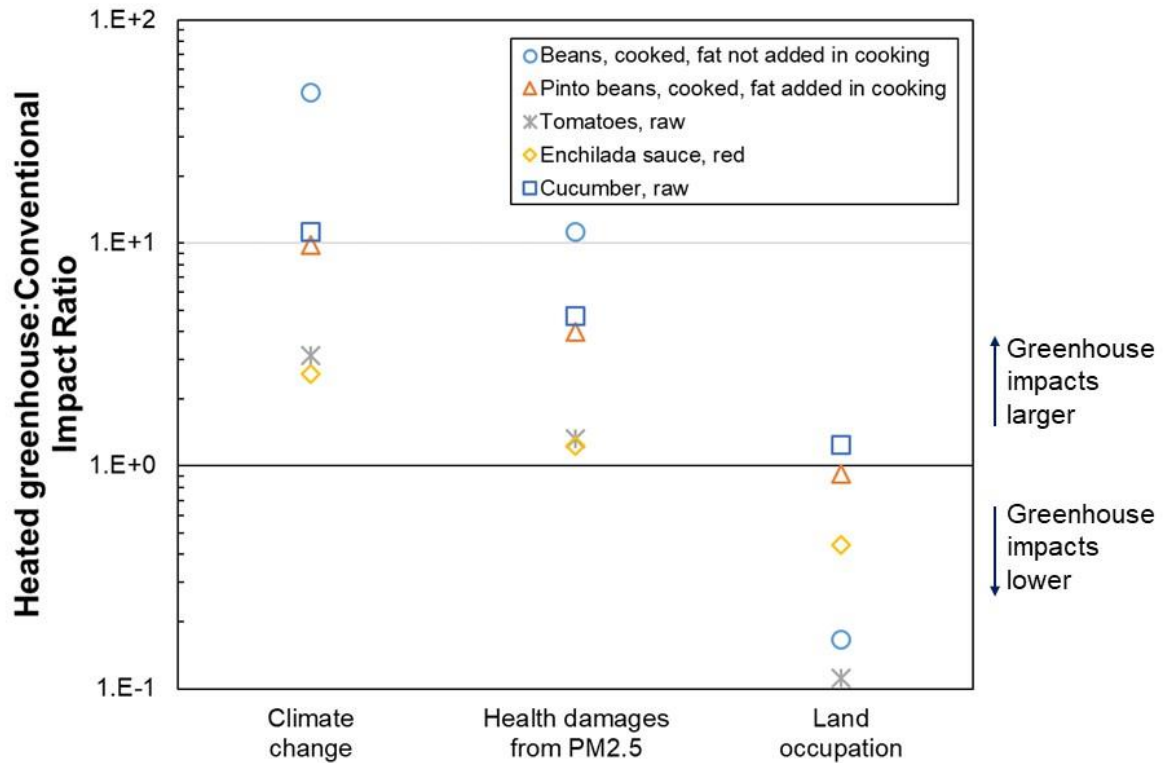
generated by the same foods between regions and production systems⁷⁹. Henderson et al. (2017) demonstrate the importance of spatial considerations when evaluating water usage and water scarcity of food systems. Regional inventories help leverage spatially explicit life cycle impact assessment methods that are becoming available⁸⁰. In the present analysis, PM_{2.5}-related health damages were determined using a spatially-explicit impact method that was designed to reflect sector-specific emission-weighted impacts for the US^{62,63}. While this method improves the impact characterization attributable to NH₃ emissions from agricultural practices, it is important to further scrutinize the spatial representativeness of food-LCIs, especially for water use.

The system boundary of this analysis was either “cradle to farm gate” or “cradle to processor gate,” depending on the ingredient. As a result, environmental impacts are likely to be underestimated as impact associated with food production, packaging, distribution, storage (refrigeration and freezing), preparation, cooking, and waste processing have not been considered. According to Poore and Nemecek (2018), post farm gate stages of the global food supply chain can increase climate change impacts by 18% (processing 4%, transportation 6%, packaging 5%, and retail 3%), with stage-specific contributions varying substantially within and between ingredients⁷⁹. Similar stage-specific climate change impact contributions from transportation (4%), packaging (6%), and retail (5%) have been estimated for US food systems^{54,81}.

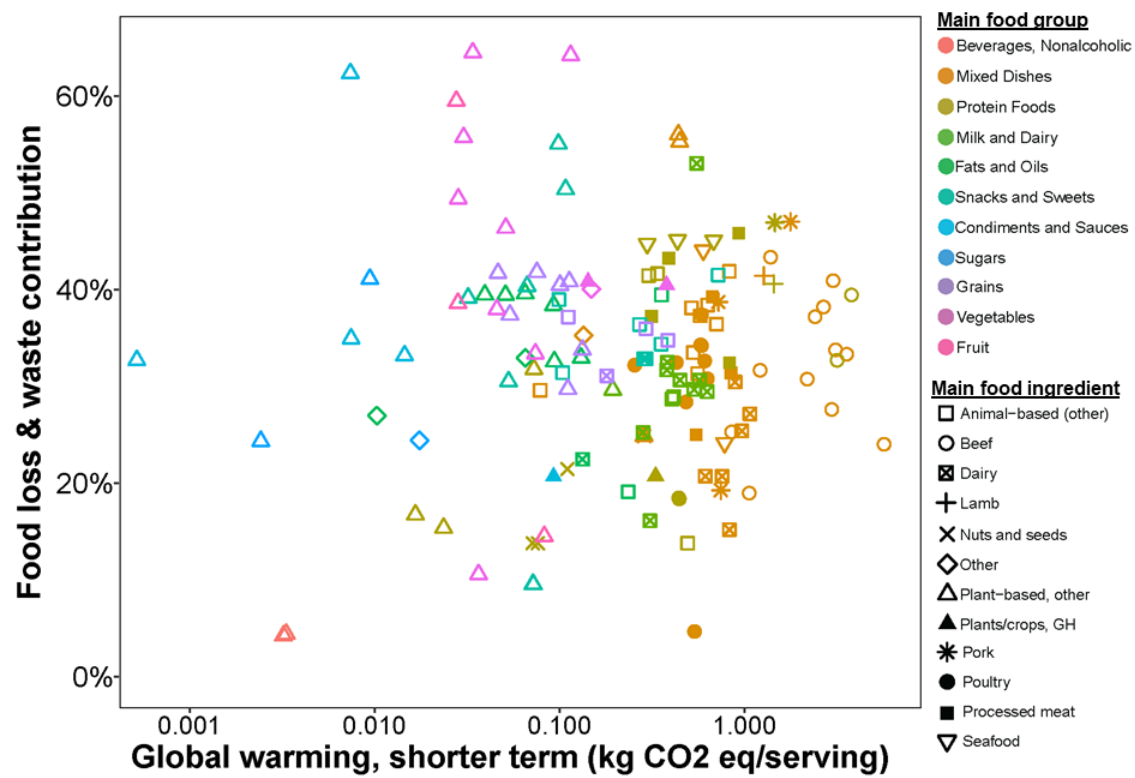
Variability in production system has not been the focus of the present paper, our analysis primarily studying conventional food production systems. Nevertheless, the influence of production systems (e.g., open field, heated/unheated greenhouse, organic) on environmental impacts has been covered in several LCAs⁸², indicating that heated crop

production is associated with substantial impacts, whereas the reduction in impacts associated with organic production might be compensated by decreases in yields^{55,83–85}. To illustrate these differences, five additional foods were considered that contain ingredients grown in heated greenhouse systems, such as beans, cucumbers, and tomatoes. The estimated impacts of these foods are different when ingredients were greenhouse-grown compared to conventionally cultivated (Suppl. Fig. 13). Overall, heated greenhouse-grown ingredients generate higher climate change impacts and health damages from PM_{2.5}-related emissions but lower land occupation impacts at the food level compared to ingredients produced in fields. These differences are enhanced when foods were compared on a 100 kcal basis (Suppl. Figs. 13 and 14).

Finally, environmental impact estimates account for loss and waste contributions. Commodity-specific loss and waste factors for the US were obtained from the LAFA database and were applied to ingredients. For certain foods with relatively low environmental impacts, such as fruits and vegetables, food loss and waste are key contributors (>60%). In contrast, contributions are around 25% for foods with high environmental impacts such as beef, pork, and lamb (Suppl. Fig. 14). Nevertheless, as loss and waste factors were applied at the ingredients level, food-level environmental impacts might be overestimated.



Suppl. Fig. 13. Ratios of the environmental impacts of foods containing heated greenhouse and field cultivated ingredients. Ratios higher than one show that foods containing heated greenhouse-grown ingredients generate a higher impact than foods containing conventionally-produced ingredients.



847

848 **Suppl. Fig. 14. Comparison of climate change impacts and food loss and food waste**

849 **impact contribution.** Each marker represents an individual food.

S4. Comparison of nutritional and environmental impacts of foods.

The results of the nutritional and environmental performance of the 167 foods considered in this analysis are available in a separate file (Data S1). Using these results, foods are classified in one out of three impact levels in each indicator (Suppl. Table 11). Depending on their HENI scores, foods can be classified as beneficial ($\text{HENI} > 0$, e.g., minutes gained), slightly detrimental ($25^{\text{th}} \text{ percentile} < \text{HENI} \leq 0$), and detrimental ($\text{HENI} \leq 0.25^{\text{th}} \text{ percentile}$). Similarly, foods could be classified as low ($< 50^{\text{th}} \text{ percentile}$), moderate (50^{th} to $75^{\text{th}} \text{ percentile}$), and high ($\geq 75^{\text{th}} \text{ percentile}$) impact foods for each environmental indicator. Based on their nutritional and environmental performance levels, foods are categorized into three color-coded zones (green, amber, red).

The red zone contains foods that are either nutritionally detrimental or have high environmental impacts, and it is recommended to decrease the consumption of these foods. The green zone, which is aligned with an increased consumption recommendation, includes foods that are nutritionally beneficial and have low environmental impacts. The remaining foods, which are either slightly detrimental or have moderate environmental impacts and do not meet the criteria for the red and green zones, are classified in the amber zone, and their consumption is tolerable for now. This classification was performed for all nutrition-environmental indicator pairs, presenting both results per serving (Suppl. Fig. 15) and per 100 kcal (Suppl. Fig. 16). Suppl. Figs. 15 and 16 illustrated all environmental impacts on a $\log_{10}$ scale to fully display uncertainties, whereas figure 5 in the main text presents results for a subset of categories in normal scale for a better comparison of foods with low versus high impacts.

Using global warming (shorter-term, between 0 and 100 years) as an environmental indicator, 60 foods are classified in the red zone based on serving size (Fig. 5A). These foods primarily contain beef, processed meat, pork, lamb, sweetened beverages, and some high environmental impact seafood such as salmon and shrimp. For the same comparison, foods classified in the amber zone (N=58) typically contain eggs, poultry, refined grains, dairy, and greenhouse-grown vegetables. From these foods, those with low caloric density get classified in the red zone when compared per 100 kcal. Finally, the serving-based green zone contains 49 foods that are primarily nuts, whole grains, vegetables, and fruits. Food classifications for other environmental impacts are mostly consistent with those of global warming, as most impact categories are highly correlated to global warming ($r > 0.90$ for most of these categories), with the exception of freshwater ecotoxicity and consumptive water use ($\sim r = 0.50$, Suppl. Fig. 17A – correlation per serving). More specifically, plant-based foods such as fruits, vegetables, and nuts get also classified in the red zones of freshwater ecotoxicity and consumptive water use due to their relatively high demand of water use. This becomes even more accentuated when considering correlations between impacts categories when foods are compared per 100 kcal (Suppl. Fig. 17B), since fruits and vegetable are low caloric foods.

Supplementary figures 15 and 16 demonstrate that for both nutrition impacts in minutes of healthy life lost and for well-studied environmental impact categories such as global warming (Suppl. Figs. 15A and 16A), uncertainties are relatively limited and the comparison between foods enable to differentiate well product between and even within the different recommendation zones. For midpoint impact categories with higher base uncertainty, such as freshwater ecotoxicity (Suppl. Figs. 15J and 16J) or human toxicity

(Suppl. Figs. 15P and 16P), and for the two aggregated damages scores for human health (Suppl. Figs. 15Q and 16Q) and ecosystems (Suppl. Figs. 15R and 16R), uncertainties are substantially higher and there is more overlap between individual food items within and even across the different zones.

It should be noted that kcal-based comparison can be misleading for certain foods. This comparison basis increases both the nutritional and environmental rankings of low caloric density foods due to larger amounts of food being evaluated, often leading to a worse classification in comparison to serving-based classifications. For example, conventionally produced and heated greenhouse-cultivated cucumbers are classified in the green and amber zones, respectively, on a serving basis. However, since 100 kcal of cucumbers is equivalent to 5.5 cups or ~10 servings, cucumbers have the best nutritional performance (40 minutes gained per 100 kcal), while their climate impacts per 100 kcal are above the 50th (amber zone) and 75th (red zone) when they are produced conventionally (0.33 kg CO₂eq per 100 kcal, GSD²=1.64) and in a heated greenhouse (3.7 kg CO₂eq per 100 kcal, GSD²=1.64), respectively. However, the opposite is observed for calorie-dense foods. Relatively lower rankings for both the environmental and nutritional indicators were observed when comparing these foods per 100 kcal. When integrated into the overall diet, these foods can easily contribute to excess daily caloric intake, which further increases body mass index, an important health risk factor¹. Overall, foods to decrease should be identified based on serving size evaluations, while foods to increase be based on kcal-based evaluations for maximum benefit per kcal.

Supplementary figure 18 compares the nutritional and environmental performance of the median values and their standard error within 19 food groups, focusing on the global

warming score and consumptive water use per serving and per 100 kcal. The 19 food groups were mainly determined based on the type of ingredient with the highest contribution to environmental impacts. For some foods, food groups were determined based on their USDA main group since it allowed for a better grouping from a nutritional perspective. Consistent with Figure 5 in the main text, the global warming scores per serving size are the highest for beef dishes, followed by lamb and pork (Suppl. Fig. 19A). The carbon footprint of the other animal food such as poultry and dairy are substantially lower and a factor 4 below beef, at the same level as greenhouse-based plant production per serving. The lowest carbon footprints per serving are observed for legumes, fruits, vegetables and fats and oil. For consumptive water use per serving, nuts and seeds as well as fruits show high usage per serving (Suppl. Fig. 19C), in the same order of magnitude as animal-based dishes. Nutritional performances show processed meat followed by red meats as most detrimental, with nuts and seed, fruits and legumes as most beneficial. Dairy and poultry dishes are close to neutral, as well as grains.

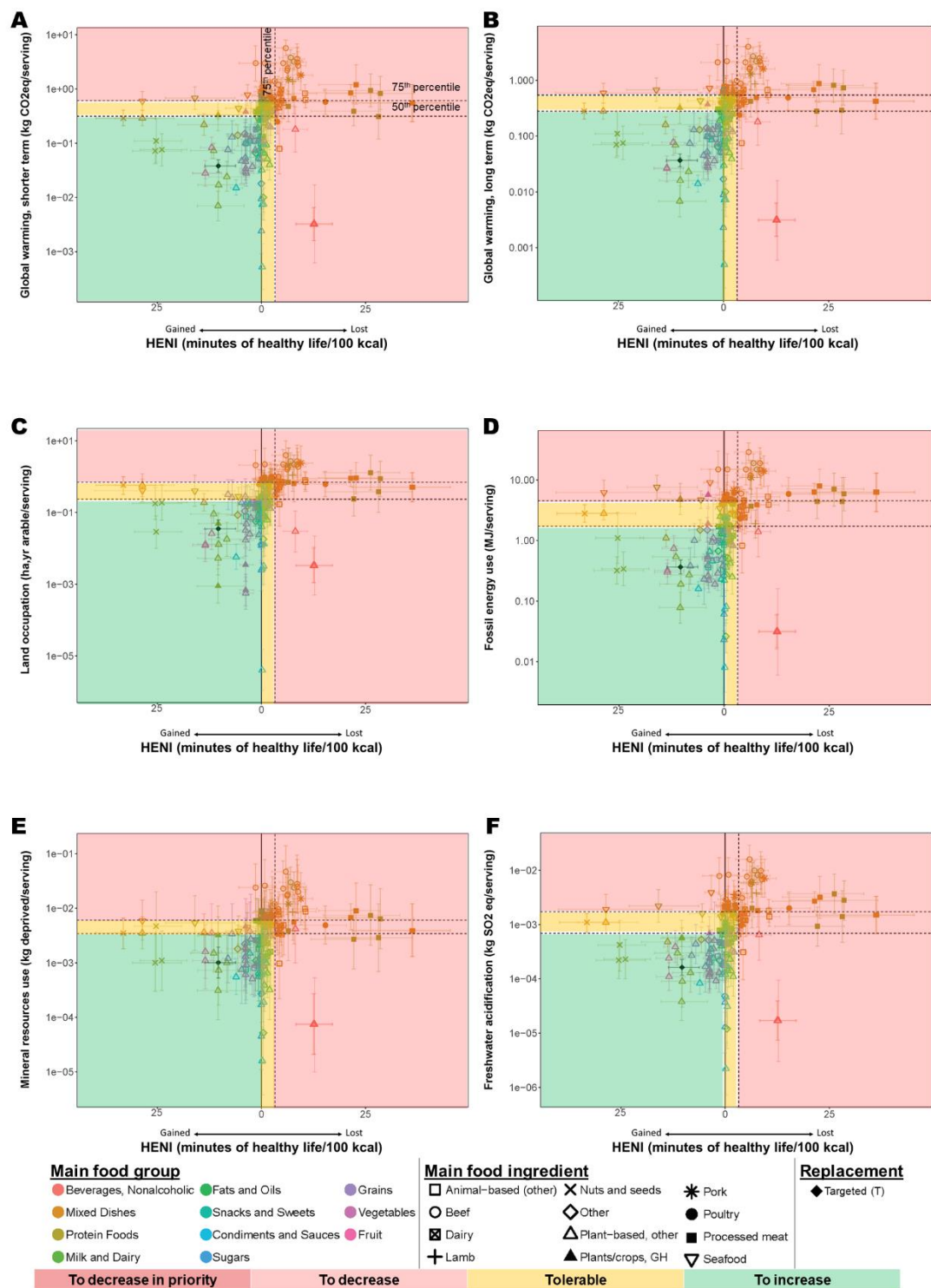
As discussed above, the representation per 100 kcal is especially interesting to consider for selecting among the beneficial foods those to use as substitution in priority (figs 19B and 19D). It clearly shows how greenhouse based production need to be avoided to limit the carbon footprint of the substitution, whereas low caloric foods such as fruits or vegetable have a substantially higher ranking than the serving representation. Selecting some of these low caloric food to substitute processed meat and red meat dishes will therefore enable a maximum benefit for a given reduction in meat kcal.

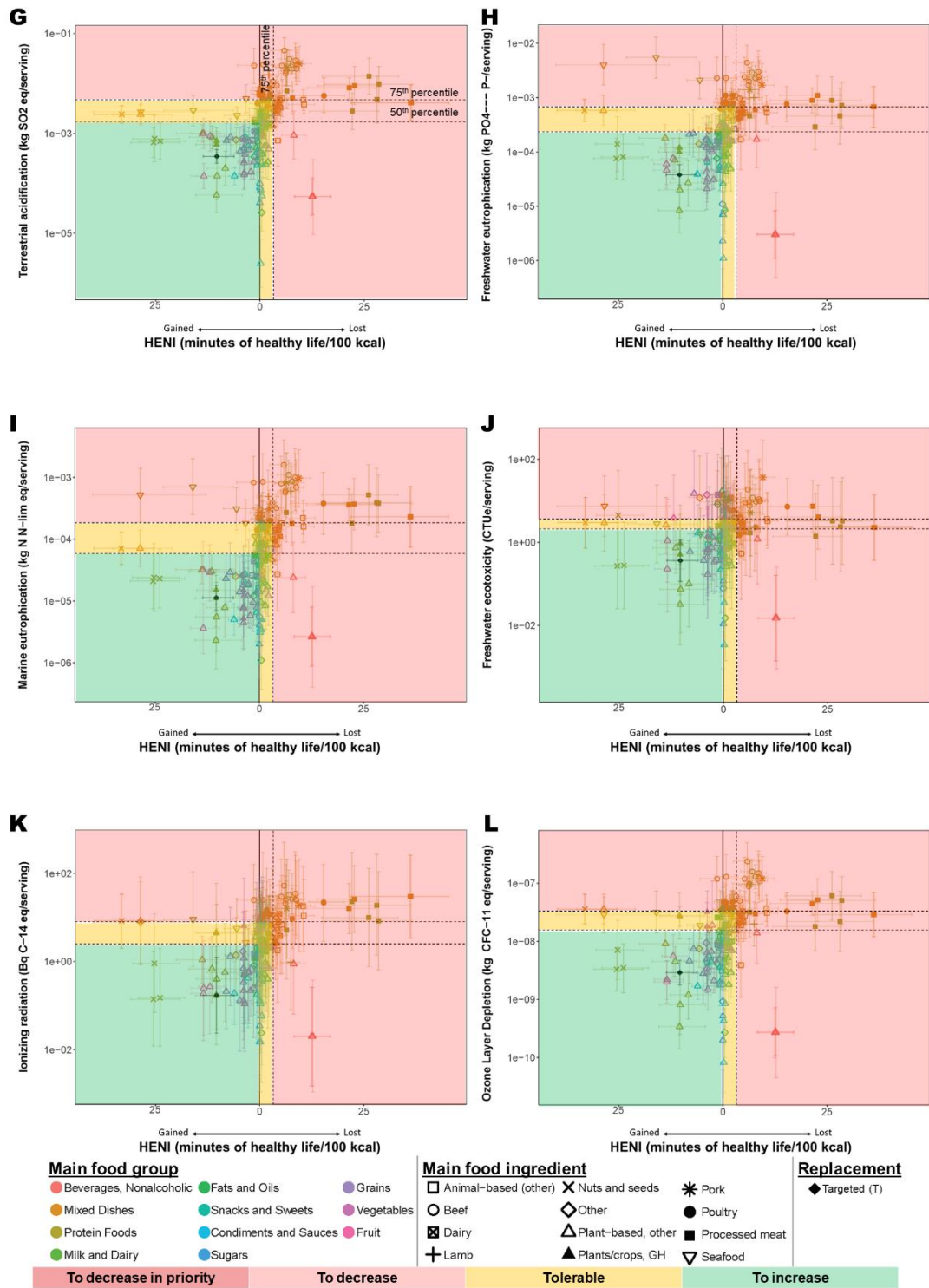
939 **Suppl. Table 11.** Definition (A) and limits (B) of recommendation zones.

A. Definition of recommendation zones	
Zone	Definition
Green (to increase)	Nutritionally beneficial AND environmentally low
Amber (tolerable)	Nutritionally slightly detrimental & environmentally low OR Nutritionally beneficial & environmentally moderate
Red (to decrease)	Nutritionally detrimental OR environmentally high

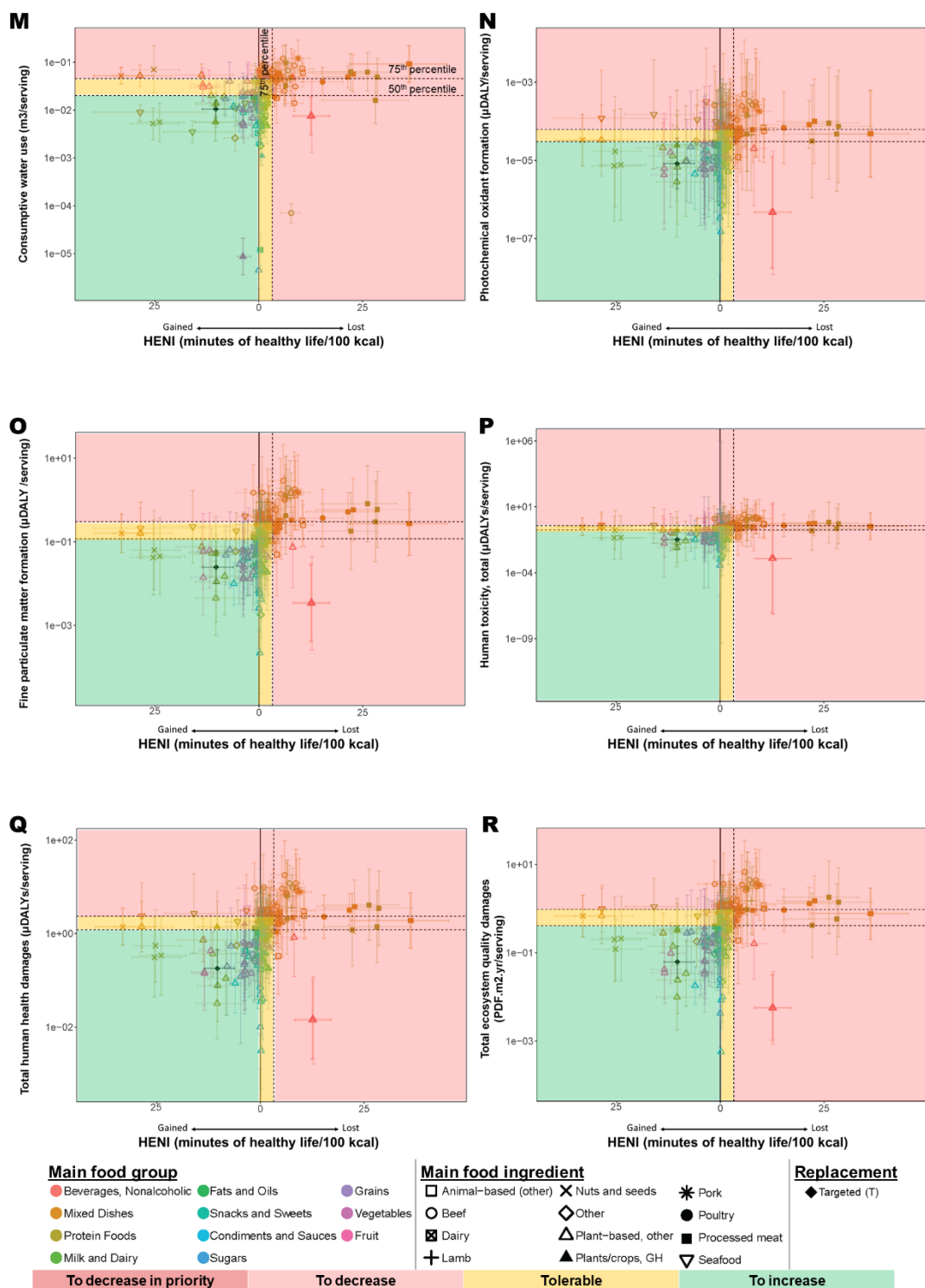
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B. Limits by indicator for recommendation zones							
		Serving size			100 kcal		
Nutrition							
Indicator	Units	Beneficial (HENI>0)	Slightly detrimental (25 th pctl to 0)	Detrimental (≤25 th pctl)	Beneficial (HENI>0)	Slightly detrimental (25 th pctl to 0)	Detrimental (≤25 th pctl)
HENI	minutes _{Healthy}	(0,∞)	(-3.2,0]	(-∞,-3.2]	(0,∞)	(-1.6,0]	(-∞,-1.6]
Environment							
		Low (<50 th pctl)	Moderate (50 th - 75 th pctl)	High (>75 th pctl)	Low (<50 th pctl)	Moderate (50 th - 75 th pctl)	High (>75 th pctl)
Global warming, shorter term	kg CO ₂ eq	[0,0.32)	[0.32,0.61)	[0.61,∞)	[0,0.17)	[0.17,0.37)	[0.37,∞)
Global warming, long term	kg CO ₂ eq	[0,0.28)	[0.28,0.54)	[0.54,∞)	[0,0.14)	[0.14,0.29)	[0.29,∞)
Land occupation, biodiversity	ha-yr arable	[0,0.24)	[0.24,0.70)	[0.70,∞)	[0,0.14)	[0.14,0.26)	[0.26,∞)
Fossil energy use	MJ	[0,1.7)	[1.7,4.5)	[4.5,∞)	[0,1.0)	[1.0,2.4)	[2.4,∞)
Mineral resources use	g _{deprived}	[0,3.4)	[3.4,6.1)	[6.1,∞)	[0,1.7)	[1.7,3.9)	[3.9,∞)
Water use	L	[0,20)	[20,45)	[45,∞)	[0,13)	[13,19)	[19,∞)
Freshwater acidification	g SO ₂ eq	[0,0.69)	[0.69,1.7)	[1.7,∞)	[0,0.39)	[0.39,0.99)	[0.99,∞)
Terrestrial acidification	g SO ₂ eq	[0,1.7)	[1.7,4.7)	[4.7,∞)	[0,0.98)	[0.98,2.3)	[2.3,∞)
Freshwater eutrophication	g PO ₄ →P- eq	[0,0.23)	[0.23,0.67)	[0.67,∞)	[0,0.14)	[0.14,0.31)	[0.31,∞)
Marine eutrophication	g N eq	[0,0.058)	[0.058,0.18)	[0.18,∞)	[0,0.036)	[0.036,0.091)	[0.091,∞)
Freshwater ecotoxicity	CTUe	[0,2.1)	[2.1,3.6)	[3.6,∞)	[0,1.1)	[1.1,2.5)	[2.5,∞)
Ionizing radiation	Bq C-14 eq	[0,2.5)	[2.5,8.0)	[8.0,∞)	[0,1.7)	[1.7,3.6)	[3.6,∞)
Ozone Layer Depletion	ng CFC-11 eq	[0,16)	[16,33)	[33,∞)	[0,7.2)	[7.2,18)	[18,∞)
Fine particulate matter formation	μDALYs	[0,0.12)	[0.12,0.30)	[0.30,∞)	[0,0.064)	[0.064,0.16)	[0.16,∞)
Photochemical oxidant formation,	pDALYs	[0,30)	[30,62)	[62,∞)	[0,16)	[16,38)	[38,∞)
Human toxicity (cancer and non-cancer)	μDALYs	[0,0.18)	[0.18,0.37)	[0.37,∞)	[0,0.091)	[0.091,0.20)	[0.20,∞)
Total human health	μDALYs	[0,1.2)	[1.2,2.4)	[2.4,∞)	[0,0.58)	[0.58,1.2)	[1.2,∞)
Total ecosystem quality	PDF.m2.yr	[0,0.41)	[0.41,0.95)	[0.95,∞)	[0,0.22)	[0.22,0.48)	[0.48,∞)



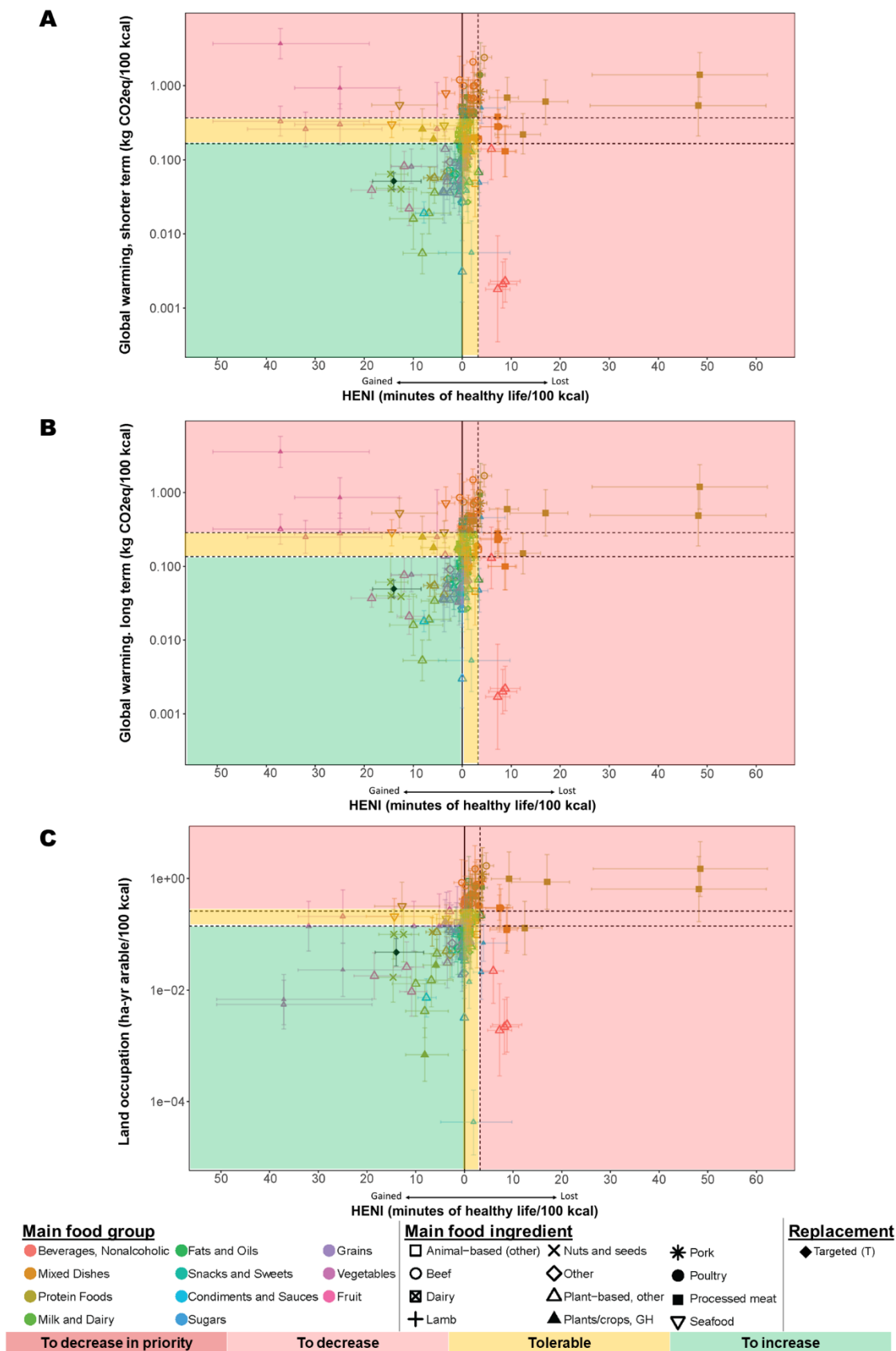


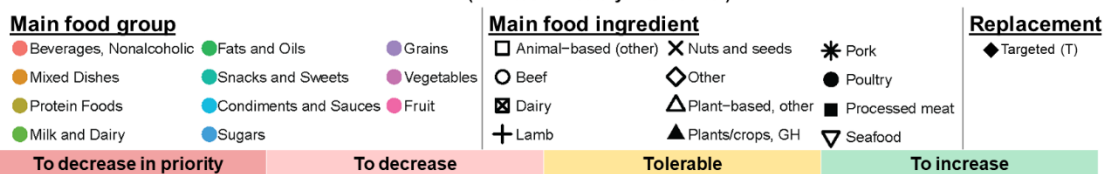
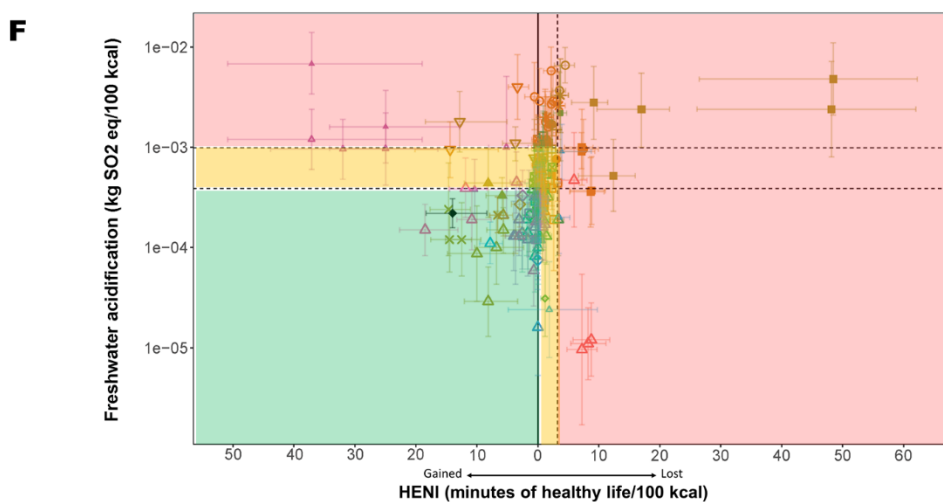
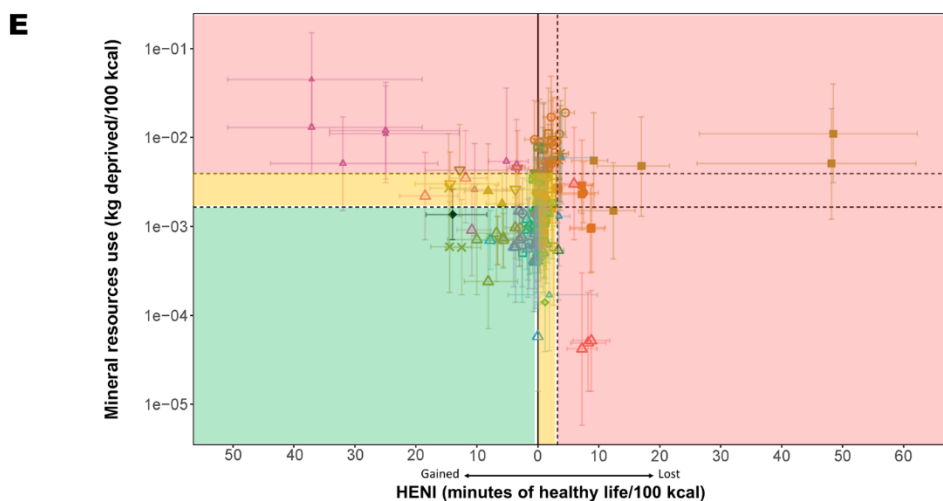
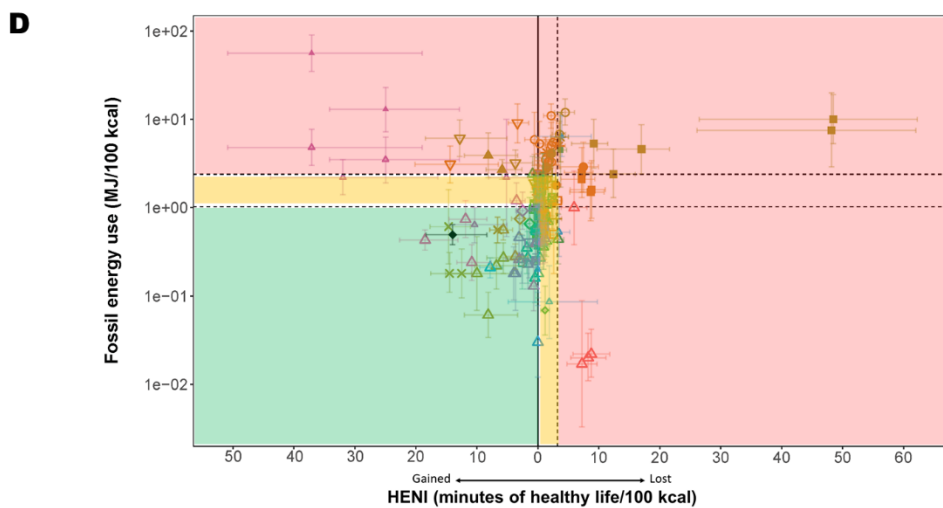
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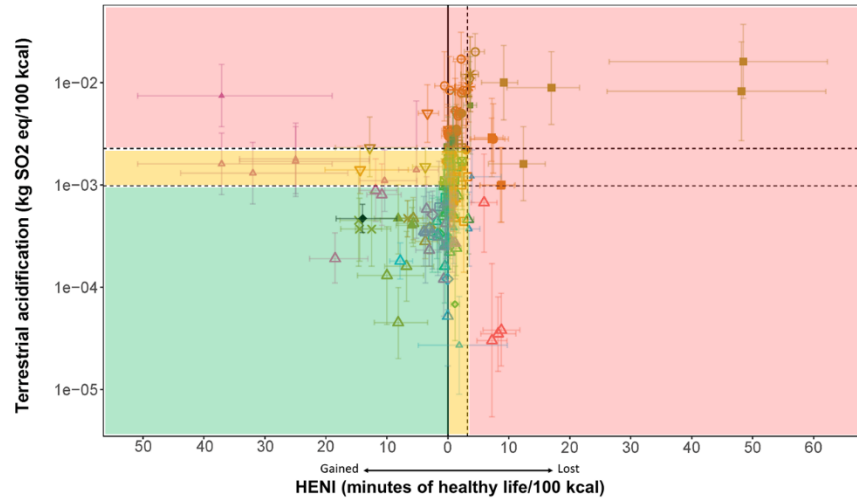
Suppl. Fig. 15. Comparison of nutritional impacts of the most popular and median HENI foods (N=167) in the US diet with their environmental impacts per serving for shorter term global warming (A), long-term global warming (B), land occupation impacts

948 (C), fossil energy use (D), mineral resources use (E), freshwater acidification (F),
949 terrestrial acidification (G), freshwater eutrophication (H), marine eutrophication (I),
950 freshwater ecotoxicity (J), ionizing radiation (K), ozone layer depletion (L), consumptive
951 water use (M), photochemical oxidation formation (N), fine particulate matter formation
952 (O), human toxicity (total, cancer and non-cancer) (P), total human health damages (Q),
953 and total ecosystem quality damages (R). Markers represent individual foods. Black
954 diamond represents a targeted mix of the most nutritionally beneficial foods from both
955 zones (T, replacement analysis). Horizontal and vertical lines represent the 95% confidence
956 interval of the nutrition and environmental indicator, respectively. GH: greenhouse.
957

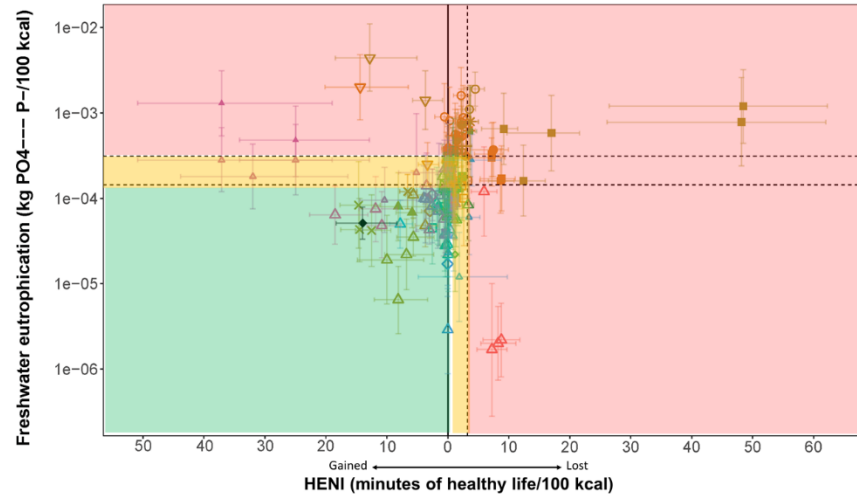




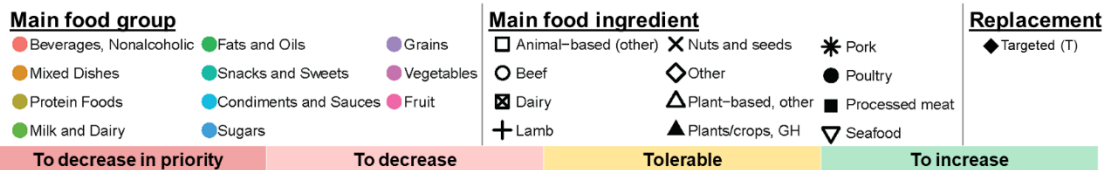
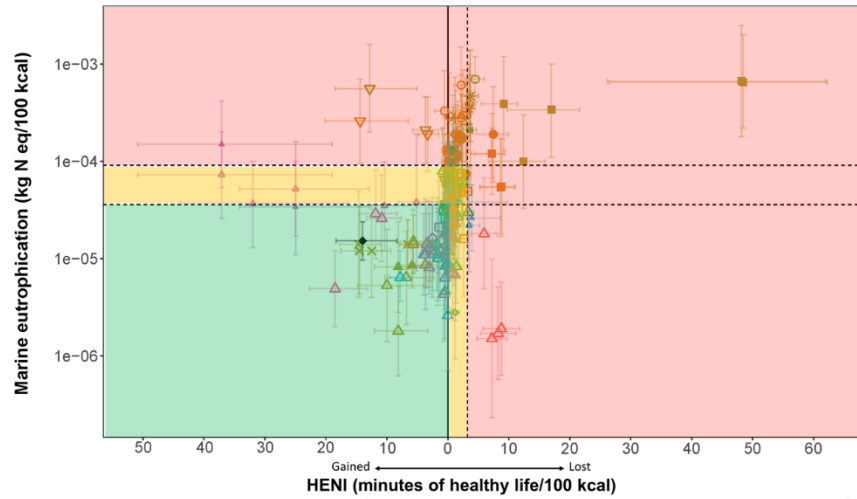
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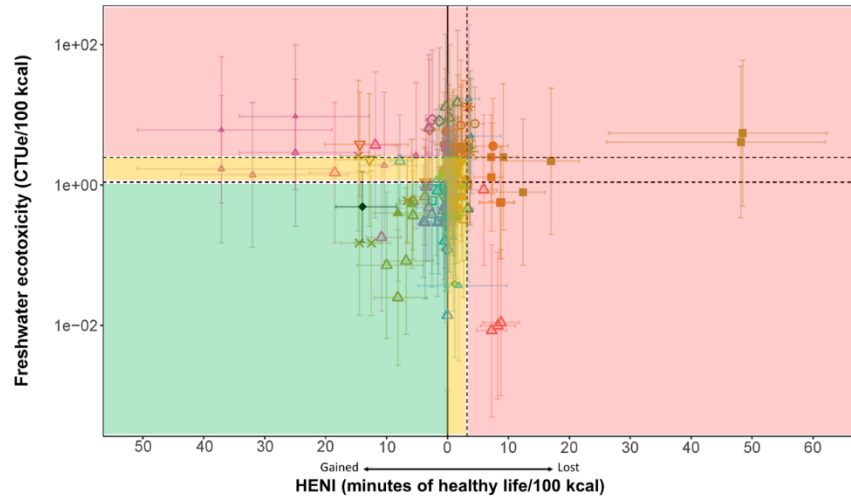
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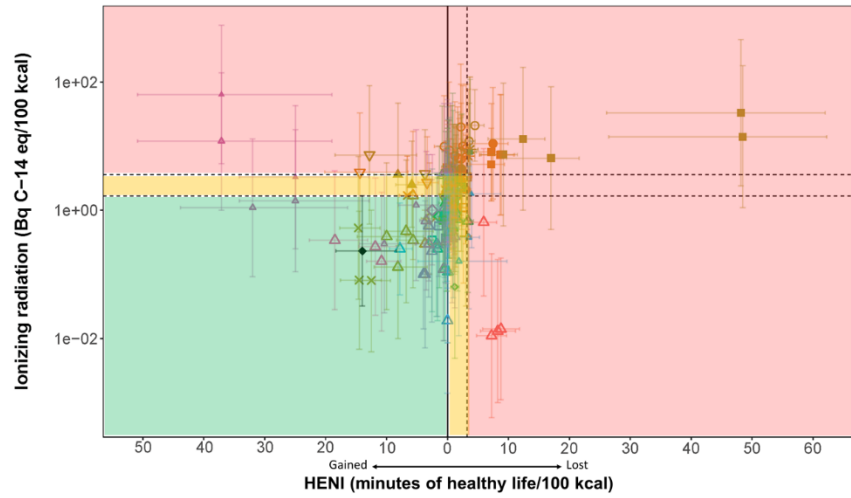
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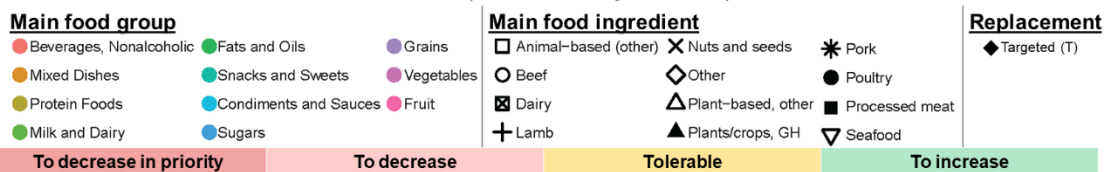
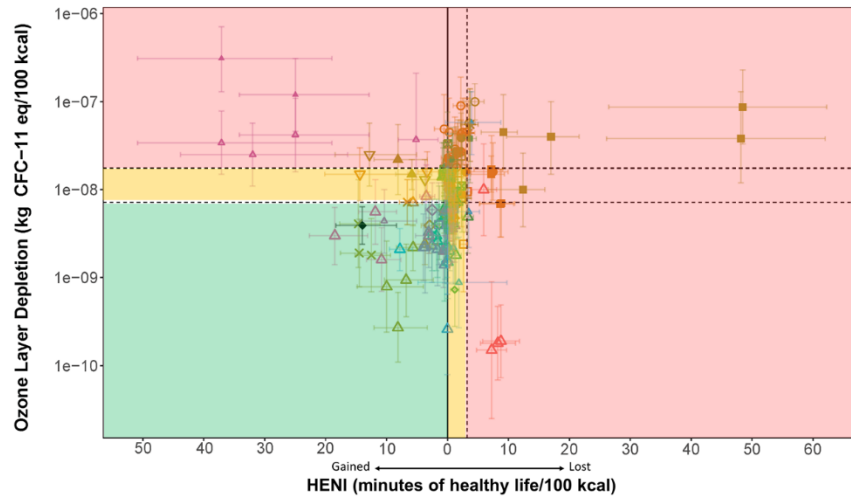
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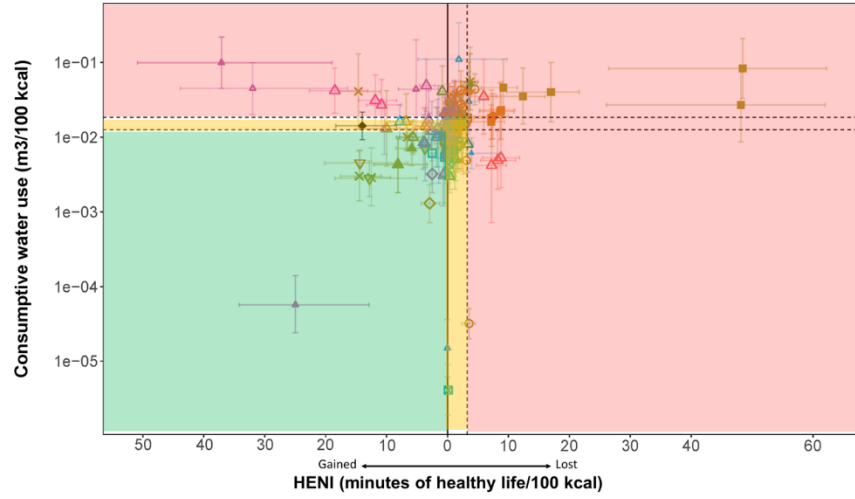
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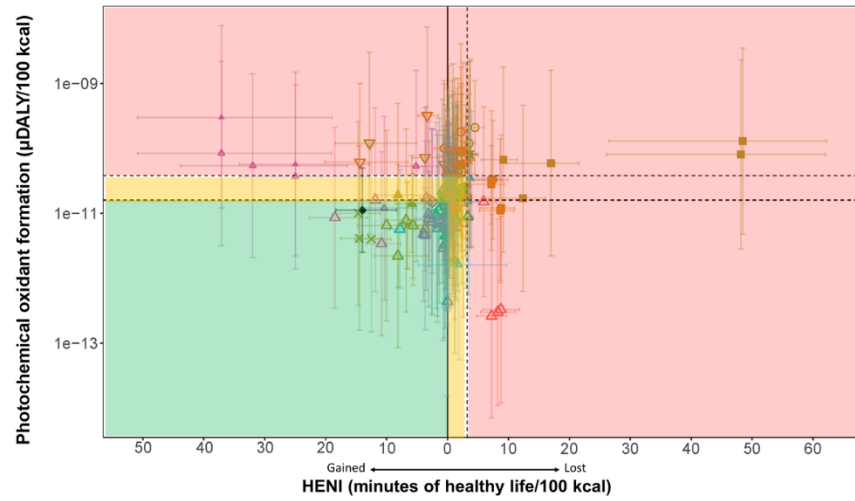
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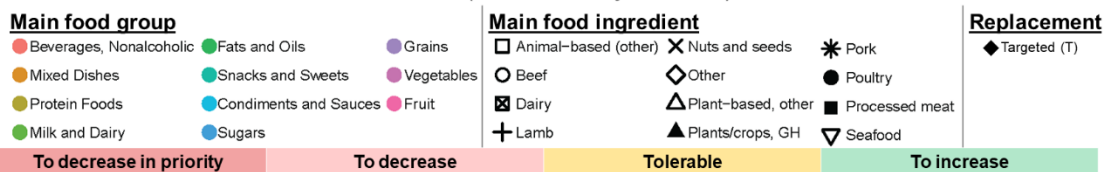
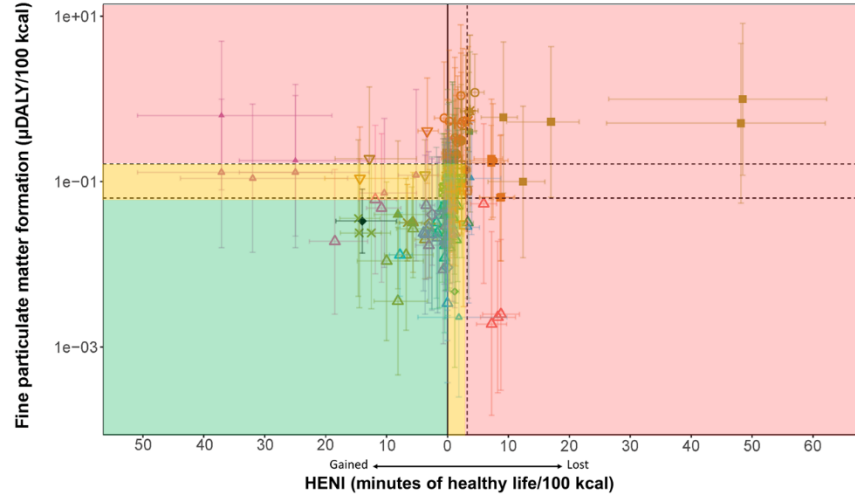
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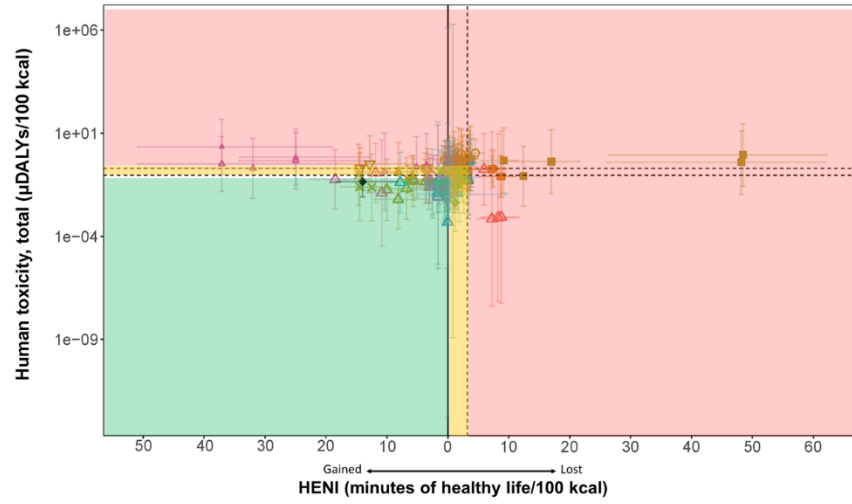
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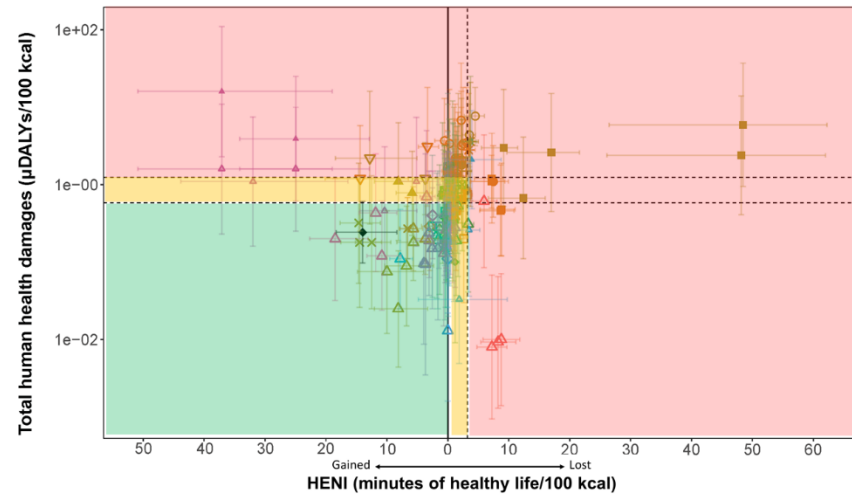
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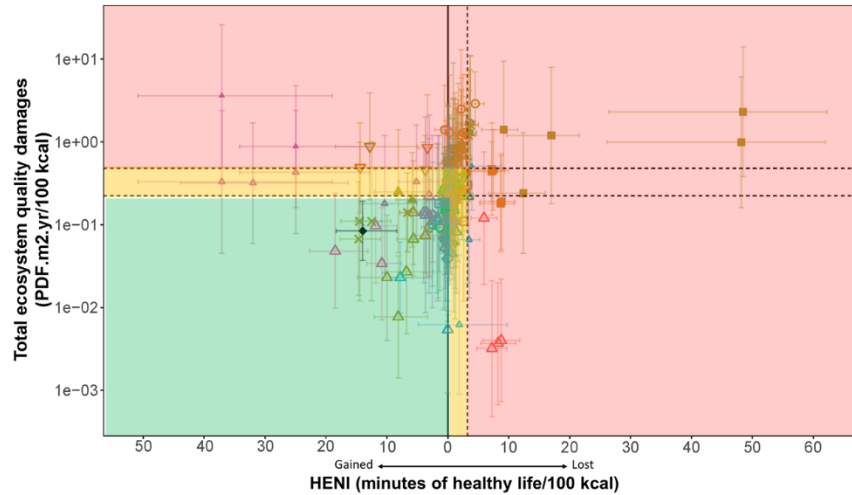
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Q



R



Main food group

- Beverages, Nonalcoholic
- Mixed Dishes
- Protein Foods
- Milk and Dairy
- Fats and Oils
- Snacks and Sweets
- Condiments and Sauces
- Sugars
- Grains
- Vegetables
- Fruit

Main food ingredient

- Animal-based (other)
- Beef
- ☒ Dairy
- + Lamb
- ✕ Nuts and seeds
- ◇ Other
- △ Plant-based, other
- ▲ Plants/crops, GH
- * Pork
- Poultry
- Processed meat
- ▽ Seafood

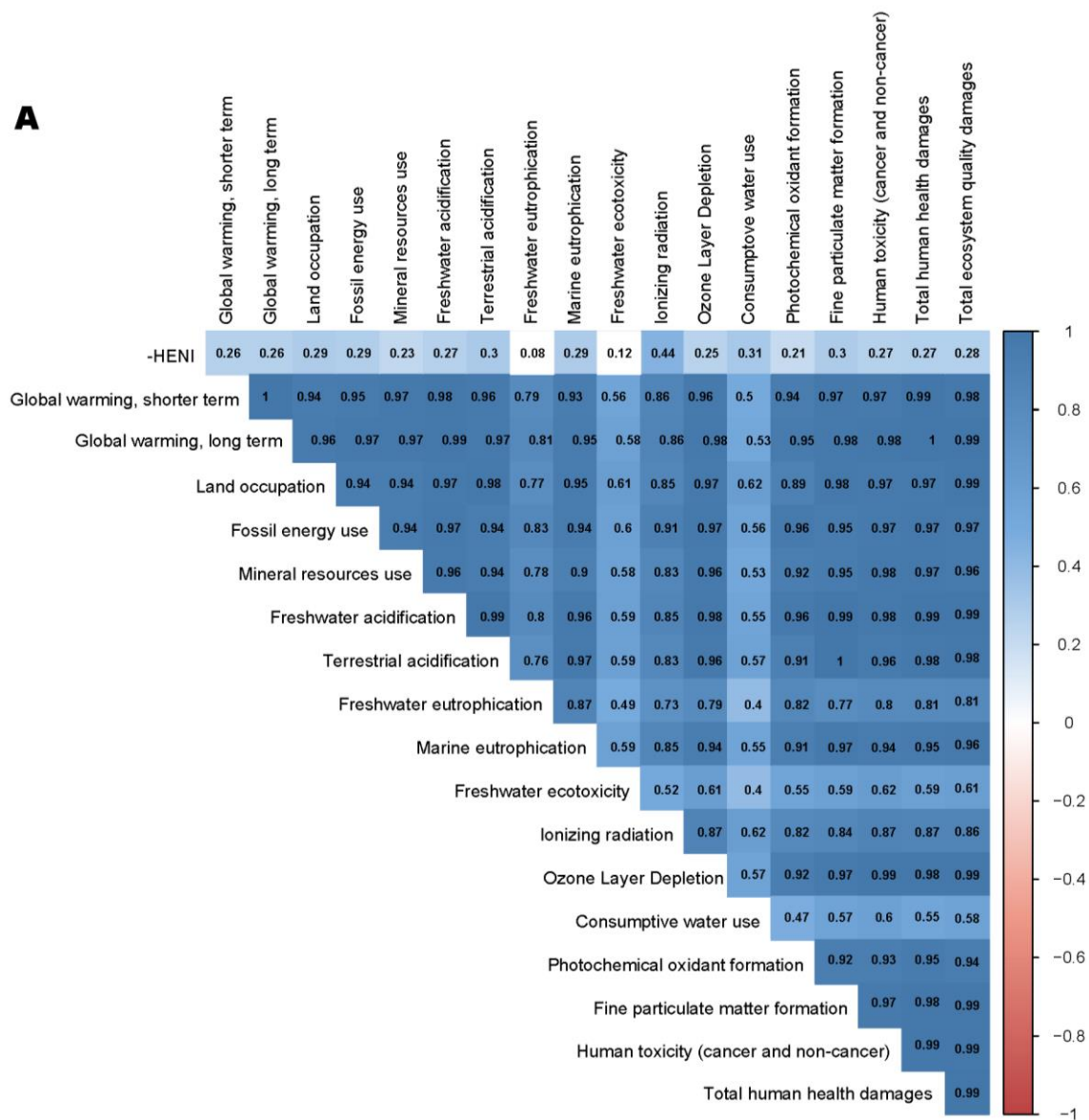
Replacement

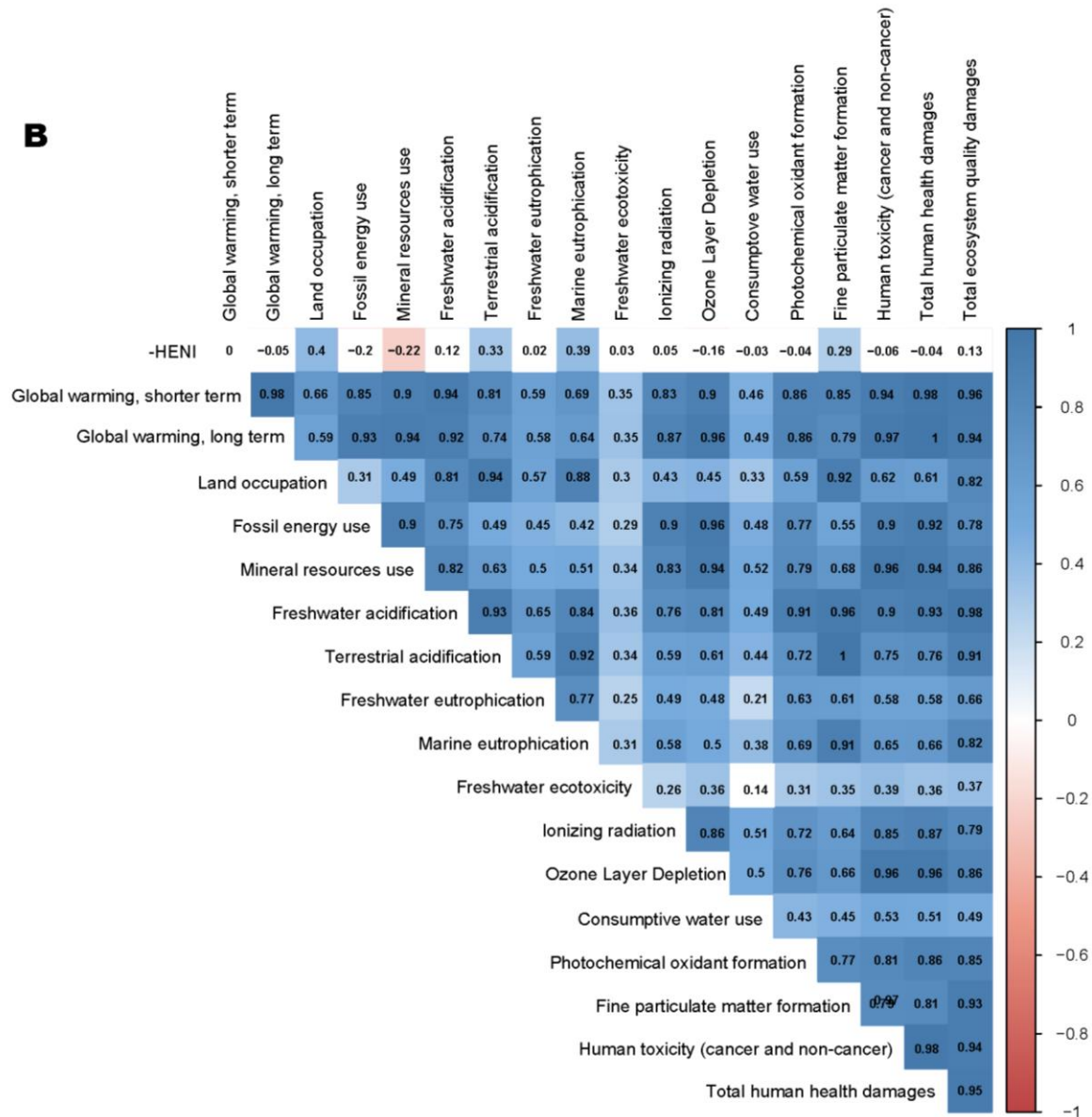
- ◆ Targeted (T)

To decrease in priority
To decrease
Tolerable
To increase

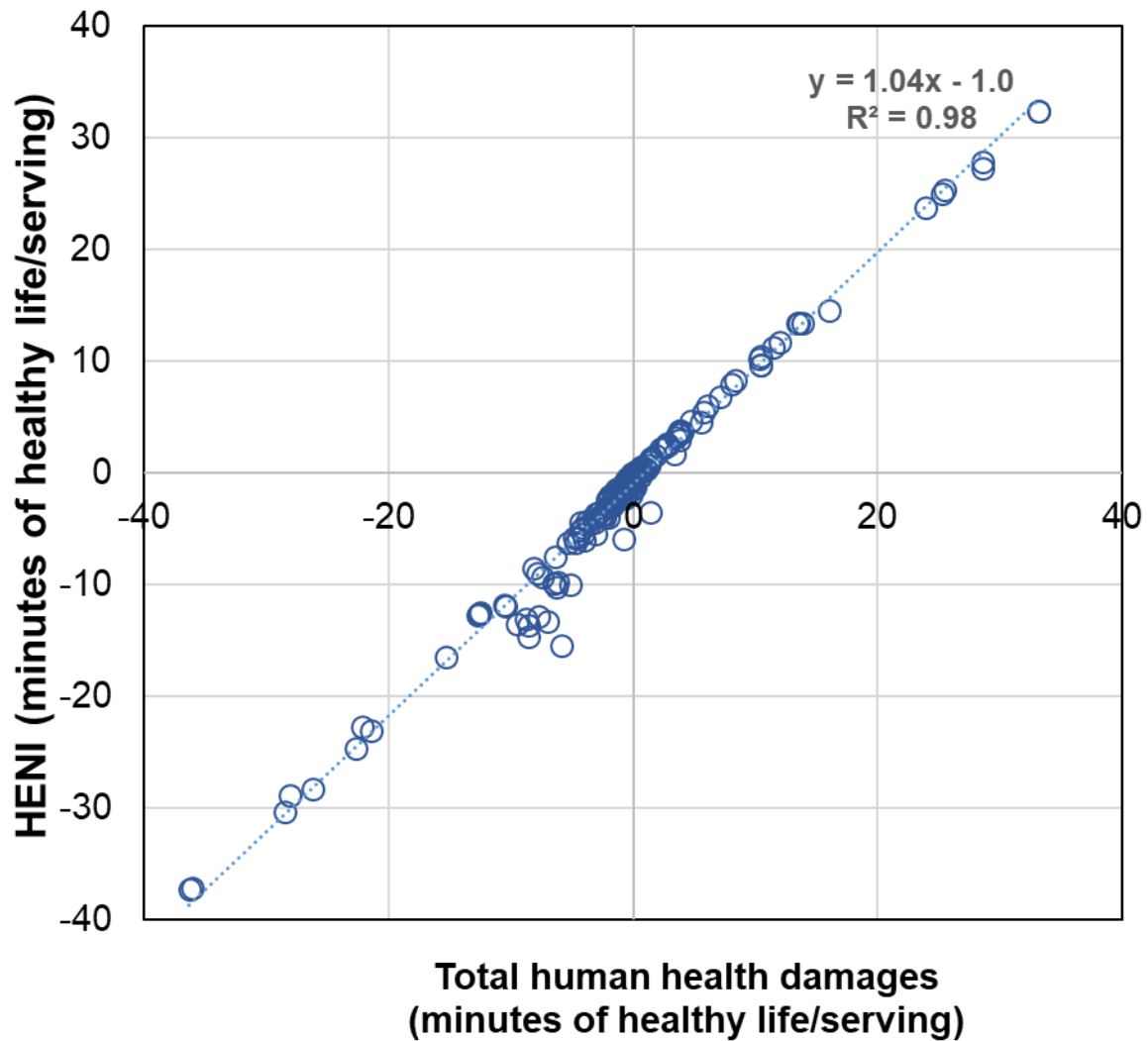
Suppl. Fig. 16. Comparison of nutritional impacts of the most popular and median HENI foods (N=167) in the US diet with their environmental impacts per 100 kcal for shorter term global warming impacts per serving (**A**), long-term global warming (**B**), land occupation impacts (**C**), fossil energy use (**D**), mineral resources use (**E**), freshwater acidification (**F**), terrestrial acidification (**G**), freshwater eutrophication (**H**), marine eutrophication (**I**), freshwater ecotoxicity (**J**), ionizing radiation (**K**), ozone layer depletion (**L**), consumptive water use (**M**), photochemical oxidation formation (**N**), fine particulate matter formation (**O**), human toxicity (total, cancer and non-cancer) (**P**), total human health damages (**Q**), and total ecosystem quality damages (**R**). Markers represent individual foods. Smaller markers indicate the foods for which 100 kcal exceed two servings. Black diamond represents a targeted mix of the most nutritionally beneficial foods from both zones (T, replacement analysis). Horizontal and vertical lines represent the 95% confidence interval of the nutrition and environmental indicator, respectively. GH: greenhouse.

A

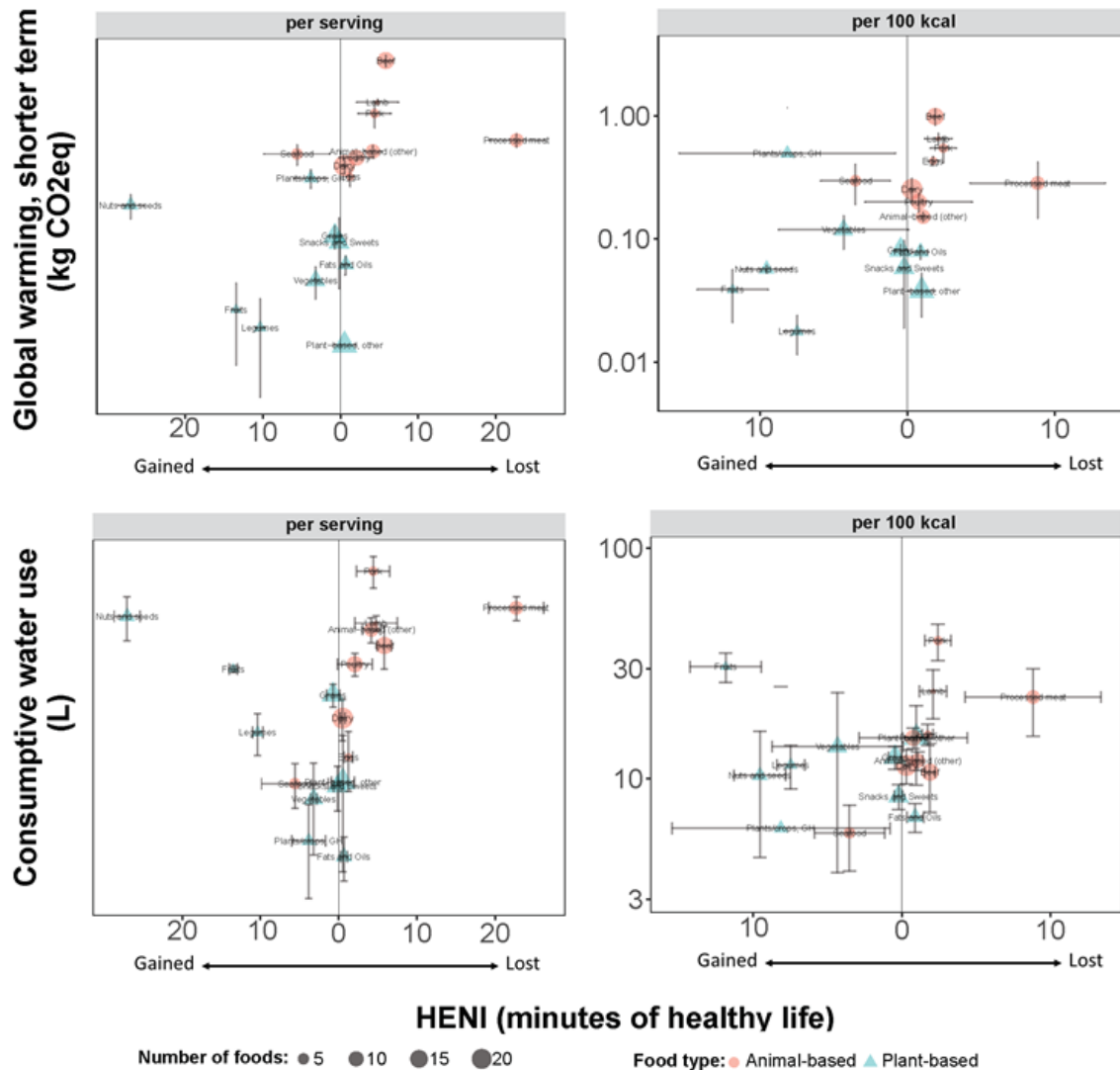




Suppl. Fig. 17. Pearson's correlation analysis of nutritional and environmental indicators per serving (A) and per 100 kcal (B). All coefficients were statistically significant ($p \leq 0.05$).



Suppl. Fig. 18. Comparison of nutritional and total human health damages per serving of food. Total human health burden includes nutritional and environmental impacts as indicated in Suppl. Table 7. Climate change impacts on human health were characterized using an uncertain association from De Schryver et al.⁸⁶.

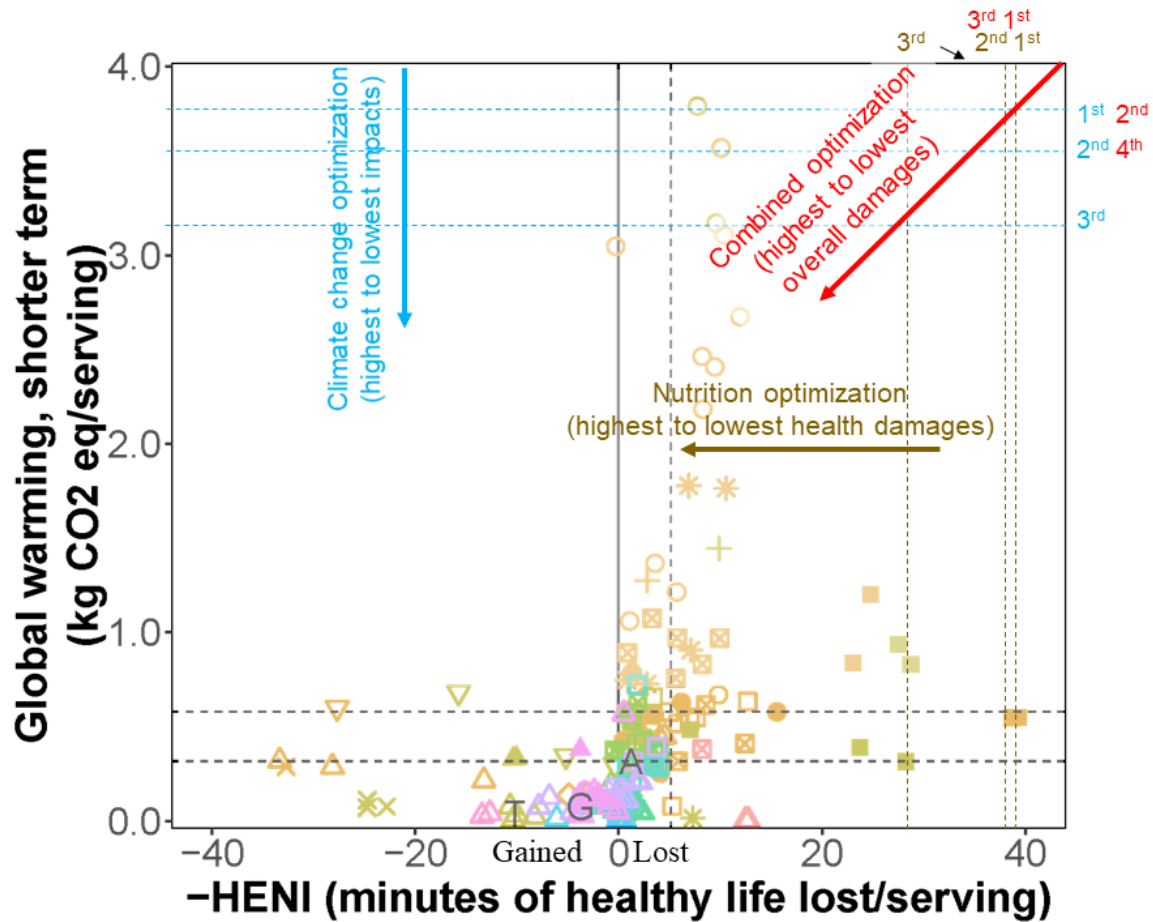


Suppl. Fig. 19. Nutritional and environmental impacts of the median HENI values and standard deviation (error bars) within 19 food groups of the 167 selected foods. Comparison with their environmental impacts for shorter-term global warming impacts per serving (A) and per 100 kcal (B), and for consumptive water use per serving (C) and per 100 kcal (D). Markers represent the median of each of the 19 food groups. The error bars indicating the standard error across foods within the food group. The size of markers represents the number of foods in each food group.

S5. Food substitutions

S5.1. Prioritization strategies

Supplementary figure 20 illustrates the three strategies developed to identify foods to be replaced in priority. In the nutrition optimization strategy, foods were replaced based on HENI scores per serving with foods generating the highest nutritional health damage substituted first (left to right substitution). If two or more foods generated the same HENI scores, priority was given to the food with the highest climate change impact and vice versa. For the climate change optimization approach, foods with the highest scores per serving were substituted first (top to bottom substitution). Based on the substitution ranking developed for the two aforementioned optimizations, a combined nutritional and environmental optimization approach (combined) was developed in which foods with the highest impact ranking were substituted first, starting with nutrition. In particular, this approach substituted first the food with the highest nutritional health damage, second the food with the highest climate change damage, third the food with the second highest nutritional health damage, and so on. This approach allowed for a ‘diagonal’ substitution from the top right (worst performing foods) to the bottom left corner (best performing foods) of the graph. Processed meat and beef make up the majority of foods to substitute in priority; however, identified foods and their substitution priority differ between strategies (Suppl. Table 12).



Suppl. Fig. 20. Graphical representation of prioritization strategies for food substitution. For nutrition, substitutions occur from left to right. For climate change, substitutions occur from top to bottom. For the combined approach, substitutions occur diagonally from the right top corner to the bottom left corner.

1022 **Suppl. Table 12.** List of food items to be substituted in priority. The nutritional and
 1023 environmental performance of each of these food items is provided in the Data S1 file.

	Optimization strategy		
	Nutrition	Climate change	Combined
1	Frankfurter or hot dog sandwich, beef, plain, on white bun	Beef burgundy (beef bourguignonne)	Frankfurter or hot dog sandwich, beef, plain, on white bun
2	Frankfurter or hot dog sandwich, beef, plain, on multigrain bun	Beef steak, broiled or baked, lean only eaten	Beef burgundy (beef bourguignonne)
3	Ham, prepackaged or deli, luncheon meat	Roast beef sandwich	Frankfurter or hot dog sandwich, beef, plain, on multigrain bun
4	Chicken or turkey loaf, prepackaged or deli, luncheon meat	Ground beef, cooked	Beef steak, broiled or baked, lean only eaten
5	Pork chop, smoked or cured, cooked, NS as to fat eaten	Meat loaf made with beef	Ham, prepackaged or deli, luncheon meat
6	Frankfurter or hot dog sandwich, with chili, on white bun	Chili con carne with beans	Roast beef sandwich
7	Pork sausage	Beef, rice, and vegetables including carrots, broccoli, and/or dark-green leafy; soy-based sauce	Chicken or turkey loaf, prepackaged or deli, luncheon meat
8	Cold cut submarine sandwich, with cheese, lettuce, tomato and spread	Meatball and spaghetti sauce submarine sandwich	Ground beef, cooked
9	Turkey submarine sandwich, with cheese, lettuce, tomato and spread	Double cheeseburger (2 patties), with tomato and/or catsup, on bun	Pork chop, smoked or cured, cooked, NS as to fat eaten
10	Soft drink, fruit flavored, caffeine containing	Cheeseburger, 1 medium patty, with condiments, on bun, from fast food / restaurant	Meat loaf made with beef

1024

S5.2. Food replacements

Foods were isocalorically substituted with one of three food mixes defined based on the nutritional performance of foods on a kcal basis. A targeted food mix was determined using the most nutritionally beneficial foods per kcal from the green and yellow zone (targeted replacement – T; Suppl. Table 13). This targeted food mix included vegetables, fruits, nuts, legumes, and select seafood with low environmental impacts. The mix composition was restricted so that the cumulative intake of each component, which is the sum of the current daily consumption for a given risk component and the amount considered in this mix, does not exceed the theoretical minimum risk level (see Suppl. Table 1). For risk components that required mixed amounts corresponding to more than one serving, various foods were considered in the mix and assumed an equal serving contribution (e.g., the 318 g of vegetables were equally divided equally into three vegetable foods from the green and yellow zone that offer the highest nutritional benefits per kcal). The performance of this targeted food mix was calculated for all environmental indicators in this analysis, assuming a joint replacement. More specifically, indicator performance per kcal was calculated as the caloric weighted average of individual foods considered. Assuming a joint replacement is realistic and addressed practical complications associated with food-by-food exchanges. Two additional food mixes were considered that represent the consumption-weighted average of foods in the green zone (green replacement - G) and the consumption-weighted average of foods in the amber zone (amber replacement – A; Suppl. Table 14).

1046 **Suppl. Table 13.** Selection of replacement foods from green and amber zone defined using climate change as the environmental indicator
1047 with maximum nutritional health gains per kcal.

Dietary component		Vegetables		Fruits		Nuts	Seafood		Legumes
Food		Lettuce, raw	Tomatoes, raw	Apple, raw	Banana, raw	Peanuts, salted	Mackerel, baked/broiled	Tuna, canned	Fava beans, cooked
Replacement	kcal/day	15	19	55	93	105	13	66	92
HENI	minutes _{healthy} /kcal	0.32	0.25	0.19	0.11	0.15	0.53	0.091	0.066
Global warming, shorter term	g CO ₂ eq/kcal	2.6	3.0	0.39	0.22	0.41	0.38	0.48	0.21
Global warming, long term	g CO ₂ eq/kcal	2.5	2.8	0.37	0.21	0.40	0.37	0.47	0.21
Land occupation, biodiversity	ha-yr arable/kcal	1.4	2.1	0.18	0.094	1.0	0.0042	0.00082	0.34
Fossil energy use	MJ/kcal	0.022	0.035	0.0043	0.0024	0.0018	0.0053	0.0068	0.0011
Mineral resources use	mg _{deprived} /kcal	51	123	22	9.1	5.9	2.6	2.8	2.6
Freshwater acidification	mg SO ₂ eq/kcal	13	17	1.9	8.0	3.7	2.8	3.5	1.3
Terrestrial acidification	mg SO ₂ eq/kcal	1.8	2.8	0.64	0.48	0.43	0.11	0.12	0.21
Freshwater eutrophication	mg PO ₄ ---P- eq/kcal	0.37	0.52	0.049	0.26	0.12	0.11	0.14	0.045
Marine eutrophication	g N eq/kcal	14	29	15	1.8	1.5	0.93	0.26	3.29
Freshwater ecotoxicity	CTUe/kcal	0.011	0.014	0.0034	0.0016	0.00082	0.00064	0.00072	0.0016
Ionizing radiation	Bq C-14 eq/kcal	2.46E-10	4.17E-10	3.00E-11	1.61E-11	1.86E-11	5.48E-12	5.73E-12	8.84E-12
Ozone Layer Depletion	mg CFC-11 eq/kcal	451	0.57	417	270	30	5.9	2.3	74
Water use	mL/kcal	9.6	9.7	1.5	1.9	1.2	2.5	3.2	0.55
Photochemical oxidant formation	nDALYs/kcal	1.1	1.3	0.19	0.48	0.24	0.24	0.31	0.087
Fine particulate matter formation	pDALYs/kcal	0.54	0.37	0.086	0.034	0.041	0.21	0.27	0.036
Human toxicity (cancer and non-cancer)	nDALYs/kcal	2.0	4.6	0.58	0.13	0.25	0.10	0.11	0.11
Total human health	nDALYs/kcal	11	16	2.0	1.2	1.8	1.7	2.1	0.88
Total ecosystem quality	PDF.m2.yr/kcal	0.0032	0.0043	0.00048	0.00034	0.0011	0.00041	0.00051	0.00044

1048

1049 **Suppl. Table 14.** Nutritional and environmental performance of selected replacement scenarios.

Scenario*		Amber (A)	Green (G)	Targeted (T)	
Description		Consumption-weighted average of foods in the amber zone	Consumption-weighted average of foods in the green zone	The caloric weighted average of foods with maximum nutritional health gains per kcal from green and amber zone (Suppl. Table 10)	
				Mean	GSD ²
HENI	minutes _{healthy} /kcal	-0.0019	0.017	0.14	(0.083,0.18) [†]
Global warming, shorter term	g CO ₂ eq/kcal	0.71	0.21	0.51	1.30
Global warming, long term	g CO ₂ eq/kcal	0.53	0.20	0.50	1.30
Land occupation, biodiversity	ha-yr arable/kcal	0.00054	0.00020	0.00048	1.76
Fossil energy use	MJ/kcal	0.0038	0.0021	0.0049	1.30
Mineral resources use	mg _{deprived} /kcal	6.5	4.0	14	1.91
Freshwater acidification	mg SO ₂ eq/kcal	1.5	0.65	2.2	1.40
Terrestrial acidification	mg SO ₂ eq/kcal	3.7	1.3	4.7	1.38
Freshwater eutrophication	mg PO ₄ ---P- eq/kcal	0.54	0.25	0.51	1.54
Marine eutrophication	mg N eq/kcal	0.14	0.043	0.15	1.57
Freshwater ecotoxicity	CTUe/kcal	0.0052	0.0060	0.0049	3.14
Ionizing radiation	Bq C-14 eq/kcal	0.0062	0.0022	0.0023	7.22
Ozone Layer Depletion	pg CFC-11 eq/kcal	31	16	39	1.63
Water use	mL/kcal	43	35	141	1.52
Photochemical oxidant formation	fDALYs/kcal	58	28	112	4.47
Fine particulate matter formation	nDALYs/kcal	0.25	0.095	0.33	2.44
Human toxicity (cancer and non-cancer)	nDALYs/kcal	0.35	0.18	0.45	5.51
Total human health	nDALYs/kcal	2.3	0.93	2.4	2.50
Total ecosystem quality	PDF.m2.yr/kcal	0.00087	0.00034	0.00084	2.28

*Foods identification based on zones defined using global warming (shorter term) as the environmental indicator

[†]95% confidence interval

S5.3. Scenario analysis

The three substitution prioritization strategies were coupled with the three replacement food mixes to investigate the potential environmental and nutritional benefits from different dietary change scenarios. The potential benefits associated with the proposed dietary changes were estimated as the cumulative difference of the nutritional and environmental performance of the replacement food mixes and the 167 foods in this analysis as a function of daily caloric intake, taking into account the food-specific reported intake in the average diet of US adults and assuming isocaloric substitutions (Fig. 6 and Suppl. Fig. 21).

As discussed in the main text, the targeted substitution strategy enables high nutritional benefits substantial cumulative reductions of dietary carbon footprint. The combined targeted strategy also enables cumulative reduction in the range of 30% reduction for most other midpoint and endpoint impact categories, similar to carbon footprint but with higher maximum uncertainty ranges, except for water use (Suppl. Table 15). The limited improvement of this strategy for water use (6% cumulative reduction, 95% CI: -9% to 25%) are due to the moderate consumptive water use required to supply the targeted food mix, which gets classified in the amber zone for this category (Fig. 6D). For all other categories, the targeted replacement generates low environmental impacts that classify the food mix in the green zone (Suppl. Fig. 16).

The benefits from a 10% daily caloric change for the nine scenarios are summarized in Suppl. Table 15. For nutrition, results have been extrapolated to reflect diet-level benefits (in minutes of healthy life saved) using a multiplier of 3.41, which corresponds to the relative difference between the average diet-level daily caloric intake of the foods

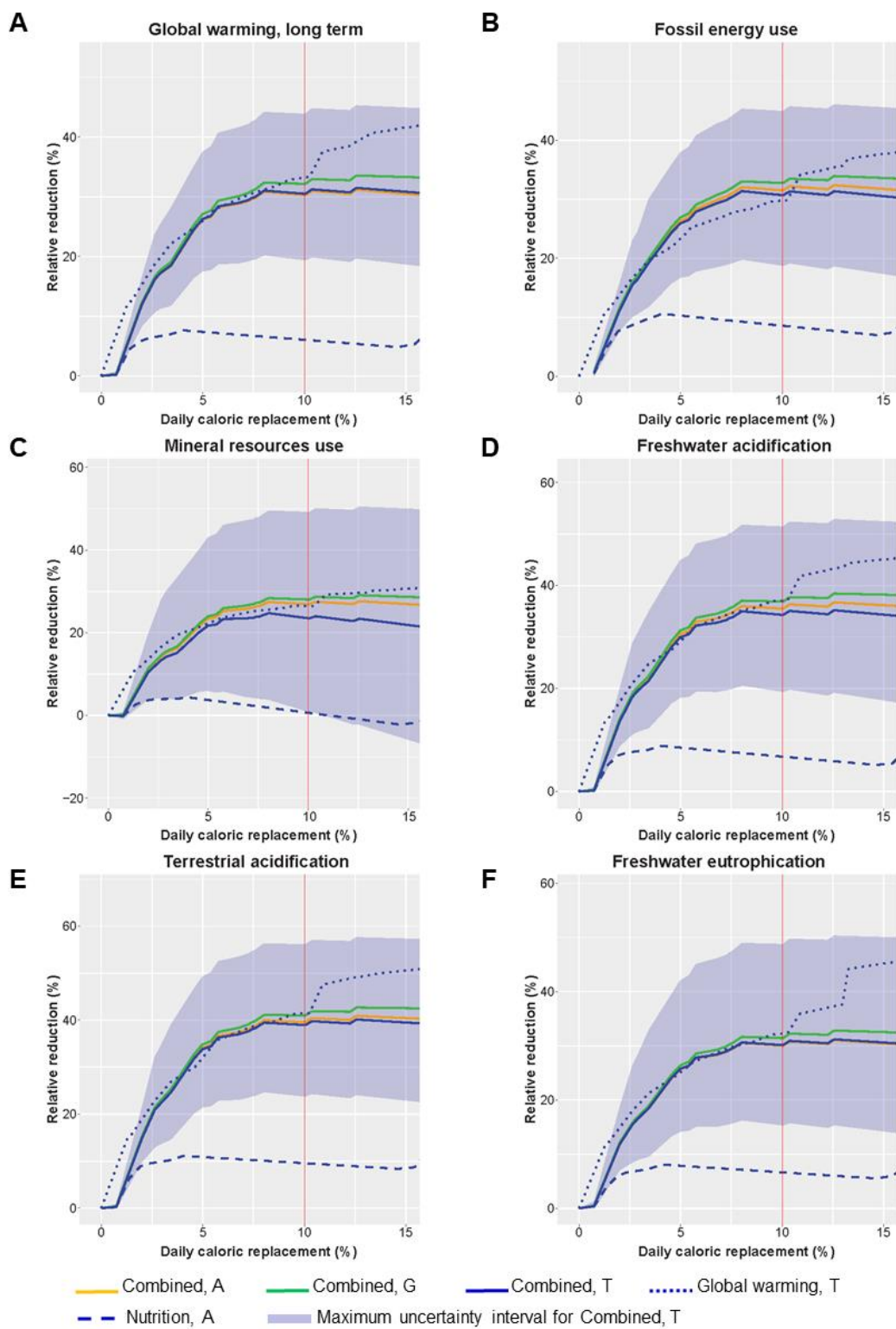
1075 considered in this analysis (1905 kcal/person/day) and the daily caloric intake of the 167
1076 foods in this analysis (570 kcal/person/day). Environmental benefits are reported as relative
1077 impact reductions between the impacts of the 167 foods and the scenario analyzed. The
1078 representativeness of this extrapolation was evaluated by applying the optimization
1079 strategy for nutrition only and substituting with the targeted food mix, to the 167 select
1080 foods compared with the same strategy applied to the entire 5,853 foods consumed in
1081 WWEIA/NHANES. The two approaches give diet-level nutritional benefits within 3
1082 minutes saved per day for a 10% daily caloric substitution (Suppl. Fig. 22). These similar
1083 results suggest that the 167 foods studied in this analysis are indeed a representative subset
1084 of the diet, at least for the first 10% daily caloric substitution.

1085 **Suppl. Table 15.** Cumulative nutritional benefits and relative environmental reductions from a 10% daily caloric change in the US
 1086 diet by substitution approach and replacement scenario.

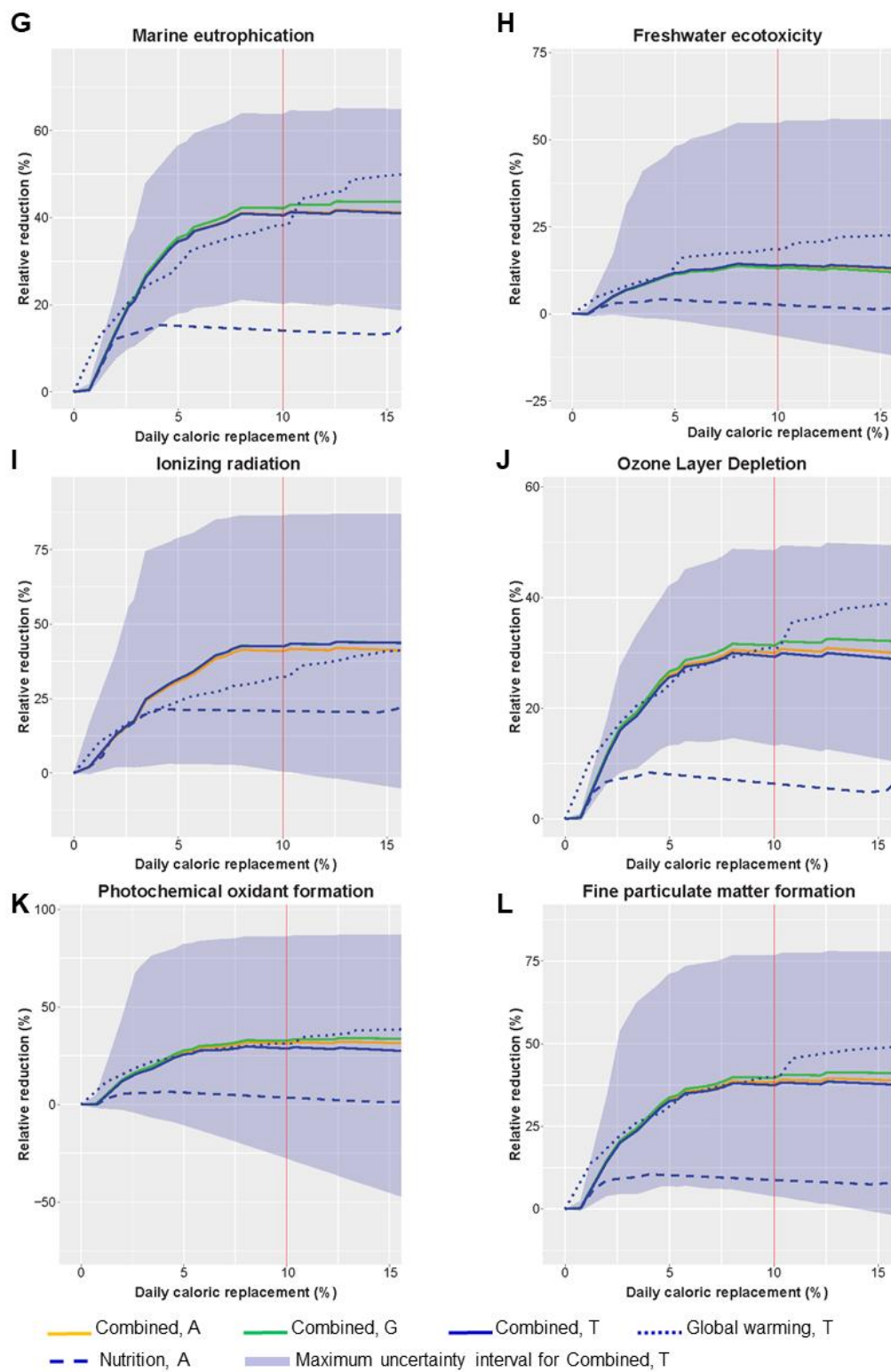
Scenario*	Nutrition			Climate change			Combined				
	A	G	T	A	G	T	A	G	T		
									Mean	Lower [†]	Upper [†]
HENI (minutes gained/person/day)	25	28	52	4.7	8.3	31	21	24	48	28	62
Global warming, shorter term	5%	7%	6%	35%	38%	36%	32%	34%	33%	22%	46%
Global warming, long term	6%	8%	6%	33%	35%	33%	30%	32%	30%	19%	44%
Land occupation	7%	9%	7%	34%	36%	34%	30%	32%	31%	13%	52%
Fossil energy use	9%	11%	9%	31%	32%	30%	32%	33%	31%	19%	45%
Mineral resources use	4%	5%	1%	30%	31%	26%	27%	28%	24%	1%	49%
Freshwater acidification	8%	9%	7%	38%	40%	37%	36%	37%	34%	19%	51%
Terrestrial acidification	10%	12%	10%	42%	44%	41%	40%	41%	39%	24%	56%
Freshwater eutrophication	7%	8%	7%	32%	34%	32%	30%	31%	30%	15%	49%
Marine eutrophication	14%	16%	14%	38%	40%	38%	41%	42%	41%	20%	64%
Freshwater ecotoxicity	2%	2%	3%	18%	18%	19%	14%	13%	14%	-6%	55%
Ionizing radiation	19%	21%	21%	31%	32%	32%	41%	43%	43%	1%	86%
Ozone Layer Depletion	7%	8%	6%	32%	33%	31%	30%	31%	29%	13%	49%
Water use	9%	10%	2%	11%	11%	3%	13%	14%	6%	-9%	26%
Photochemical oxidant formation	6%	8%	4%	34%	35%	31%	31%	33%	29%	-27%	86%
Fine particulate matter formation	10%	11%	9%	41%	42%	40%	38%	40%	37%	4%	77%
Human toxicity (cancer & non-cancer)	6%	8%	6%	31%	32%	30%	29%	30%	28%	-22%	90%
Total human health	6%	8%	6%	34%	35%	33%	31%	33%	31%	4%	65%
Total ecosystem quality	7%	9%	7%	34%	36%	34%	31%	33%	31%	6%	64%

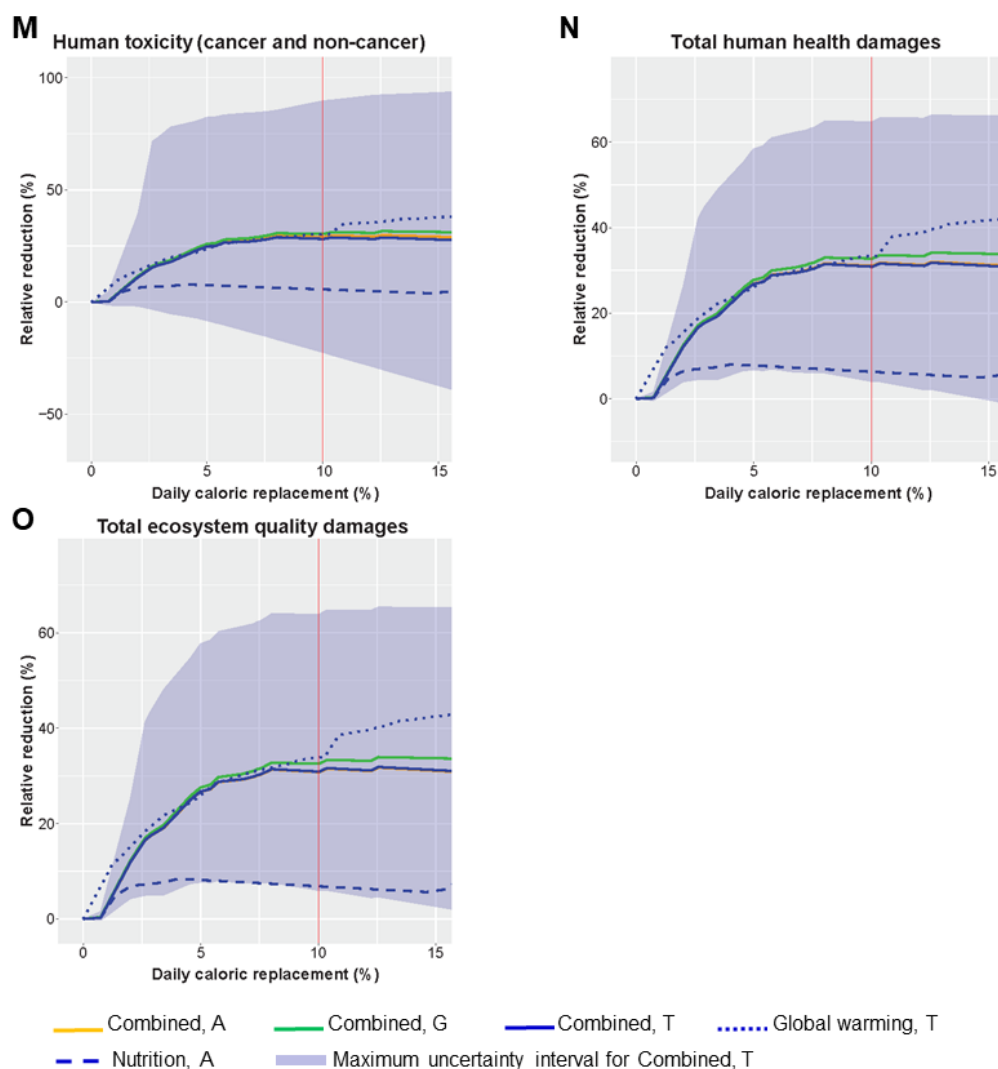
1087 *Nutrition, Climate change, and combined scenarios indicate the food substitution prioritization criterion applied. A, G, and T indicate the isocaloric replacement considered.
 1088 A=Amber; G- Green; T= Targeted

1089 [†]Lower and upper estimated for the “Combined, T” scenario indicate the 95% confidence interval limits of the cumulative estimates that were obtained under the assumption that
 1090 all substitutions follow the same uncertainty direction while the replacement follows the opposite uncertainty direction.
 1091



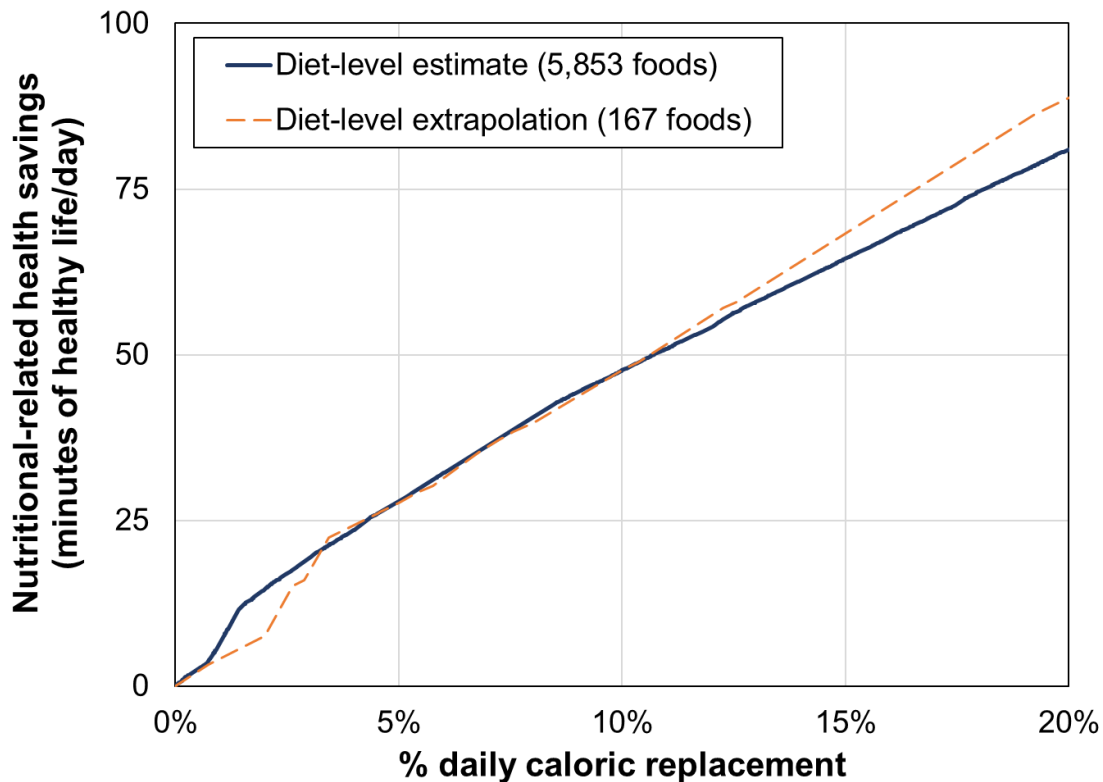
1092





Suppl. Fig. 21. Benefits from isocaloric substitutions of the most impactful foods in the US diet by targeted replacements, scaled at the diet level for global warming, long term (A), fossil energy use (B), mineral resources use (C), freshwater acidification (D), terrestrial acidification (E), freshwater eutrophication (F), marine eutrophication (G), freshwater ecotoxicity (H), ionizing radiation (I), ozone layer depletion (J), photochemical oxidation formation (K), fine particulate matter formation (L), human toxicity (total, cancer and non-cancer) (M), total human health damages (N), and total ecosystem quality damages (O). Foods were substituted in priority based on nutrition only (dashed curve), climate change only (dotted curve), or combined performances per serving (solid curve)

with food mixes from the amber zone (A, amber curve), the green zone (G, green curve), or a targeted mix of the most nutritionally beneficial foods per kcal from both zones (T, blue curve). The blue shaded area denotes the maximum uncertainty interval of the combined replacement of foods by a targeted food mix, under the assumption that all substitutions follow the same uncertainty direction while the replacement follows the opposite uncertainty direction.



Suppl. Fig. 22. Nutritional health benefits from diet-level isocaloric substitutions with the targeted food mix when optimizing only for nutrition. Benefits were estimates based on 5,853 foods in the US diet (solid blue line) and a caloric extrapolation from the 167 commonly consumed foods in the US diet (dashed orange line).

Data S1. Data in spreadsheet format (separate file)

This file contains the underlying nutritional and environmental data used to estimate the corresponding impacts of the 167 commonly used food in the average US diet. Impacts are reported per serving. The substitution order used in the replacement analysis is also reported.

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